Mathematics

Class 3 (English Medium)

Solved Exercises — Sindh Textbook Board

7 lessons — 777 questions

Unit 1 NUMBERS

Activity — Complete: (Roman Numbers)

Question: 8 = 5 + 1 + 1 + 1 = _____

Answer: Step 1: 5 is written as V.

Step 2: 1 + 1 + 1 is written as III.

Step 3: Join them: V + III = VIII.

Final answer: 8 = VIII.

Question: 13 = _____

Answer: Step 1: 13 = 10 + 3.

Step 2: 10 is X and 3 is III.

Step 3: Join them: X + III = XIII.

Final answer: 13 = XIII.

Question: 14 = _____

Answer: Step 1: 14 = 10 + 4.

Step 2: 10 is X and 4 is IV (one less than 5).

Step 3: Join them: X + IV = XIV.

Final answer: 14 = XIV.

Question: 16 = _____

Answer: Step 1: 16 = 10 + 5 + 1.

Step 2: 10 is X, 5 is V and 1 is I.

Step 3: Join them: X + V + I = XVI.

Final answer: 16 = XVI.

Question: 19 = _____

Answer: Step 1: 19 = 10 + 9.

Step 2: 10 is X and 9 is IX (one less than 10).

Step 3: Join them: X + IX = XIX.

Final answer: 19 = XIX.

Question: 3 = _____

Answer: Step 1: 3 = 1 + 1 + 1.

Step 2: Write I three times: III.

Final answer: 3 = III.

Question: 4 = _____

Answer: Step 1: 4 is one less than 5.

Step 2: Write I before V: IV.

Final answer: 4 = IV.

Question: 6 = _____

Answer: Step 1: 6 = 5 + 1.

Step 2: 5 is V and 1 is I.

Step 3: Join them: V + I = VI.

Final answer: 6 = VI.

Question: 8 = _____

Answer: Step 1: 8 = 5 + 1 + 1 + 1.

Step 2: 5 is V and 1 + 1 + 1 is III.

Step 3: Join them: V + III = VIII.

Final answer: 8 = VIII.

Question: 12 = _____

Answer: Step 1: 12 = 10 + 1 + 1.

Step 2: 10 is X and 1 + 1 is II.

Step 3: Join them: X + II = XII.

Final answer: 12 = XII.

EXERCISE 1 — 1. Write down the following in Roman numbers.

Question: (1) 5 _____

Answer: Step 1: 5 is a basic Roman symbol.

Final answer: 5 = V.

Question: (2) 7 _____

Answer: Step 1: 7 = 5 + 1 + 1.

Step 2: 5 is V and 1 + 1 is II.

Step 3: Join them: V + II = VII.

Final answer: 7 = VII.

Question: (3) 11 _____

Answer: Step 1: 11 = 10 + 1.

Step 2: 10 is X and 1 is I.

Step 3: Join them: X + I = XI.

Final answer: 11 = XI.

Question: (4) 9 _____

Answer: Step 1: 9 is one less than 10.

Step 2: Write I before X: IX.

Final answer: 9 = IX.

Question: (5) 8 _____

Answer: Step 1: 8 = 5 + 1 + 1 + 1.

Step 2: Join V and III: VIII.

Final answer: 8 = VIII.

Question: (6) 16 _____

Answer: Step 1: 16 = 10 + 5 + 1.

Step 2: Join X, V and I: XVI.

Final answer: 16 = XVI.

Question: (7) 17 _____

Answer: Step 1: 17 = 10 + 5 + 1 + 1.

Step 2: Join X, V and II: XVII.

Final answer: 17 = XVII.

Question: (8) 20 _____

Answer: Step 1: 20 = 10 + 10.

Step 2: Write X two times: XX.

Final answer: 20 = XX.

EXERCISE 1 — 2. Write the following in Hindi Arabic numbers.

Question: (1) III _____

Answer: Step 1: III = 1 + 1 + 1.

Step 2: 1 + 1 + 1 = 3.

Final answer: III = 3.

Question: (2) IV _____

Answer: Step 1: I before V means one less than 5.

Step 2: 5 - 1 = 4.

Final answer: IV = 4.

Question: (3) VI _____

Answer: Step 1: VI = 5 + 1.

Step 2: 5 + 1 = 6.

Final answer: VI = 6.

Question: (4) X _____

Answer: Step 1: X is a basic Roman symbol.

Final answer: X = 10.

Question: (5) II _____

Answer: Step 1: II = 1 + 1 = 2.

Final answer: II = 2.

Question: (6) XV _____

Answer: Step 1: XV = 10 + 5.

Step 2: 10 + 5 = 15.

Final answer: XV = 15.

Question: (7) XII _____

Answer: Step 1: XII = 10 + 1 + 1.

Step 2: 10 + 1 + 1 = 12.

Final answer: XII = 12.

Question: (8) XIX _____

Answer: Step 1: XIX = 10 + 9.

Step 2: 10 + 9 = 19.

Final answer: XIX = 19.

EXERCISE 1 — 3. Write the missing numbers.

Question: Complete the table: I, _____, III, _____, _____, VI, _____, _____, _____, X and _____, _____, XIII, _____, _____, _____, XVII, _____, _____, _____

Answer: Step 1: The numbers go in counting order from I (1) to XX (20).

Step 2: After I comes II, after III comes IV and V, after VI comes VII, VIII and IX, then X.

Step 3: After X comes XI and XII, after XIII comes XIV, XV and XVI, after XVII comes XVIII, XIX and XX.

Final answer: The missing numbers are II, IV, V, VII, VIII, IX and XI, XII, XIV, XV, XVI, XVIII, XIX, XX.

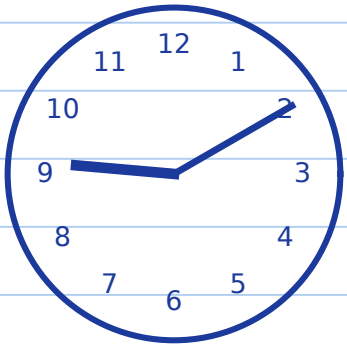
EXERCISE 1 — 4. Write the missing Roman numbers of the given clocks.

Question: (a) Clock (a) shows XI, I, III, V, VI and IX. Write the missing Roman numbers.

Answer: Step 1: A clock face has the numbers I to XII in order.

Step 2: The numbers not shown are II, IV, VII, VIII, X and XII.

Final answer: The missing numbers on clock (a) are II, IV, VII, VIII, X and XII.

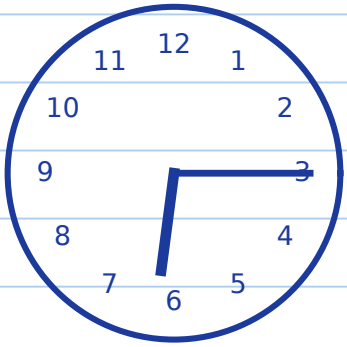


Question: (b) Clock (b) shows XII, X, II, VIII, IV and VI. Write the missing Roman numbers.

Answer: Step 1: A clock face has the numbers I to XII in order.

Step 2: The numbers not shown are I, III, V, VII, IX and XI.

Final answer: The missing numbers on clock (b) are I, III, V, VII, IX and XI.

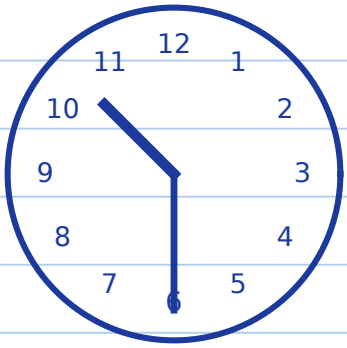


Question: (c) Clock (c) shows XI, X, II, III, VII and VI. Write the missing Roman numbers.

Answer: Step 1: A clock face has the numbers I to XII in order.

Step 2: The numbers not shown are I, IV, V, VIII, IX and XII.

Final answer: The missing numbers on clock (c) are I, IV, V, VIII, IX and XII.

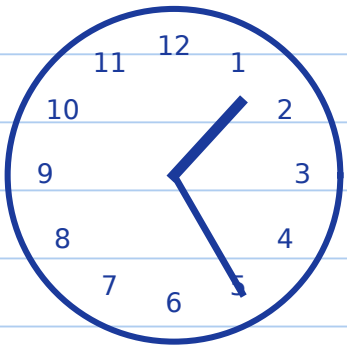


Question: (d) Clock (d) shows XI, X, I, II, III, VIII, VII and V. Write the missing Roman numbers.

Answer: Step 1: A clock face has the numbers I to XII in order.

Step 2: The numbers not shown are IV, VI, IX and XII.

Final answer: The missing numbers on clock (d) are IV, VI, IX and XII.



EXERCISE 1 — 5. Match the following.

Question: Match: V, XII, IX, VII, II with 9, 2, 5, 12, 7.

Answer: Step 1: V = 5.

Step 2: XII = 10 + 2 = 12.

Step 3: IX = 10 - 1 = 9.

Step 4: VII = 5 + 2 = 7.

Step 5: II = 2.

Final answer: V matches 5, XII matches 12, IX matches 9, VII matches 7

and II matches 2.

Question: Match: III, X, XI, IV, VIII with 11, 8, 3, 10, 4.

Answer: Step 1: III = 3.

Step 2: X = 10.

Step 3: XI = 10 + 1 = 11.

Step 4: IV = 5 - 1 = 4.

Step 5: VIII = 5 + 3 = 8.

Final answer: III matches 3, X matches 10, XI matches 11, IV matches 4 and VIII matches 8.

EXERCISE 1 — 6. Write Roman numbers from 1 to 20.

Question: Write Roman numbers from 1 to 20.

Answer: Step 1: Use the symbols I = 1, V = 5 and X = 10.

Step 2: Add I after a symbol to make it bigger and write I before V or X to make it smaller.

Final answer: 1 = I, 2 = II, 3 = III, 4 = IV, 5 = V, 6 = VI, 7 = VII, 8 = VIII, 9 = IX, 10 = X, 11 = XI, 12 = XII, 13 = XIII, 14 = XIV, 15 = XV, 16 = XVI, 17 = XVII, 18 = XVIII, 19 = XIX, 20 = XX.

EXERCISE 2 — Even and Odd Numbers

Question: 1. Encircle (O) the even numbers from the following: 80, 83, 15, 98, 67, 78, 42, 27, 45.

Answer: Step 1: Even numbers have 0, 2, 4, 6 or 8 at the ones place.

Step 2: 80 ends in 0, 98 ends in 8, 78 ends in 8 and 42 ends in 2, so they are even.

Step 3: 83, 15, 67, 27 and 45 end in odd digits, so they are odd.

Final answer: The even numbers are 80, 98, 78 and 42.

Question: 2. Encircle (O) the odd numbers from the following: 30, 38, 71, 32, 64, 69, 70, 11, 95.

Answer: Step 1: Odd numbers have 1, 3, 5, 7 or 9 at the ones place.

Step 2: 71 ends in 1, 69 ends in 9, 11 ends in 1 and 95 ends in 5, so they are odd.

Step 3: 30, 38, 32, 64 and 70 end in even digits, so they are even.

Final answer: The odd numbers are 71, 69, 11 and 95.

Question: 3. Write even numbers between 1 to 20.

Answer: Step 1: Start from 2 and keep adding 2 each time.

Step 2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20.

Final answer: The even numbers between 1 and 20 are 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20.

Question: 4. Write odd numbers from 1 to 21.

Answer: Step 1: Start from 1 and keep adding 2 each time.

Step 2: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21.

Final answer: The odd numbers from 1 to 21 are 1, 3, 5, 7, 9, 11, 13, 15, 17, 19 and 21.

Question: 5. After 30, write first five even numbers.

Answer: Step 1: The first even number after 30 is 32.

Step 2: Keep adding 2: 32, 34, 36, 38, 40.

Final answer: The first five even numbers after 30 are 32, 34, 36, 38 and

40.

Question: 6. After 50, write first four odd numbers.

Answer: Step 1: The first odd number after 50 is 51.

Step 2: Keep adding 2: 51, 53, 55, 57.

Final answer: The first four odd numbers after 50 are 51, 53, 55 and 57.

Question: 7. Write even numbers from 50 to 80.

Answer: Step 1: Start at 50, which is even.

Step 2: Keep adding 2 until 80: 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80.

Final answer: The even numbers from 50 to 80 are 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78 and 80.

Question: 8. Write odd numbers from 70 to 90.

Answer: Step 1: The first odd number from 70 is 71.

Step 2: Keep adding 2 until 89: 71, 73, 75, 77, 79, 81, 83, 85, 87, 89.

Final answer: The odd numbers from 70 to 90 are 71, 73, 75, 77, 79, 81, 83, 85, 87 and 89.

Question: 9(i). Write (T) for true and (F) for false: 23 is an even number.

Answer: Step 1: 23 has 3 at the ones place.

Step 2: 3 is an odd digit, so 23 is odd, not even.

Final answer: (F) False. 23 is an odd number.

Question: 9(ii). Write (T) for true and (F) for false: 52 is an odd number.

Answer: Step 1: 52 has 2 at the ones place.

Step 2: 2 is an even digit, so 52 is even, not odd.

Final answer: (F) False. 52 is an even number.

Question: 9(iii). Write (T) for true and (F) for false: 15 is an odd number.

Answer: Step 1: 15 has 5 at the ones place.

Step 2: 5 is an odd digit, so 15 is odd.

Final answer: (T) True.

Question: 9(iv). Write (T) for true and (F) for false: 30 is an even number.

Answer: Step 1: 30 has 0 at the ones place.

Step 2: 0 is an even digit, so 30 is even.

Final answer: (T) True.

Activity — Write the place value of coloured digit in the following numbers

Question: (i) In 245 (H T O), the coloured digit is 5. Write its place value.

Answer: Step 1: 5 is at the ones place.

Step 2: Its place value is 5 ones = 5.

Final answer: The place value of 5 in 245 is 5 ones.



Question: (ii) In 8062 (Th H T O), the coloured digit is 8. Write its place value.

Answer: Step 1: 8 is at the thousands place.

Step 2: Its place value is 8 thousands = 8000.

Final answer: The place value of 8 in 8062 is 8 thousands, that is 8000.

8	0	6	2

Question: (iii) In 49674 (T-Th Th H T O), the coloured digit is 4 at the ten thousands place. Write its place value.

Answer: Step 1: The first 4 is at the ten thousands place.

Step 2: Its place value is 4 ten thousands = 40000.

Final answer: The place value of 4 in 49674 is 4 ten thousands, that is 40000.

4	9	6	7	4

EXERCISE 3 — 1. Fill the boxes.

Question: (i) 5806 = _____ thousands + _____ hundreds + _____ tens + _____ ones

Answer: Step 1: In 5806, 5 is at the thousands place.

Step 2: 8 is at the hundreds place.

Step 3: 0 is at the tens place.

Step 4: 6 is at the ones place.

Final answer: 5806 = 5 thousands + 8 hundreds + 0 tens + 6 ones.

Question: (ii) 51001 = _____ ten thousands + _____ thousands + _____ hundreds + _____ tens + _____ ones

Answer: Step 1: In 51001, 5 is at the ten thousands place and 1 is at the thousands place.

Step 2: 0 is at the hundreds place and 0 is at the tens place.

Step 3: 1 is at the ones place.

Final answer: 51001 = 5 ten thousands + 1 thousands + 0 hundreds + 0 tens + 1 ones.

Question: (iii) 12902 = _____ ten thousands + _____ thousands + _____ hundreds + _____ tens + _____ ones

Answer: Step 1: In 12902, 1 is at the ten thousands place and 2 is at the thousands place.

Step 2: 9 is at the hundreds place and 0 is at the tens place.

Step 3: 2 is at the ones place.

Final answer: 12902 = 1 ten thousands + 2 thousands + 9 hundreds + 0 tens + 2 ones.

Question: (iv) 442609 = _____ hundred thousands + _____ ten thousands + _____ thousands + _____ hundreds + _____ tens + _____ ones

Answer: Step 1: In 442609, 4 is at the hundred thousands place and 4 is at the ten thousands place.

Step 2: 2 is at the thousands place and 6 is at the hundreds place.

Step 3: 0 is at the tens place and 9 is at the ones place.

Final answer: 442609 = 4 hundred thousands + 4 ten thousands + 2 thousands + 6 hundreds + 0 tens + 9 ones.

EXERCISE 3 — 2. Identify the place value of coloured digit number given in boxes.

Question: (i) 2436 — the digit 3 is in a box. Write its place value.

Answer: Step 1: 3 is at the tens place in 2436.

Step 2: 3 tens = 30.

Final answer: The place value of 3 is 30 (3 tens).

Question: (ii) 54123 — the digit 4 is in a box. Write its place value.

Answer: Step 1: 4 is at the thousands place in 54123.

Step 2: 4 thousands = 4000.

Final answer: The place value of 4 is 4000 (4 thousands).

Question: (iii) 612378 — the digit 8 is in a box. Write its place value.

Answer: Step 1: 8 is at the ones place in 612378.

Step 2: 8 ones = 8.

Final answer: The place value of 8 is 8 (8 ones).

Question: (iv) 4129 — the digit 4 is in a box. Write its place value.

Answer: Step 1: 4 is at the thousands place in 4129.

Step 2: 4 thousands = 4000.

Final answer: The place value of 4 is 4000 (4 thousands).

Question: (v) 335566 — the digit 5 is in a box. Write its place value.

Answer: Step 1: The boxed 5 is at the hundreds place in 335566.

Step 2: 5 hundreds = 500.

Final answer: The place value of 5 is 500 (5 hundreds).

Question: (vi) 72081 — the digit 2 is in a box. Write its place value.

Answer: Step 1: 2 is at the hundreds place in 72081.

Step 2: 2 hundreds = 200.

Final answer: The place value of 2 is 200 (2 hundreds).

Question: (vii) 94123 — the digit 9 is in a box. Write its place value.

Answer: Step 1: 9 is at the ten thousands place in 94123.

Step 2: 9 ten thousands = 90000.

Final answer: The place value of 9 is 90000 (9 ten thousands).

Question: (viii) 567812 — the digit 8 is in a box. Write its place value.

Answer: Step 1: 8 is at the hundreds place in 567812.

Step 2: 8 hundreds = 800.

Final answer: The place value of 8 is 800 (8 hundreds).

Question: (ix) 810703 — the digit 0 is in a box. Write its place value.

Answer: Step 1: The boxed 0 is at the thousands place in 810703.

Step 2: 0 thousands = 0.

Final answer: The place value of 0 is 0.

EXERCISE 3 — 3. Complete the following table (H-Th, T-Th, Th, H, T, O).

Question: (i) Six thousand four hundred forty two — Th = 6, H = 4, O = 2 are given. Complete H-Th, T-Th and T.

Answer: Step 1: Six thousand four hundred forty two is 6442.

Step 2: 6442 has no hundred thousands and no ten thousands, so H-Th = 0 and T-Th = 0.

Step 3: The tens digit is 4, so T = 4.

Final answer: H-Th = 0, T-Th = 0, Th = 6, H = 4, T = 4, O = 2. The number is 6442.

6	4	4	2

Question: (ii) Twenty five thousand six hundred thirty one — T-Th = 2, T = 3, O = 1 are given. Complete H-Th, Th and H.

Answer: Step 1: Twenty five thousand six hundred thirty one is 25631.

Step 2: 25631 has no hundred thousands, so H-Th = 0.

Step 3: The thousands digit is 5, so Th = 5.

Step 4: The hundreds digit is 6, so H = 6.

Final answer: H-Th = 0, T-Th = 2, Th = 5, H = 6, T = 3, O = 1. The number is 25631.

2	5	6	3	1

Question: (iii) Ninety two thousand two — T-Th = 9, H = 0, O = 2 are given. Complete H-Th, Th and T.

Answer: Step 1: Ninety two thousand two is 92002.

Step 2: 92002 has no hundred thousands, so H-Th = 0.

Step 3: The thousands digit is 2, so Th = 2.

Step 4: The tens digit is 0, so T = 0.

Final answer: H-Th = 0, T-Th = 9, Th = 2, H = 0, T = 0, O = 2. The number

is 92002.

9	2	0	0	2

Question: (iv) Five hundred twenty three thousand five hundred — H-Th = 5, Th = 3, O = 0 are given. Complete T-Th, H and T.

Answer: Step 1: Five hundred twenty three thousand five hundred is 523500.

Step 2: The ten thousands digit is 2, so T-Th = 2.

Step 3: The hundreds digit is 5, so H = 5.

Step 4: The tens digit is 0, so T = 0.

Final answer: H-Th = 5, T-Th = 2, Th = 3, H = 5, T = 0, O = 0. The number is 523500.

5	2	3	5	0	0

Activity 3 — Fill in the blanks.

Question: (1) The greatest 4-digit number is _____.

Answer: Step 1: The greatest digit is 9.

Step 2: Write 9 four times: 9999.

Final answer: The greatest 4-digit number is 9999.

Question: (2) The smallest 5-digit number is _____.

Answer: Step 1: The number after 9999 is 9999 + 1 = 10000.

Final answer: The smallest 5-digit number is 10000.

Question: (3) The greatest 5-digit number is _____.

Answer: Step 1: The greatest digit is 9.

Step 2: Write 9 five times: 99999.

Final answer: The greatest 5-digit number is 99999.

Question: (4) The greatest 6-digit number is _____.

Answer: Step 1: The greatest digit is 9.

Step 2: Write 9 six times: 999999.

Final answer: The greatest 6-digit number is 999999.

EXERCISE 4 — 1. Read and write each of the following numbers in words.

Question: (i) 4156

Answer: Step 1: 4 is at the thousands place, 1 at hundreds, 5 at tens and 6 at ones.

Step 2: Read place by place: four thousand, one hundred, fifty, six.

Final answer: 4156 = Four thousand one hundred fifty six.

Question: (ii) 35678

Answer: Step 1: 35 is the thousands part and 678 is the rest.

Step 2: Read: thirty five thousand, six hundred, seventy eight.

Final answer: 35678 = Thirty five thousand six hundred seventy eight.

Question: (iii) 981254

Answer: Step 1: 981 is the thousands part and 254 is the rest.

Step 2: Read: nine hundred eighty one thousand, two hundred, fifty four.

Final answer: 981254 = Nine hundred eighty one thousand two hundred fifty four.

Question: (iv) 5060

Answer: Step 1: 5 is at the thousands place and 6 is at the tens place.

Step 2: There are no hundreds and no ones.

Final answer: 5060 = Five thousand sixty.

Question: (v) 7120

Answer: Step 1: 7 is at the thousands place, 1 at hundreds and 2 at tens.

Step 2: There are no ones.

Final answer: 7120 = Seven thousand one hundred twenty.

Question: (vi) 356782

Answer: Step 1: 356 is the thousands part and 782 is the rest.

Step 2: Read: three hundred fifty six thousand, seven hundred, eighty two.

Final answer: 356782 = Three hundred fifty six thousand seven hundred eighty two.

Question: (vii) 41524

Answer: Step 1: 41 is the thousands part and 524 is the rest.

Step 2: Read: forty one thousand, five hundred, twenty four.

Final answer: 41524 = Forty one thousand five hundred twenty four.

Question: (viii) 123456

Answer: Step 1: 123 is the thousands part and 456 is the rest.

Step 2: Read: one hundred twenty three thousand, four hundred, fifty six.

Final answer: 123456 = One hundred twenty three thousand four hundred fifty six.

EXERCISE 4 — 2. Write the following in figures.

Question: (i) Eight thousand four hundred sixty one.

Answer: Step 1: Eight thousand = 8000.

Step 2: Four hundred = 400 and sixty one = 61.

Step 3: Add: 8000 + 400 + 61 = 8461.

Final answer: Eight thousand four hundred sixty one = 8461.

Question: (ii) Seventy two thousand six hundred ninety five.

Answer: Step 1: Seventy two thousand = 72000.

Step 2: Six hundred = 600 and ninety five = 95.

Step 3: Add: 72000 + 600 + 95 = 72695.

Final answer: Seventy two thousand six hundred ninety five = 72695.

Question: (iii) Fifty five thousand one.

Answer: Step 1: Fifty five thousand = 55000.

Step 2: One = 1.

Step 3: Add: 55000 + 1 = 55001.

Final answer: Fifty five thousand one = 55001.

EXERCISE 4 — 3. Fill in the blanks.

Question: (i) Number before 951 is _____

Answer: Step 1: The number before means one less.

Step 2: 951 - 1 = 950.

Final answer: The number before 951 is 950.

Question: (ii) Number after 25645 is _____

Answer: Step 1: The number after means one more.

Step 2: 25645 + 1 = 25646.

Final answer: The number after 25645 is 25646.

Question: (iii) Number between 5420 and 5422 is _____

Answer: Step 1: Count on from 5420: 5420, 5421, 5422.

Step 2: The middle number is 5421.

Final answer: The number between 5420 and 5422 is 5421.

EXERCISE 4 — 4. Continue the patterns.

Question: (i) 2124, 3124, 4124, _____, _____, _____

Answer: Step 1: Each number is 1000 more than the number before it.

Step 2: 4124 + 1000 = 5124.

Step 3: 5124 + 1000 = 6124.

Step 4: 6124 + 1000 = 7124.

Final answer: 2124, 3124, 4124, 5124, 6124, 7124.

Question: (ii) 51427, 51437, 51447, _____, _____, _____

Answer: Step 1: Each number is 10 more than the number before it.

Step 2: 51447 + 10 = 51457.

Step 3: 51457 + 10 = 51467.

Step 4: 51467 + 10 = 51477.

Final answer: 51427, 51437, 51447, 51457, 51467, 51477.

Question: (iii) 123456, 223456, 323456, _____, _____, _____

Answer: Step 1: Each number is 100000 more than the number before it.

Step 2: 323456 + 100000 = 423456.

Step 3: 423456 + 100000 = 523456.

Step 4: 523456 + 100000 = 623456.

Final answer: 123456, 223456, 323456, 423456, 523456, 623456.

Activity — Use symbols < or >.

Question: (i) 5 _____ 3

Answer: Step 1: 5 is more than 3.

Step 2: The symbol for more than (greater than) is >.

Final answer: 5 > 3.

Question: (ii) 20 _____ 16

Answer: Step 1: 20 is more than 16.

Final answer: 20 > 16.

Question: (iii) 36 _____ 47

Answer: Step 1: 36 is less than 47.

Step 2: The symbol for less than is <.

Final answer: 36 < 47.

Question: (iv) 60 _____ 80

Answer: Step 1: 60 is less than 80.

Final answer: 60 < 80.

Question: (v) 325 _____ 300

Answer: Step 1: Both are 3-digit numbers. Compare the tens: 2 is more than 0.

Final answer: 325 > 300.

Question: (vi) 960 _____ 980

Answer: Step 1: Both are 3-digit numbers with 9 hundreds.

Step 2: Compare the tens: 6 is less than 8.

Final answer: 960 < 980.

Activity — Compare the numbers.

Question: Compare 98 and 123. 98 has _____ digits, 123 has _____ digits, so 98 _____ 123.

Answer: Step 1: 98 has 2 digits.

Step 2: 123 has 3 digits.

Step 3: A number with more digits is bigger, so 98 is less than 123.

Final answer: 98 has 2 digits, 123 has 3 digits, so 98 < 123.

Question: Compare 504 and 99. 504 has _____ digits, 99 has _____ digits, so 504 _____ 99.

Answer: Step 1: 504 has 3 digits.

Step 2: 99 has 2 digits.

Step 3: A number with more digits is bigger, so 504 is greater than 99.

Final answer: 504 has 3 digits, 99 has 2 digits, so 504 > 99.

Question: Compare 67835 and 67541. Both are _____ digit numbers. We compare first extreme left different digits which are _____ and _____. Here 8 _____ 5, so 67835 _____ 67541.

Answer: Step 1: Both numbers have 5 digits.

Step 2: The first different digits from the left are 8 and 5 (at the hundreds place).

Step 3: 8 is greater than 5.

Final answer: Both are 5-digit numbers, the different digits are 8 and 5, 8 > 5, so 67835 > 67541.

EXERCISE 5 — (1) Fill in the blanks by using symbols <, > or =.

Question: (i) 245 _____ 3167

Answer: Step 1: 245 has 3 digits and 3167 has 4 digits.

Step 2: The number with fewer digits is smaller.

Final answer: 245 < 3167.

Question: (ii) 54231 _____ 964

Answer: Step 1: 54231 has 5 digits and 964 has 3 digits.

Step 2: The number with more digits is bigger.

Final answer: 54231 > 964.

Question: (iii) 3105 _____ 3105

Answer: Step 1: Both numbers are the same, digit by digit.

Final answer: 3105 = 3105.

Question: (iv) 5421 _____ 5418

Answer: Step 1: Both are 4-digit numbers with 5 thousands and 4 hundreds.

Step 2: Compare the tens: 2 is more than 1.

Final answer: 5421 > 5418.

Question: (v) 672315 _____ 871237

Answer: Step 1: Both are 6-digit numbers.

Step 2: Compare the first digits: 6 is less than 8.

Final answer: 672315 < 871237.

Question: (vi) 120001 _____ 105552

Answer: Step 1: Both are 6-digit numbers with 1 hundred thousand.

Step 2: Compare the ten thousands: 2 is more than 0.

Final answer: 120001 > 105552.

EXERCISE 5 — (2) Compare the numbers by using symbols <, > or =.

Question: (i) 542 and 6712

Answer: Step 1: 542 has 3 digits and 6712 has 4 digits.

Final answer: 542 < 6712.

Question: (ii) 94562 and 4392

Answer: Step 1: 94562 has 5 digits and 4392 has 4 digits.

Final answer: 94562 > 4392.

Question: (iii) 6324 and 6324

Answer: Step 1: Both numbers are exactly the same.

Final answer: 6324 = 6324.

Question: (iv) 6421 and 6578

Answer: Step 1: Both are 4-digit numbers with 6 thousands.

Step 2: Compare the hundreds: 4 is less than 5.

Final answer: 6421 < 6578.

Question: (v) 94561 and 94271

Answer: Step 1: Both are 5-digit numbers with 9 ten thousands and 4 thousands.

Step 2: Compare the hundreds: 5 is more than 2.

Final answer: 94561 > 94271.

Question: (vi) 345671 and 345921

Answer: Step 1: Both are 6-digit numbers and their first three digits (345) are the same.

Step 2: Compare the next (hundreds) digits: 6 is less than 9.

Final answer: 345671 < 345921.

EXERCISE 6 — 1. Write the following numbers in ascending order in given boxes.

Question: (i) 476, 52341, 7881, 5034

Answer: Step 1: Ascending order means smallest to largest.

Step 2: 476 has 3 digits, so it is smallest.

Step 3: 5034 and 7881 have 4 digits; 5 thousands is less than 7 thousands, so 5034 comes before 7881.

Step 4: 52341 has 5 digits, so it is largest.

Final answer: 476, 5034, 7881, 52341.

Question: (ii) 2346, 4632, 2354, 1778

Answer: Step 1: All are 4-digit numbers. Compare the thousands: 1, 2, 2,

4.

Step 2: 1778 is smallest and 4632 is largest.

Step 3: Between 2346 and 2354, compare the tens: 4 is less than 5, so 2346 comes first.

Final answer: 1778, 2346, 2354, 4632.

Question: (iii) 43513, 3451, 53314, 41353

Answer: Step 1: 3451 has 4 digits, so it is smallest.

Step 2: Among the 5-digit numbers, compare the ten thousands: 4, 5, 4.

Step 3: Between 43513 and 41353, compare the thousands: 3 is more than 1, so 41353 comes first.

Step 4: 53314 has 5 ten thousands, so it is largest.

Final answer: 3451, 41353, 43513, 53314.

EXERCISE 6 — 2. Write the following numbers in descending orders in given boxes.

Question: (i) 6432, 3213, 4343, 7120

Answer: Step 1: Descending order means largest to smallest.

Step 2: Compare the thousands: 7, 6, 4, 3.

Final answer: 7120, 6432, 4343, 3213.

Question: (ii) 49231, 12349, 94321, 31249

Answer: Step 1: All are 5-digit numbers. Compare the ten thousands: 9, 4, 3, 1.

Final answer: 94321, 49231, 31249, 12349.

Question: (iii) 120451, 12345, 57401, 10000

Answer: Step 1: 120451 has 6 digits, so it is largest.

Step 2: Among the 5-digit numbers, 57401 is bigger than 12345 and 10000.

Step 3: 12345 is bigger than 10000.

Final answer: 120451, 57401, 12345, 10000.

Activity — Number line

Question: Which number has been ticked (✓) at the number line 0 to 11?

Answer: Step 1: Count the marks from 0 along the line.

Step 2: The tick is on the mark named 5.

Final answer: The ticked number is 5.



Question: Which number has been crossed (x) at the number line 0 to 11?

Answer: Step 1: Count the marks from 0 up to the cross.

Step 2: The cross is at 8.

Final answer: The crossed number is 8.



EXERCISE 7 — Number Line

Question: 1. Represent 6 on a number line (0 to 7).

Answer: Step 1: Draw a line and mark 0, 1, 2, 3, 4, 5, 6, 7 at equal

distances.

Step 2: Count 6 steps from 0.

Step 3: Mark and circle the point at 6.

Final answer: 6 is represented at the sixth mark from 0.

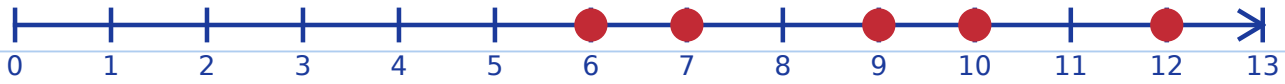


Question: 2. Represent the following numbers on given number line: 6, 7, 9, 10 and 12 (number line 0 to 13).

Answer: Step 1: Find each number on the line 0 to 13.

Step 2: Mark 6, then 7, then 9, then 10 and then 12 at their own points.

Final answer: The numbers 6, 7, 9, 10 and 12 are marked at their points on the number line.



Question: 3. Fill in the blanks with before and after numbers on the given number line (the middle circle shows 5, with four circles before it and four circles after it).

Answer: Step 1: The numbers before 5 are 4, 3, 2, 1, counting backwards.

Step 2: The numbers after 5 are 6, 7, 8, 9, counting forwards.

Final answer: The circles are filled as 1, 2, 3, 4, 5, 6, 7, 8, 9.



Question: 4. Encircle the following numbers on the given number line: 14,

16 and 18 (number line 11 to 20).

Answer: Step 1: Start counting from 11: 11, 12, 13, 14. Circle 14.

Step 2: Count on: 15, 16. Circle 16.

Step 3: Count on: 17, 18. Circle 18.

Final answer: The numbers 14, 16 and 18 are encircled on the number line.



Unit 2 NUMBER OPERATIONS

EXERCISE 8 — Solve. (Addition)

Question: (1) 43 + 24 = _____

Answer: Step 1: Add the ones: 3 + 4 = 7.

Step 2: Add the tens: 4 + 2 = 6.

Final answer: 43 + 24 = 67.

Question: (2) 431 + 24 = _____

Answer: Step 1: Add the ones: 1 + 4 = 5.

Step 2: Add the tens: 3 + 2 = 5.

Step 3: Add the hundreds: 4 + 0 = 4.

Final answer: 431 + 24 = 455.

Question: (3) 65 + 24 = _____

Answer: Step 1: Add the ones: 5 + 4 = 9.

Step 2: Add the tens: 6 + 2 = 8.

Final answer: 65 + 24 = 89.

Question: (4) 114 + 85 = _____

Answer: Step 1: Add the ones: 4 + 5 = 9.

Step 2: Add the tens: 1 + 8 = 9.

Step 3: Add the hundreds: 1 + 0 = 1.

Final answer: 114 + 85 = 199.

Question: (5) 582 + 137 = _____

Answer: Step 1: Add the ones: 2 + 7 = 9.

Step 2: Add the tens: 8 + 3 = 11. Write 1 and carry 1.

Step 3: Add the hundreds: 5 + 1 = 6, plus the carried 1 makes 7.

Final answer: 582 + 137 = 719.

Question: (6) 905 + 182 = _____

Answer: Step 1: Add the ones: 5 + 2 = 7.

Step 2: Add the tens: 0 + 8 = 8.

Step 3: Add the hundreds: 9 + 1 = 10. Write 0 and carry 1 to make the thousands 1.

Final answer: 905 + 182 = 1087.

Question: (7) 631 + 289 = _____

Answer: Step 1: Add the ones: 1 + 9 = 10. Write 0 and carry 1.

Step 2: Add the tens: 3 + 8 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 6 + 2 = 8, plus the carried 1 makes 9.

Final answer: 631 + 289 = 920.

Question: (8) 599 + 324 = _____

Answer: Step 1: Add the ones: 9 + 4 = 13. Write 3 and carry 1.

Step 2: Add the tens: 9 + 2 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 5 + 3 = 8, plus the carried 1 makes 9.

Final answer: 599 + 324 = 923.

Question: (9) 488 + 325 = _____

Answer: Step 1: Add the ones: 8 + 5 = 13. Write 3 and carry 1.

Step 2: Add the tens: 8 + 2 = 10, plus the carried 1 makes 11. Write 1 and carry 1.

Step 3: Add the hundreds: 4 + 3 = 7, plus the carried 1 makes 8.

Final answer: 488 + 325 = 813.

Question: (10) 27 + 22 = _____

Answer: Step 1: Add the ones: 7 + 2 = 9.

Step 2: Add the tens: 2 + 2 = 4.

Final answer: 27 + 22 = 49.

Question: (11) 84 + 101 = _____

Answer: Step 1: Add the ones: 4 + 1 = 5.

Step 2: Add the tens: 8 + 0 = 8.

Step 3: Add the hundreds: 0 + 1 = 1.

Final answer: 84 + 101 = 185.

Question: (12) 850 + 88 = _____

Answer: Step 1: Add the ones: 0 + 8 = 8.

Step 2: Add the tens: 5 + 8 = 13. Write 3 and carry 1.

Step 3: Add the hundreds: 8 + 0 = 8, plus the carried 1 makes 9.

Final answer: 850 + 88 = 938.

Question: (13) 104 + 645 = _____

Answer: Step 1: Add the ones: 4 + 5 = 9.

Step 2: Add the tens: 0 + 4 = 4.

Step 3: Add the hundreds: 1 + 6 = 7.

Final answer: 104 + 645 = 749.

Question: (14) 542 + 227 = _____

Answer: Step 1: Add the ones: 2 + 7 = 9.

Step 2: Add the tens: 4 + 2 = 6.

Step 3: Add the hundreds: 5 + 2 = 7.

Final answer: 542 + 227 = 769.

Question: (15) 300 + 200 = _____

Answer: Step 1: Add the ones: 0 + 0 = 0.

Step 2: Add the tens: 0 + 0 = 0.

Step 3: Add the hundreds: 3 + 2 = 5.

Final answer: 300 + 200 = 500.

EXERCISE 9 — (A) Solve: (Add numbers up to 4-digits without carrying)

Question: (1) 2564 + 1321 = _____

Answer: Step 1: Add the ones: 4 + 1 = 5.

Step 2: Add the tens: 6 + 2 = 8.

Step 3: Add the hundreds: 5 + 3 = 8.

Step 4: Add the thousands: 2 + 1 = 3.

Final answer: 2564 + 1321 = 3885.

Question: (2) 3572 + 1223 = _____

Answer: Step 1: Add the ones: 2 + 3 = 5.

Step 2: Add the tens: 7 + 2 = 9.

Step 3: Add the hundreds: 5 + 2 = 7.

Step 4: Add the thousands: 3 + 1 = 4.

Final answer: 3572 + 1223 = 4795.

Question: (3) 7425 + 2353 = _____

Answer: Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 2 + 5 = 7.

Step 3: Add the hundreds: 4 + 3 = 7.

Step 4: Add the thousands: 7 + 2 = 9.

Final answer: 7425 + 2353 = 9778.

Question: (4) 4653 + 4233 = _____

Answer: Step 1: Add the ones: 3 + 3 = 6.

Step 2: Add the tens: 5 + 3 = 8.

Step 3: Add the hundreds: 6 + 2 = 8.

Step 4: Add the thousands: 4 + 4 = 8.

Final answer: 4653 + 4233 = 8886.

Question: (5) 7245 + 2433 = _____

Answer: Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 4 + 3 = 7.

Step 3: Add the hundreds: 2 + 4 = 6.

Step 4: Add the thousands: 7 + 2 = 9.

Final answer: 7245 + 2433 = 9678.

Question: (6) 2847 + 3152 = _____

Answer: Step 1: Add the ones: 7 + 2 = 9.

Step 2: Add the tens: 4 + 5 = 9.

Step 3: Add the hundreds: 8 + 1 = 9.

Step 4: Add the thousands: 2 + 3 = 5.

Final answer: 2847 + 3152 = 5999.

Question: (7) 3604 + 4242 + 0123 = _____

Answer: Step 1: Add the ones: 4 + 2 + 3 = 9.

Step 2: Add the tens: 0 + 4 + 2 = 6.

Step 3: Add the hundreds: 6 + 2 + 1 = 9.

Step 4: Add the thousands: 3 + 4 + 0 = 7.

Final answer: 3604 + 4242 + 123 = 7969.

Question: (8) 7464 + 1012 + 1421 = _____

Answer: Step 1: Add the ones: 4 + 2 + 1 = 7.

Step 2: Add the tens: 6 + 1 + 2 = 9.

Step 3: Add the hundreds: 4 + 0 + 4 = 8.

Step 4: Add the thousands: 7 + 1 + 1 = 9.

Final answer: 7464 + 1012 + 1421 = 9897.

Question: (9) 5093 + 2704 + 1102 = _____

Answer: Step 1: Add the ones: 3 + 4 + 2 = 9.

Step 2: Add the tens: 9 + 0 + 0 = 9.

Step 3: Add the hundreds: 0 + 7 + 1 = 8.

Step 4: Add the thousands: 5 + 2 + 1 = 8.

Final answer: 5093 + 2704 + 1102 = 8899.

EXERCISE 9 — (B) Solve the following:

Question: (1) 2215 + 1322 = _____

Answer: Step 1: Add the ones: 5 + 2 = 7.

Step 2: Add the tens: 1 + 2 = 3.

Step 3: Add the hundreds: 2 + 3 = 5.

Step 4: Add the thousands: 2 + 1 = 3.

Final answer: 2215 + 1322 = 3537.

Question: (2) 4325 + 3210 = _____

Answer: Step 1: Add the ones: 5 + 0 = 5.

Step 2: Add the tens: 2 + 1 = 3.

Step 3: Add the hundreds: 3 + 2 = 5.

Step 4: Add the thousands: 4 + 3 = 7.

Final answer: 4325 + 3210 = 7535.

Question: (3) 4625 + 1234 = _____

Answer: Step 1: Add the ones: 5 + 4 = 9.

Step 2: Add the tens: 2 + 3 = 5.

Step 3: Add the hundreds: 6 + 2 = 8.

Step 4: Add the thousands: 4 + 1 = 5.

Final answer: 4625 + 1234 = 5859.

Question: (4) 3624 + 4172 = _____

Answer: Step 1: Add the ones: 4 + 2 = 6.

Step 2: Add the tens: 2 + 7 = 9.

Step 3: Add the hundreds: 6 + 1 = 7.

Step 4: Add the thousands: 3 + 4 = 7.

Final answer: 3624 + 4172 = 7796.

Question: (5) 7383 + 2514 = _____

Answer: Step 1: Add the ones: 3 + 4 = 7.

Step 2: Add the tens: 8 + 1 = 9.

Step 3: Add the hundreds: 3 + 5 = 8.

Step 4: Add the thousands: 7 + 2 = 9.

Final answer: 7383 + 2514 = 9897.

Question: (6) 4502 + 1224 + 4113 = _____

Answer: Step 1: Add the ones: 2 + 4 + 3 = 9.

Step 2: Add the tens: 0 + 2 + 1 = 3.

Step 3: Add the hundreds: 5 + 2 + 1 = 8.

Step 4: Add the thousands: 4 + 1 + 4 = 9.

Final answer: 4502 + 1224 + 4113 = 9839.

Question: (7) 1102 + 1323 + 1340 = _____

Answer: Step 1: Add the ones: 2 + 3 + 0 = 5.

Step 2: Add the tens: 0 + 2 + 4 = 6.

Step 3: Add the hundreds: 1 + 3 + 3 = 7.

Step 4: Add the thousands: 1 + 1 + 1 = 3.

Final answer: 1102 + 1323 + 1340 = 3765.

EXERCISE 10 — (A) Solve: (Add numbers up to 4-digits with carrying)

Question: (1) 4626 + 3394 = _____

Answer: Step 1: Add the ones: 6 + 4 = 10. Write 0 and carry 1.

Step 2: Add the tens: 2 + 9 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 6 + 3 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 4: Add the thousands: 4 + 3 = 7, plus the carried 1 makes 8.

Final answer: 4626 + 3394 = 8020.

Question: (2) 3097 + 4974 = _____

Answer: Step 1: Add the ones: 7 + 4 = 11. Write 1 and carry 1.

Step 2: Add the tens: 9 + 7 = 16, plus the carried 1 makes 17. Write 7 and carry 1.

Step 3: Add the hundreds: 0 + 9 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 4: Add the thousands: 3 + 4 = 7, plus the carried 1 makes 8.

Final answer: 3097 + 4974 = 8071.

Question: (3) 7385 + 2798 = _____

Answer: Step 1: Add the ones: 5 + 8 = 13. Write 3 and carry 1.

Step 2: Add the tens: 8 + 9 = 17, plus the carried 1 makes 18. Write 8 and carry 1.

Step 3: Add the hundreds: 3 + 7 = 10, plus the carried 1 makes 11. Write 1 and carry 1.

Step 4: Add the thousands: 7 + 2 = 9, plus the carried 1 makes 10. Write 0 and carry 1 to make a new ten thousand 1.

Final answer: 7385 + 2798 = 10183.

Question: (4) 3487 + 932 + 1472 = _____

Answer: Step 1: Add the ones: 7 + 2 + 2 = 11. Write 1 and carry 1.

Step 2: Add the tens: 8 + 3 + 7 = 18, plus the carried 1 makes 19. Write 9 and carry 1.

Step 3: Add the hundreds: 4 + 9 + 4 = 17, plus the carried 1 makes 18. Write 8 and carry 1.

Step 4: Add the thousands: 3 + 0 + 1 = 4, plus the carried 1 makes 5.

Final answer: 3487 + 932 + 1472 = 5891.

Question: (5) 5705 + 2899 + 102 = _____

Answer: Step 1: Add the ones: 5 + 9 + 2 = 16. Write 6 and carry 1.

Step 2: Add the tens: 0 + 9 + 0 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 3: Add the hundreds: 7 + 8 + 1 = 16, plus the carried 1 makes 17. Write 7 and carry 1.

Step 4: Add the thousands: 5 + 2 + 0 = 7, plus the carried 1 makes 8.

Final answer: 5705 + 2899 + 102 = 8706.

Question: (6) 3789 + 2668 + 202 = _____

Answer: Step 1: Add the ones: 9 + 8 + 2 = 19. Write 9 and carry 1.

Step 2: Add the tens: 8 + 6 + 0 = 14, plus the carried 1 makes 15. Write 5 and carry 1.

Step 3: Add the hundreds: 7 + 6 + 2 = 15, plus the carried 1 makes 16. Write 6 and carry 1.

Step 4: Add the thousands: 3 + 2 + 0 = 5, plus the carried 1 makes 6.

Final answer: 3789 + 2668 + 202 = 6659.

EXERCISE 10 — (B) Add the following:

Question: (1) 4866 and 2154

Answer: Step 1: Add the ones: 6 + 4 = 10. Write 0 and carry 1.

Step 2: Add the tens: 6 + 5 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 8 + 1 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 4: Add the thousands: 4 + 2 = 6, plus the carried 1 makes 7.

Final answer: 4866 + 2154 = 7020.

Question: (2) 3239 and 2453

Answer: Step 1: Add the ones: 9 + 3 = 12. Write 2 and carry 1.

Step 2: Add the tens: 3 + 5 = 8, plus the carried 1 makes 9.

Step 3: Add the hundreds: 2 + 4 = 6.

Step 4: Add the thousands: 3 + 2 = 5.

Final answer: 3239 + 2453 = 5692.

Question: (3) 8370, 3745 and 235

Answer: Step 1: Add the ones: 0 + 5 + 5 = 10. Write 0 and carry 1.

Step 2: Add the tens: 7 + 4 + 3 = 14, plus the carried 1 makes 15. Write 5 and carry 1.

Step 3: Add the hundreds: 3 + 7 + 2 = 12, plus the carried 1 makes 13. Write 3 and carry 1.

Step 4: Add the thousands: 8 + 3 + 0 = 11, plus the carried 1 makes 12. Write 2 and carry 1 to make a new ten thousand 1.

Final answer: 8370 + 3745 + 235 = 12350.

Question: (4) 2555, 3876 and 1000

Answer: Step 1: Add the ones: 5 + 6 + 0 = 11. Write 1 and carry 1.

Step 2: Add the tens: 5 + 7 + 0 = 12, plus the carried 1 makes 13. Write 3 and carry 1.

Step 3: Add the hundreds: 5 + 8 + 0 = 13, plus the carried 1 makes 14. Write 4 and carry 1.

Step 4: Add the thousands: 2 + 3 + 1 = 6, plus the carried 1 makes 7.

Final answer: 2555 + 3876 + 1000 = 7431.

EXERCISE 10 — (C) Solve the following:

Question: (1) 3994 + 1089 = _____

Answer: Step 1: Add the ones: 4 + 9 = 13. Write 3 and carry 1.

Step 2: Add the tens: 9 + 8 = 17, plus the carried 1 makes 18. Write 8 and carry 1.

Step 3: Add the hundreds: 9 + 0 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 4: Add the thousands: 3 + 1 = 4, plus the carried 1 makes 5.

Final answer: 3994 + 1089 = 5083.

Question: (2) 3456 + 1268 = _____

Answer: Step 1: Add the ones: 6 + 8 = 14. Write 4 and carry 1.

Step 2: Add the tens: 5 + 6 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 4 + 2 = 6, plus the carried 1 makes 7.

Step 4: Add the thousands: 3 + 1 = 4.

Final answer: 3456 + 1268 = 4724.

Question: (3) 2658 + 3846 = _____

Answer: Step 1: Add the ones: 8 + 6 = 14. Write 4 and carry 1.

Step 2: Add the tens: 5 + 4 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 3: Add the hundreds: 6 + 8 = 14, plus the carried 1 makes 15. Write 5 and carry 1.

Step 4: Add the thousands: 2 + 3 = 5, plus the carried 1 makes 6.

Final answer: 2658 + 3846 = 6504.

Question: (4) 3477 + 2955 = _____

Answer: Step 1: Add the ones: 7 + 5 = 12. Write 2 and carry 1.

Step 2: Add the tens: 7 + 5 = 12, plus the carried 1 makes 13. Write 3 and carry 1.

Step 3: Add the hundreds: 4 + 9 = 13, plus the carried 1 makes 14. Write 4 and carry 1.

Step 4: Add the thousands: 3 + 2 = 5, plus the carried 1 makes 6.

Final answer: 3477 + 2955 = 6432.

EXERCISE 11 — (A) Add the following numbers by using more than one strategies.

Question: (1) 14 + 5

Answer: Step 1: Split 5 as 5. Add the ones: 4 + 5 = 9.

Step 2: Keep the tens: 10 + 9 = 19.

Final answer: 14 + 5 = 19.

Question: (2) 15 + 6

Answer: Step 1: Split 6 into 5 + 1.

Step 2: 15 + 5 = 20.

Step 3: 20 + 1 = 21.

Final answer: 15 + 6 = 21.

Question: (3) 16 + 8

Answer: Step 1: Split 8 into 4 + 4.

Step 2: 16 + 4 = 20.

Step 3: 20 + 4 = 24.

Final answer: 16 + 8 = 24.

Question: (4) 21 + 11

Answer: Step 1: Add the tens: 20 + 10 = 30.

Step 2: Add the ones: 1 + 1 = 2.

Step 3: Join: 30 + 2 = 32.

Final answer: 21 + 11 = 32.

Question: (5) 22 + 6

Answer: Step 1: Add the ones: 2 + 6 = 8.

Step 2: Keep the tens: 20 + 8 = 28.

Final answer: 22 + 6 = 28.

Question: (6) 11 + 32

Answer: Step 1: Add the tens: 10 + 30 = 40.

Step 2: Add the ones: 1 + 2 = 3.

Step 3: Join: 40 + 3 = 43.

Final answer: 11 + 32 = 43.

EXERCISE 11 — (B) Solve mentally.

Question: (1) 25 + 20 = _____

Answer: Step 1: Add the tens: 2 tens + 2 tens = 4 tens = 40.

Step 2: Add the ones: 5 + 0 = 5.

Step 3: Join: 40 + 5 = 45.

Final answer: 25 + 20 = 45.

Question: (2) 22 + 21 = _____

Answer: Step 1: Add the tens: 20 + 20 = 40.

Step 2: Add the ones: 2 + 1 = 3.

Step 3: Join: 40 + 3 = 43.

Final answer: 22 + 21 = 43.

Question: (3) 46 + 14 = _____

Answer: Step 1: Add the tens: 40 + 10 = 50.

Step 2: Add the ones: 6 + 4 = 10.

Step 3: Join: 50 + 10 = 60.

Final answer: 46 + 14 = 60.

Question: (4) 32 + 4 = _____

Answer: Step 1: Add the ones: 2 + 4 = 6.

Step 2: Keep the tens: 30 + 6 = 36.

Final answer: 32 + 4 = 36.

Question: (5) 42 + 28 = _____

Answer: Step 1: Add the tens: 40 + 20 = 60.

Step 2: Add the ones: 2 + 8 = 10.

Step 3: Join: 60 + 10 = 70.

Final answer: 42 + 28 = 70.

Question: (6) 54 + 26 = _____

Answer: Step 1: Add the tens: 50 + 20 = 70.

Step 2: Add the ones: 4 + 6 = 10.

Step 3: Join: 70 + 10 = 80.

Final answer: 54 + 26 = 80.

EXERCISE 11 — (C) Add mentally and match your answer (answers are in the cloud: 44, 35, 86, 30, 92, 68, 90, 69).

Question: (1) 10 + 20 = _____

Answer: Step 1: Add the tens: 1 ten + 2 tens = 3 tens.

Final answer: 10 + 20 = 30.

Question: (2) 45 + 23 = _____

Answer: Step 1: Add the tens: 40 + 20 = 60.

Step 2: Add the ones: 5 + 3 = 8.

Step 3: Join: 60 + 8 = 68.

Final answer: 45 + 23 = 68.

Question: (3) 23 + 12 = _____

Answer: Step 1: Add the tens: 20 + 10 = 30.

Step 2: Add the ones: 3 + 2 = 5.

Step 3: Join: 30 + 5 = 35.

Final answer: 23 + 12 = 35.

Question: (4) 22 + 22 = _____

Answer: Step 1: Add the tens: 20 + 20 = 40.

Step 2: Add the ones: 2 + 2 = 4.

Step 3: Join: 40 + 4 = 44.

Final answer: 22 + 22 = 44.

Question: (5) 10 + 50 + 30 = _____

Answer: Step 1: Add the tens: 1 + 5 + 3 = 9 tens.

Final answer: 10 + 50 + 30 = 90.

Question: (6) 18 + 38 + 36 = _____

Answer: Step 1: Add the ones: 8 + 8 + 6 = 22. Write 2 and carry 2 tens.

Step 2: Add the tens: 1 + 3 + 3 = 7, plus the carried 2 makes 9 tens.

Final answer: 18 + 38 + 36 = 92.

Question: (7) 8 + 46 + 32 = _____

Answer: Step 1: Add the ones: 8 + 6 + 2 = 16. Write 6 and carry 1 ten.

Step 2: Add the tens: 0 + 4 + 3 = 7, plus the carried 1 makes 8 tens.

Final answer: 8 + 46 + 32 = 86.

Question: (8) 20 + 13 + 36 = _____

Answer: Step 1: Add the ones: 0 + 3 + 6 = 9.

Step 2: Add the tens: 2 + 1 + 3 = 6 tens.

Final answer: 20 + 13 + 36 = 69.

EXERCISE 12 — Solve real life problems involving addition

Question: (1) Rehana bought two books. One book has 328 pages and other has 1034 pages. How many pages in both?

Answer: Step 1: Add the ones: 8 + 4 = 12. Write 2 and carry 1.

Step 2: Add the tens: 2 + 3 = 5, plus the carried 1 makes 6.

Step 3: Add the hundreds: 3 + 0 = 3.

Step 4: Add the thousands: 0 + 1 = 1.

Final answer: There are 1362 pages in both books.

Question: (2) Aashir walks 800 m in the morning and 1400 m in the evening. How far does he walk altogether?

Answer: Step 1: Add the ones and tens: 0 + 0 = 0.

Step 2: Add the hundreds: 8 + 4 = 12. Write 2 and carry 1.

Step 3: Add the thousands: 0 + 1 = 1, plus the carried 1 makes 2.

Final answer: He walks 2200 m altogether.

Question: (3) In a village there are 3321 men, 2843 women and 1822 children. What is the total population of the village?

Answer: Step 1: Add the ones: 1 + 3 + 2 = 6.

Step 2: Add the tens: 2 + 4 + 2 = 8.

Step 3: Add the hundreds: 3 + 8 + 8 = 19. Write 9 and carry 1.

Step 4: Add the thousands: 3 + 2 + 1 = 6, plus the carried 1 makes 7.

Final answer: The total population of the village is 7986.

Question: (4) In a Zoo 3451 visitors visited on Sunday, 2342 on Monday. How many visitors visited in these two days?

Answer: Step 1: Add the ones: 1 + 2 = 3.

Step 2: Add the tens: 5 + 4 = 9.

Step 3: Add the hundreds: 4 + 3 = 7.

Step 4: Add the thousands: 3 + 2 = 5.

Final answer: 5793 visitors visited in these two days.

Question: (5) Azhar reads a story book. On Saturday, he reads 236 pages and on Sunday 142 pages. How many pages does he read in both days?

Answer: Step 1: Add the ones: 6 + 2 = 8.

Step 2: Add the tens: 3 + 4 = 7.

Step 3: Add the hundreds: 2 + 1 = 3.

Final answer: He reads 378 pages in both days.

Question: (6) In a primary school there are 436 girls and 341 boys. How many students are altogether?

Answer: Step 1: Add the ones: 6 + 1 = 7.

Step 2: Add the tens: 3 + 4 = 7.

Step 3: Add the hundreds: 4 + 3 = 7.

Final answer: There are 777 students altogether.

Question: (7) In a city 4798 soft drinks sold on Friday and 3202 on Saturday. How many soft drinks were sold on both days?

Answer: Step 1: Add the ones: 8 + 2 = 10. Write 0 and carry 1.

Step 2: Add the tens: 9 + 0 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 3: Add the hundreds: 7 + 2 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 4: Add the thousands: 4 + 3 = 7, plus the carried 1 makes 8.

Final answer: 8000 soft drinks were sold on both days.

EXERCISE 13 — Solve. (Subtraction)

Question: (1) 88 - 47 = _____

Answer: Step 1: Subtract the ones: 8 - 7 = 1.

Step 2: Subtract the tens: 8 - 4 = 4.

Final answer: 88 - 47 = 41.

Question: (2) 76 - 54 = _____

Answer: Step 1: Subtract the ones: 6 - 4 = 2.

Step 2: Subtract the tens: 7 - 5 = 2.

Final answer: 76 - 54 = 22.

Question: (3) 639 - 38 = _____

Answer: Step 1: Subtract the ones: 9 - 8 = 1.

Step 2: Subtract the tens: 3 - 3 = 0.

Step 3: Subtract the hundreds: 6 - 0 = 6.

Final answer: 639 - 38 = 601.

Question: (4) 473 - 151 = _____

Answer: Step 1: Subtract the ones: 3 - 1 = 2.

Step 2: Subtract the tens: 7 - 5 = 2.

Step 3: Subtract the hundreds: 4 - 1 = 3.

Final answer: 473 - 151 = 322.

Question: (5) 378 - 56 = _____

Answer: Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Subtract the tens: 7 - 5 = 2.

Step 3: Subtract the hundreds: 3 - 0 = 3.

Final answer: 378 - 56 = 322.

Question: (6) 297 - 74 = _____

Answer: Step 1: Subtract the ones: 7 - 4 = 3.

Step 2: Subtract the tens: 9 - 7 = 2.

Step 3: Subtract the hundreds: 2 - 0 = 2.

Final answer: 297 - 74 = 223.

Question: (7) 356 - 178 = _____

Answer: Step 1: Subtract the ones: 6 cannot take away 8, so borrow 1 ten: 16 - 8 = 8.

Step 2: Subtract the tens: 5 tens became 4 tens; 4 cannot take away 7, so borrow 1 hundred: 14 - 7 = 7.

Step 3: Subtract the hundreds: 3 hundreds became 2 hundreds; 2 - 1 = 1.

Final answer: 356 - 178 = 178.

Question: (8) 625 - 478 = _____

Answer: Step 1: Subtract the ones: 5 cannot take away 8, so borrow 1 ten: 15 - 8 = 7.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 cannot take away 7, so borrow 1 hundred: 11 - 7 = 4.

Step 3: Subtract the hundreds: 6 hundreds became 5 hundreds; 5 - 4 = 1.

Final answer: 625 - 478 = 147.

Question: (9) 243 - 184 = _____

Answer: Step 1: Subtract the ones: 3 cannot take away 4, so borrow 1 ten: 13 - 4 = 9.

Step 2: Subtract the tens: 4 tens became 3 tens; 3 cannot take away 8, so borrow 1 hundred: 13 - 8 = 5.

Step 3: Subtract the hundreds: 2 hundreds became 1 hundred; 1 - 1 = 0.

Final answer: 243 - 184 = 59.

Question: (10) 702 - 189 = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 9; there is no ten, so borrow 1 hundred as 9 tens and 10 ones: 12 - 9 = 3.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 7 hundreds became 6 hundreds; 6 - 1 = 5.

Final answer: 702 - 189 = 513.

Question: (11) 958 - 199 = _____

Answer: Step 1: Subtract the ones: 8 cannot take away 9, so borrow 1 ten: 18 - 9 = 9.

Step 2: Subtract the tens: 5 tens became 4 tens; 4 cannot take away 9, so borrow 1 hundred: 14 - 9 = 5.

Step 3: Subtract the hundreds: 9 hundreds became 8 hundreds; 8 - 1 = 7.

Final answer: 958 - 199 = 759.

Question: (12) 342 - 155 = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 5, so borrow 1 ten: 12 - 5 = 7.

Step 2: Subtract the tens: 4 tens became 3 tens; 3 cannot take away 5, so borrow 1 hundred: 13 - 5 = 8.

Step 3: Subtract the hundreds: 3 hundreds became 2 hundreds; 2 - 1 = 1.

Final answer: 342 - 155 = 187.

EXERCISE 14 — (A) Solve: (Subtract 4-digit numbers without borrowing)

Question: (1) 4269 - 3032 = _____

Answer: Step 1: Subtract the ones: 9 - 2 = 7.

Step 2: Subtract the tens: 6 - 3 = 3.

Step 3: Subtract the hundreds: 2 - 0 = 2.

Step 4: Subtract the thousands: 4 - 3 = 1.

Final answer: 4269 - 3032 = 1237.

Question: (2) 9968 - 2734 = _____

Answer: Step 1: Subtract the ones: 8 - 4 = 4.

Step 2: Subtract the tens: 6 - 3 = 3.

Step 3: Subtract the hundreds: 9 - 7 = 2.

Step 4: Subtract the thousands: 9 - 2 = 7.

Final answer: 9968 - 2734 = 7234.

Question: (3) 7695 - 3253 = _____

Answer: Step 1: Subtract the ones: 5 - 3 = 2.

Step 2: Subtract the tens: 9 - 5 = 4.

Step 3: Subtract the hundreds: 6 - 2 = 4.

Step 4: Subtract the thousands: 7 - 3 = 4.

Final answer: 7695 - 3253 = 4442.

Question: (4) 8756 - 3234 = _____

Answer: Step 1: Subtract the ones: 6 - 4 = 2.

Step 2: Subtract the tens: 5 - 3 = 2.

Step 3: Subtract the hundreds: 7 - 2 = 5.

Step 4: Subtract the thousands: 8 - 3 = 5.

Final answer: 8756 - 3234 = 5522.

Question: (5) 6598 - 4386 = _____

Answer: Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 5 - 3 = 2.

Step 4: Subtract the thousands: 6 - 4 = 2.

Final answer: 6598 - 4386 = 2212.

Question: (6) 6976 - 2343 = _____

Answer: Step 1: Subtract the ones: 6 - 3 = 3.

Step 2: Subtract the tens: 7 - 4 = 3.

Step 3: Subtract the hundreds: 9 - 3 = 6.

Step 4: Subtract the thousands: 6 - 2 = 4.

Final answer: 6976 - 2343 = 4633.

Question: (7) 5754 - 2352 = _____

Answer: Step 1: Subtract the ones: 4 - 2 = 2.

Step 2: Subtract the tens: 5 - 5 = 0.

Step 3: Subtract the hundreds: 7 - 3 = 4.

Step 4: Subtract the thousands: 5 - 2 = 3.

Final answer: 5754 - 2352 = 3402.

Question: (8) 3879 - 2346 = _____

Answer: Step 1: Subtract the ones: 9 - 6 = 3.

Step 2: Subtract the tens: 7 - 4 = 3.

Step 3: Subtract the hundreds: 8 - 3 = 5.

Step 4: Subtract the thousands: 3 - 2 = 1.

Final answer: 3879 - 2346 = 1533.

Question: (9) 4589 - 3466 = _____

Answer: Step 1: Subtract the ones: 9 - 6 = 3.

Step 2: Subtract the tens: 8 - 6 = 2.

Step 3: Subtract the hundreds: 5 - 4 = 1.

Step 4: Subtract the thousands: 4 - 3 = 1.

Final answer: 4589 - 3466 = 1123.

EXERCISE 14 — (B) Solve:

Question: (1) 3455 - 1343 = _____

Answer: Step 1: Subtract the ones: 5 - 3 = 2.

Step 2: Subtract the tens: 5 - 4 = 1.

Step 3: Subtract the hundreds: 4 - 3 = 1.

Step 4: Subtract the thousands: 3 - 1 = 2.

Final answer: 3455 - 1343 = 2112.

Question: (2) 6657 - 3246 = _____

Answer: Step 1: Subtract the ones: 7 - 6 = 1.

Step 2: Subtract the tens: 5 - 4 = 1.

Step 3: Subtract the hundreds: 6 - 2 = 4.

Step 4: Subtract the thousands: 6 - 3 = 3.

Final answer: 6657 - 3246 = 3411.

Question: (3) 3689 - 1345 = _____

Answer: Step 1: Subtract the ones: 9 - 5 = 4.

Step 2: Subtract the tens: 8 - 4 = 4.

Step 3: Subtract the hundreds: 6 - 3 = 3.

Step 4: Subtract the thousands: 3 - 1 = 2.

Final answer: 3689 - 1345 = 2344.

Question: (4) 8672 - 4350 = _____

Answer: Step 1: Subtract the ones: 2 - 0 = 2.

Step 2: Subtract the tens: 7 - 5 = 2.

Step 3: Subtract the hundreds: 6 - 3 = 3.

Step 4: Subtract the thousands: 8 - 4 = 4.

Final answer: 8672 - 4350 = 4322.

Question: (5) 9563 - 5341 = _____

Answer: Step 1: Subtract the ones: 3 - 1 = 2.

Step 2: Subtract the tens: 6 - 4 = 2.

Step 3: Subtract the hundreds: 5 - 3 = 2.

Step 4: Subtract the thousands: 9 - 5 = 4.

Final answer: 9563 - 5341 = 4222.

EXERCISE 15 — (A) Solve: (Subtract 4-digit numbers with borrowing)

Question: (1) 5714 - 3546 = _____

Answer: Step 1: Subtract the ones: 4 cannot take away 6, so borrow 1 ten: 14 - 6 = 8.

Step 2: Subtract the tens: 1 ten became 0 tens; 0 cannot take away 4, so borrow 1 hundred: 10 - 4 = 6.

Step 3: Subtract the hundreds: 7 hundreds became 6 hundreds; 6 - 5 = 1.

Step 4: Subtract the thousands: 5 - 3 = 2.

Final answer: 5714 - 3546 = 2168.

Question: (2) 6608 - 3247 = _____

Answer: Step 1: Subtract the ones: 8 - 7 = 1.

Step 2: Subtract the tens: 0 cannot take away 4, so borrow 1 hundred: 10 - 4 = 6.

Step 3: Subtract the hundreds: 6 hundreds became 5 hundreds; 5 - 2 = 3.

Step 4: Subtract the thousands: 6 - 3 = 3.

Final answer: 6608 - 3247 = 3361.

Question: (3) 5423 - 3987 = _____

Answer: Step 1: Subtract the ones: 3 cannot take away 7, so borrow 1 ten: 13 - 7 = 6.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 cannot take away 8, so borrow 1 hundred: 11 - 8 = 3.

Step 3: Subtract the hundreds: 4 hundreds became 3 hundreds; 3 cannot take away 9, so borrow 1 thousand: 13 - 9 = 4.

Step 4: Subtract the thousands: 5 thousands became 4 thousands; 4 - 3 = 1.

Final answer: 5423 - 3987 = 1436.

Question: (4) 3940 - 2735 = _____

Answer: Step 1: Subtract the ones: 0 cannot take away 5, so borrow 1 ten: 10 - 5 = 5.

Step 2: Subtract the tens: 4 tens became 3 tens; 3 - 3 = 0.

Step 3: Subtract the hundreds: 9 - 7 = 2.

Step 4: Subtract the thousands: 3 - 2 = 1.

Final answer: 3940 - 2735 = 1205.

Question: (5) 5705 - 2934 = _____

Answer: Step 1: Subtract the ones: 5 - 4 = 1.

Step 2: Subtract the tens: 0 cannot take away 3, so borrow 1 hundred: 10 - 3 = 7.

Step 3: Subtract the hundreds: 7 hundreds became 6 hundreds; 6 cannot take away 9, so borrow 1 thousand: 16 - 9 = 7.

Step 4: Subtract the thousands: 5 thousands became 4 thousands; 4 - 2 = 2.

Final answer: 5705 - 2934 = 2771.

Question: (6) 8004 - 4035 = _____

Answer: Step 1: Subtract the ones: 4 cannot take away 5; borrow through the zeros: the thousands gives 1 hundred, the hundred gives 1 ten, so 14 - 5 = 9.

Step 2: Subtract the tens: 9 - 3 = 6.

Step 3: Subtract the hundreds: 9 - 0 = 9.

Step 4: Subtract the thousands: 8 became 7; 7 - 4 = 3.

Final answer: 8004 - 4035 = 3969.

Question: (7) 9012 - 7654 = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 4, so borrow 1 ten: 12 - 4 = 8. The tens 1 becomes 0.

Step 2: Subtract the tens: 0 cannot take away 5, so borrow 1 hundred. The hundreds digit is also 0, so first borrow 1 thousand: thousands 9 becomes

8 and hundreds 0 becomes 10. Then the hundreds give 1 ten: hundreds become 9 and tens become 10. Now 10 - 5 = 5.

Step 3: Subtract the hundreds: 9 - 6 = 3.

Step 4: Subtract the thousands: 8 - 7 = 1.

Final answer: 9012 - 7654 = 1358.

Question: (8) 3452 - 2328 = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 8, so borrow 1 ten: 12 - 8 = 4.

Step 2: Subtract the tens: 5 tens became 4 tens; 4 - 2 = 2.

Step 3: Subtract the hundreds: 4 - 3 = 1.

Step 4: Subtract the thousands: 3 - 2 = 1.

Final answer: 3452 - 2328 = 1124.

Question: (9) 3381 - 2645 = _____

Answer: Step 1: Subtract the ones: 1 cannot take away 5, so borrow 1 ten: 11 - 5 = 6.

Step 2: Subtract the tens: 8 tens became 7 tens; 7 - 4 = 3.

Step 3: Subtract the hundreds: 3 cannot take away 6, so borrow 1 thousand: 13 - 6 = 7.

Step 4: Subtract the thousands: 3 thousands became 2 thousands; 2 - 2 = 0.

Final answer: 3381 - 2645 = 736.

EXERCISE 15 — (B) Complete the following:

Question: (1) 6572 - 1459 = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 9, so borrow 1 ten: 12 - 9 = 3.

Step 2: Subtract the tens: 7 tens became 6 tens; 6 - 5 = 1.

Step 3: Subtract the hundreds: 5 - 4 = 1.

Step 4: Subtract the thousands: 6 - 1 = 5.

Final answer: 6572 - 1459 = 5113.

Question: (2) 8095 - 7251 = _____

Answer: Step 1: Subtract the ones: 5 - 1 = 4.

Step 2: Subtract the tens: 9 - 5 = 4.

Step 3: Subtract the hundreds: 0 cannot take away 2, so borrow 1 thousand: 10 - 2 = 8.

Step 4: Subtract the thousands: 8 thousands became 7 thousands; 7 - 7 = 0.

Final answer: 8095 - 7251 = 844.

Question: (3) 4562 - 3571 = _____

Answer: Step 1: Subtract the ones: 2 - 1 = 1.

Step 2: Subtract the tens: 6 - 7 cannot be done, so borrow 1 hundred: 16 - 7 = 9.

Step 3: Subtract the hundreds: 5 hundreds became 4 hundreds; 4 - 5 cannot be done, so borrow 1 thousand: 14 - 5 = 9.

Step 4: Subtract the thousands: 4 thousands became 3 thousands; 3 - 3 = 0.

Final answer: 4562 - 3571 = 991.

Question: (4) 6559 - 2382 = _____

Answer: Step 1: Subtract the ones: 9 - 2 = 7.

Step 2: Subtract the tens: 5 - 8 cannot be done, so borrow 1 hundred: 15 - 8 = 7.

Step 3: Subtract the hundreds: 5 hundreds became 4 hundreds; 4 - 3 = 1.

Step 4: Subtract the thousands: 6 - 2 = 4.

Final answer: 6559 - 2382 = 4177.

EXERCISE 15 — (C) Subtract horizontally:

Question: (1) Subtract 3254 from 7293.

Answer: Step 1: Write it as 7293 - 3254. Subtract the ones: 3 cannot take away 4, so borrow 1 ten: 13 - 4 = 9.

Step 2: Subtract the tens: 9 tens became 8 tens; 8 - 5 = 3.

Step 3: Subtract the hundreds: 2 - 2 = 0.

Step 4: Subtract the thousands: 7 - 3 = 4.

Final answer: 7293 - 3254 = 4039.

Question: (2) Subtract 4972 from 6743.

Answer: Step 1: Write it as 6743 - 4972. Subtract the ones: 3 - 2 = 1.

Step 2: Subtract the tens: 4 - 7 cannot be done, so borrow 1 hundred: 14 - 7 = 7.

Step 3: Subtract the hundreds: 7 hundreds became 6 hundreds; 6 - 9 cannot be done, so borrow 1 thousand: 16 - 9 = 7.

Step 4: Subtract the thousands: 6 thousands became 5 thousands; 5 - 4 =

1.

Final answer: 6743 - 4972 = 1771.

Question: (3) Subtract 2484 from 6512.

Answer: Step 1: Write it as 6512 - 2484. Subtract the ones: 2 cannot take away 4, so borrow 1 ten: 12 - 4 = 8.

Step 2: Subtract the tens: 1 ten became 0 tens; 0 cannot take away 8, so borrow 1 hundred: 10 - 8 = 2.

Step 3: Subtract the hundreds: 5 hundreds became 4 hundreds; 4 - 4 = 0.

Step 4: Subtract the thousands: 6 - 2 = 4.

Final answer: 6512 - 2484 = 4028.

Question: (4) Subtract 5328 from 8672.

Answer: Step 1: Write it as 8672 - 5328. Subtract the ones: 2 cannot take away 8, so borrow 1 ten: 12 - 8 = 4.

Step 2: Subtract the tens: 7 tens became 6 tens; 6 - 2 = 4.

Step 3: Subtract the hundreds: 6 - 3 = 3.

Step 4: Subtract the thousands: 8 - 5 = 3.

Final answer: 8672 - 5328 = 3344.

Activity — Subtract. (Mental subtraction chains)

Question: (1) Start at 50. Subtract 5, then subtract 10, then subtract 2. Fill the boxes.

Answer: Step 1: 50 - 5 = 45.

Step 2: 45 - 10 = 35.

Step 3: 35 - 2 = 33.

Final answer: 50, 45, 35, 33.

Question: (2) Start at 100. Subtract 30, then subtract 5, then subtract 20. Fill the boxes.

Answer: Step 1: 100 - 30 = 70.

Step 2: 70 - 5 = 65.

Step 3: 65 - 20 = 45.

Final answer: 100, 70, 65, 45.

EXERCISE 16 — 1. Subtract the following numbers by using more than one strategies.

Question: (i) 15 - 6

Answer: Step 1: Split 6 into 5 + 1.

Step 2: 15 - 5 = 10.

Step 3: 10 - 1 = 9.

Final answer: 15 - 6 = 9.

Question: (ii) 70 - 8

Answer: Step 1: Split 70 into 60 + 10.

Step 2: 10 - 8 = 2.

Step 3: 60 + 2 = 62.

Final answer: 70 - 8 = 62.

Question: (iii) 78 - 12

Answer: Step 1: Subtract the tens: 78 - 10 = 68.

Step 2: Subtract the ones: 68 - 2 = 66.

Final answer: 78 - 12 = 66.

Question: (iv) 89 - 26

Answer: Step 1: Subtract the tens: 89 - 20 = 69.

Step 2: Subtract the ones: 69 - 6 = 63.

Final answer: 89 - 26 = 63.

Question: (v) 56 - 35

Answer: Step 1: Subtract the tens: 56 - 30 = 26.

Step 2: Subtract the ones: 26 - 5 = 21.

Final answer: 56 - 35 = 21.

Question: (vi) 93 - 27

Answer: Step 1: Subtract the tens: 93 - 20 = 73.

Step 2: Subtract the ones: 73 - 7 = 66.

Final answer: 93 - 27 = 66.

EXERCISE 16 — 2. Solve.

Question: (i) 47 - 16 = _____

Answer: Step 1: Subtract the ones: 7 - 6 = 1.

Step 2: Subtract the tens: 4 - 1 = 3.

Final answer: 47 - 16 = 31.

Question: (ii) 19 - 13 = _____

Answer: Step 1: Subtract the ones: 9 - 3 = 6.

Step 2: Subtract the tens: 1 - 1 = 0.

Final answer: 19 - 13 = 6.

Question: (iii) 50 - 12 = _____

Answer: Step 1: Subtract the ones: 0 cannot take away 2, so borrow 1 ten: 10 - 2 = 8.

Step 2: Subtract the tens: 5 tens became 4 tens; 4 - 1 = 3.

Final answer: 50 - 12 = 38.

Question: (iv) 46 - 21 = _____

Answer: Step 1: Subtract the ones: 6 - 1 = 5.

Step 2: Subtract the tens: 4 - 2 = 2.

Final answer: 46 - 21 = 25.

Question: (v) 89 - 40 = _____

Answer: Step 1: Subtract the ones: 9 - 0 = 9.

Step 2: Subtract the tens: 8 - 4 = 4.

Final answer: 89 - 40 = 49.

Question: (vi) 67 - 14 = _____

Answer: Step 1: Subtract the ones: 7 - 4 = 3.

Step 2: Subtract the tens: 6 - 1 = 5.

Final answer: 67 - 14 = 53.

Question: (vii) 97 - 18 = _____

Answer: Step 1: Subtract the ones: 7 cannot take away 8, so borrow 1 ten: 17 - 8 = 9.

Step 2: Subtract the tens: 9 tens became 8 tens; 8 - 1 = 7.

Final answer: 97 - 18 = 79.

Question: (viii) 100 - 87 = _____

Answer: Step 1: Subtract the ones: borrow through the zeros: 10 - 7 = 3.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 1 became 0.

Final answer: 100 - 87 = 13.

EXERCISE 16 — 3. Solve the following mentally.

Question: (i) There are 9 birds sitting on a tree. 3 flew away. How many birds remain there?

Answer: Step 1: Birds left = 9 - 3.

Step 2: 9 - 3 = 6.

Final answer: 6 birds remain there.

Question: (ii) Abdul Raheem has 10 sheep and he has 5 cows. How many more sheep than cows does he have?

Answer: Step 1: More sheep = 10 - 5.

Step 2: 10 - 5 = 5.

Final answer: He has 5 more sheep than cows.

Question: (iii) Sania has 42 kg of rice. She gave 20 kg to her sister. How much rice is left with her?

Answer: Step 1: Rice left = 42 - 20.

Step 2: Subtract the tens: 4 tens - 2 tens = 2 tens, and the ones stay 2.

Final answer: 22 kg of rice is left with her.

EXERCISE 17 — Solve real life problems involving subtraction

Question: (1) A shopkeeper has 850 chocolates of milk. He sold 480

chocolates. How many chocolates are left?

Answer: Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 5 cannot take away 8, so borrow 1 hundred: 15 - 8 = 7.

Step 3: Subtract the hundreds: 8 hundreds became 7 hundreds; 7 - 4 = 3.

Final answer: 370 chocolates are left.

Question: (2) Ali has two notes in his wallet. Their value together is Rs 600. One note is of Rs 100. What is the value of other note?

Answer: Step 1: Other note = 600 - 100.

Step 2: Subtract the hundreds: 6 - 1 = 5 hundreds.

Final answer: The value of the other note is Rs 500.

Question: (3) A factory manufactures 2460 cars in a year. 1946 cars are sold. How many cars are yet to be sold?

Answer: Step 1: Subtract the ones: 0 cannot take away 6, so borrow 1 ten: 10 - 6 = 4.

Step 2: Subtract the tens: 6 tens became 5 tens; 5 - 4 = 1.

Step 3: Subtract the hundreds: 4 cannot take away 9, so borrow 1 thousand: 14 - 9 = 5.

Step 4: Subtract the thousands: 2 became 1; 1 - 1 = 0.

Final answer: 514 cars are yet to be sold.

Question: (4) Shamim collected Rs 4950 as donation for flood affected. She purchased grocery and utensils Rs 2895 for the affected villagers. How much amount is still left with her?

Answer: Step 1: Subtract the ones: 0 cannot take away 5, so borrow 1 ten:

10 - 5 = 5.

Step 2: Subtract the tens: 5 tens became 4 tens; 4 cannot take away 9, so borrow 1 hundred: 14 - 9 = 5.

Step 3: Subtract the hundreds: 9 hundreds became 8 hundreds; 8 - 8 = 0.

Step 4: Subtract the thousands: 4 - 2 = 2.

Final answer: Rs 2055 is still left with her.

Question: (5) A bakery had 4500 packets of cookies. 1862 were sold. How many were left?

Answer: Step 1: Subtract the ones: 0 cannot take away 2. The tens digit is also 0, so borrow 1 hundred: hundreds 5 becomes 4 and tens become 10. The tens give 1 ten to the ones: tens become 9 and ones become 10. Now 10 - 2 = 8.

Step 2: Subtract the tens: 9 - 6 = 3.

Step 3: Subtract the hundreds: 4 cannot take away 8, so borrow 1 thousand: thousands 4 becomes 3 and hundreds become 14. Now 14 - 8 = 6.

Step 4: Subtract the thousands: 3 - 1 = 2.

Final answer: 2638 packets were left.

Question: (6) The population of a village comprises 9015 adult persons. 4386 are men, find the number of women.

Answer: Step 1: Subtract the ones: 5 cannot take away 6, so borrow 1 ten: tens 1 becomes 0 and ones become 15. Now 15 - 6 = 9.

Step 2: Subtract the tens: 0 cannot take away 8, so borrow 1 hundred. The hundreds digit is also 0, so first borrow 1 thousand: thousands 9 becomes

8 and hundreds 0 becomes 10. Then the hundreds give 1 ten: hundreds become 9 and tens become 10. Now 10 - 8 = 2.

Step 3: Subtract the hundreds: 9 - 3 = 6.

Step 4: Subtract the thousands: 8 - 4 = 4.

Final answer: The number of women is 4629.

EXERCISE 18 — Count the objects in groups and complete the multiplication

Question: (A)(i) 3 × 4 = _____

Answer: Step 1: There are 3 groups and each group has 4 objects.

Step 2: Count in fours: 4, 8, 12.

Final answer: 3 × 4 = 12.

Question: (A)(ii) 2 × 6 = _____

Answer: Step 1: There are 2 groups and each group has 6 objects.

Step 2: Count in sixes: 6, 12.

Final answer: 2 × 6 = 12.

Question: (A)(iii) 3 × 3 = _____

Answer: Step 1: There are 3 groups and each group has 3 objects.

Step 2: Count in threes: 3, 6, 9.

Final answer: 3 × 3 = 9.

Question: (A)(iv) 7 × 5 = _____

Answer: Step 1: There are 7 groups and each group has 5 objects.

Step 2: Count in fives: 5, 10, 15, 20, 25, 30, 35.

Final answer: 7 × 5 = 35.

Question: (A)(v) 5 × 9 = _____

Answer: Step 1: There are 5 groups and each group has 9 objects.

Step 2: Count in nines: 9, 18, 27, 36, 45.

Final answer: 5 × 9 = 45.

Question: (A)(vi) 8 × 3 = _____

Answer: Step 1: There are 8 groups and each group has 3 objects.

Step 2: Count in threes: 3, 6, 9, 12, 15, 18, 21, 24.

Final answer: 8 × 3 = 24.

Question: (B)(i) 5 × 4 = _____

Answer: Step 1: There are 5 groups and each group has 4 objects.

Step 2: Count in fours: 4, 8, 12, 16, 20.

Final answer: 5 × 4 = 20.

Question: (B)(ii) 2 × 8 = _____

Answer: Step 1: There are 2 groups and each group has 8 objects.

Step 2: Count in eights: 8, 16.

Final answer: 2 × 8 = 16.

Question: (B)(iii) 5 × 6 = _____

Answer: Step 1: There are 5 groups and each group has 6 objects.

Step 2: Count in sixes: 6, 12, 18, 24, 30.

Final answer: 5 × 6 = 30.

ACTIVITY — Count in 6 and fill in the missing numbers (table of 6)

Question: Complete the counting in 6: 6, _____, _____, 24, _____, _____, 42, _____, _____, _____, 66, _____

Answer: Step 1: Counting in 6 means we add 6 every time.

Step 2: 6 + 6 = 12, 12 + 6 = 18, 18 + 6 = 24.

Step 3: 24 + 6 = 30, 30 + 6 = 36, 36 + 6 = 42.

Step 4: 42 + 6 = 48, 48 + 6 = 54, 54 + 6 = 60, 60 + 6 = 66, 66 + 6 = 72.

Final answer: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72.

ACTIVITY — Complete the table of 7

Question: (i) 3 × 7 = _____

Answer: Step 1: Count in sevens 3 times: 7, 14, 21.

Final answer: 3 × 7 = 21.

Question: (ii) 6 × 7 = _____

Answer: Step 1: Count in sevens 6 times: 7, 14, 21, 28, 35, 42.

Final answer: 6 × 7 = 42.

Question: (iii) 8 × 7 = _____

Answer: Step 1: Count in sevens 8 times: 7, 14, 21, 28, 35, 42, 49, 56.

Final answer: 8 × 7 = 56.

Question: (iv) 9 × 7 = _____

Answer: Step 1: Count in sevens 9 times: 7, 14, 21, 28, 35, 42, 49, 56, 63.

Final answer: 9 × 7 = 63.

Question: (v) 7 × 7 = _____

Answer: Step 1: Count in sevens 7 times: 7, 14, 21, 28, 35, 42, 49.

Final answer: 7 × 7 = 49.

ACTIVITY — Count in 8 and complete the table of 8

Question: Write the table of 8 from 1 × 8 to 10 × 8.

Answer: Step 1: Keep adding 8 each time: 8, 16, 24, 32, 40.

Step 2: Continue adding 8: 48, 56, 64, 72, 80.

Final answer: 1 × 8 = 8, 2 × 8 = 16, 3 × 8 = 24, 4 × 8 = 32, 5 × 8 = 40, 6 × 8 = 48, 7 × 8 = 56, 8 × 8 = 64, 9 × 8 = 72, 10 × 8 = 80.

ACTIVITY — Count in 9 and complete the table of 9

Question: Write the table of 9 from 1 × 9 to 10 × 9.

Answer: Step 1: Keep adding 9 each time: 9, 18, 27, 36, 45.

Step 2: Continue adding 9: 54, 63, 72, 81, 90.

Final answer: 1 × 9 = 9, 2 × 9 = 18, 3 × 9 = 27, 4 × 9 = 36, 5 × 9 = 45, 6 × 9 = 54, 7 × 9 = 63, 8 × 9 = 72, 9 × 9 = 81, 10 × 9 = 90.

EXERCISE 19 — (A) Multiply the following

Question: (1) 14 × 2 = _____

Answer: Step 1: Multiply the ones: 4 × 2 = 8.

Step 2: Multiply the tens: 1 × 2 = 2.

Final answer: 14 × 2 = 28.

Question: (2) 31 × 2 = _____

Answer: Step 1: Multiply the ones: 1 × 2 = 2.

Step 2: Multiply the tens: 3 × 2 = 6.

Final answer: 31 × 2 = 62.

Question: (3) 21 × 4 = _____

Answer: Step 1: Multiply the ones: 1 × 4 = 4.

Step 2: Multiply the tens: 2 × 4 = 8.

Final answer: 21 × 4 = 84.

Question: (4) 12 × 4 = _____

Answer: Step 1: Multiply the ones: 2 × 4 = 8.

Step 2: Multiply the tens: 1 × 4 = 4.

Final answer: 12 × 4 = 48.

Question: (5) 23 × 3 = _____

Answer: Step 1: Multiply the ones: 3 × 3 = 9.

Step 2: Multiply the tens: 2 × 3 = 6.

Final answer: 23 × 3 = 69.

Question: (6) 35 × 2 = _____

Answer: Step 1: Multiply the ones: 5 × 2 = 10. Write 0 and carry 1.

Step 2: Multiply the tens: 3 × 2 = 6, plus the carried 1 makes 7.

Final answer: 35 × 2 = 70.

Question: (7) 11 × 6 = _____

Answer: Step 1: Multiply the ones: 1 × 6 = 6.

Step 2: Multiply the tens: 1 × 6 = 6.

Final answer: 11 × 6 = 66.

Question: (8) 53 × 6 = _____

Answer: Step 1: Multiply the ones: 3 × 6 = 18. Write 8 and carry 1.

Step 2: Multiply the tens: 5 × 6 = 30, plus the carried 1 makes 31.

Final answer: 53 × 6 = 318.

Question: (9) 83 × 4 = _____

Answer: Step 1: Multiply the ones: 3 × 4 = 12. Write 2 and carry 1.

Step 2: Multiply the tens: 8 × 4 = 32, plus the carried 1 makes 33.

Final answer: 83 × 4 = 332.

EXERCISE 19 — (B) Multiply the following

Question: (1) 13 × 2 = _____

Answer: Step 1: Multiply the ones: 3 × 2 = 6.

Step 2: Multiply the tens: 1 × 2 = 2.

Final answer: 13 × 2 = 26.

Question: (2) 64 × 4 = _____

Answer: Step 1: Multiply the ones: 4 × 4 = 16. Write 6 and carry 1.

Step 2: Multiply the tens: 6 × 4 = 24, plus the carried 1 makes 25.

Final answer: 64 × 4 = 256.

Question: (3) 23 × 5 = _____

Answer: Step 1: Multiply the ones: 3 × 5 = 15. Write 5 and carry 1.

Step 2: Multiply the tens: 2 × 5 = 10, plus the carried 1 makes 11.

Final answer: 23 × 5 = 115.

Question: (4) 22 × 6 = _____

Answer: Step 1: Multiply the ones: 2 × 6 = 12. Write 2 and carry 1.

Step 2: Multiply the tens: 2 × 6 = 12, plus the carried 1 makes 13.

Final answer: 22 × 6 = 132.

Question: (5) 21 × 7 = _____

Answer: Step 1: Multiply the ones: 1 × 7 = 7.

Step 2: Multiply the tens: 2 × 7 = 14.

Final answer: 21 × 7 = 147.

Question: (6) 15 × 3 = _____

Answer: Step 1: Multiply the ones: 5 × 3 = 15. Write 5 and carry 1.

Step 2: Multiply the tens: 1 × 3 = 3, plus the carried 1 makes 4.

Final answer: 15 × 3 = 45.

Question: (7) 37 × 5 = _____

Answer: Step 1: Multiply the ones: 7 × 5 = 35. Write 5 and carry 3.

Step 2: Multiply the tens: 3 × 5 = 15, plus the carried 3 makes 18.

Final answer: 37 × 5 = 185.

Question: (8) 14 × 6 = _____

Answer: Step 1: Multiply the ones: 4 × 6 = 24. Write 4 and carry 2.

Step 2: Multiply the tens: 1 × 6 = 6, plus the carried 2 makes 8.

Final answer: 14 × 6 = 84.

EXERCISE 20 — Multiply the following (multiplication by 0)

Question: (1) 0 × 5 = _____

Answer: Step 1: When any number is multiplied by 0, the answer is always

0.

Final answer: 0 × 5 = 0.

Question: (2) 7 × 0 = _____

Answer: Step 1: When any number is multiplied by 0, the answer is always 0.

Final answer: 7 × 0 = 0.

Question: (3) 126 × 0 = _____

Answer: Step 1: When any number is multiplied by 0, the answer is always 0.

Final answer: 126 × 0 = 0.

Question: (4) 0 × 213 = _____

Answer: Step 1: When 0 is multiplied by any number, the answer is always 0.

Final answer: 0 × 213 = 0.

Question: (5) 0 × 500 = _____

Answer: Step 1: When 0 is multiplied by any number, the answer is always 0.

Final answer: 0 × 500 = 0.

Question: (6) 0 × 950 = _____

Answer: Step 1: When 0 is multiplied by any number, the answer is always 0.

Final answer: 0 × 950 = 0.

Question: (7) 0 × 2000 = _____

Answer: Step 1: When 0 is multiplied by any number, the answer is always 0.

Final answer: 0 × 2000 = 0.

Question: (8) 400 × 0 = _____

Answer: Step 1: When any number is multiplied by 0, the answer is always 0.

Final answer: 400 × 0 = 0.

Question: (9) 2010 × 0 = _____

Answer: Step 1: When any number is multiplied by 0, the answer is always 0.

Final answer: 2010 × 0 = 0.

Question: (10) 0 × 3075 = _____

Answer: Step 1: When 0 is multiplied by any number, the answer is always 0.

Final answer: 0 × 3075 = 0.

Question: (11) 6009 × 0 = _____

Answer: Step 1: When any number is multiplied by 0, the answer is always 0.

Final answer: 6009 × 0 = 0.

Question: (12) 0 × 8000 = _____

Answer: Step 1: When 0 is multiplied by any number, the answer is always 0.

Final answer: 0 × 8000 = 0.

EXERCISE 21 — Multiply the following

Question: (1) 4 × 5 = _____

Answer: Step 1: Count in fives 4 times: 5, 10, 15, 20.

Final answer: 4 × 5 = 20.

Question: (2) 6 × 5 = _____

Answer: Step 1: Count in fives 6 times: 5, 10, 15, 20, 25, 30.

Final answer: 6 × 5 = 30.

Question: (3) 7 × 6 = _____

Answer: Step 1: Count in sixes 7 times: 6, 12, 18, 24, 30, 36, 42.

Final answer: 7 × 6 = 42.

Question: (4) 9 × 6 = _____

Answer: Step 1: Count in sixes 9 times: 6, 12, 18, 24, 30, 36, 42, 48, 54.

Final answer: 9 × 6 = 54.

Question: (5) 9 × 12 = _____

Answer: Step 1: Split 12 into 10 and 2.

Step 2: 9 × 10 = 90.

Step 3: 9 × 2 = 18.

Step 4: Add the two parts: 90 + 18 = 108.

Final answer: 9 × 12 = 108.

Question: (6) 8 × 14 = _____

Answer: Step 1: Split 14 into 10 and 4.

Step 2: 8 × 10 = 80.

Step 3: 8 × 4 = 32.

Step 4: Add the two parts: 80 + 32 = 112.

Final answer: 8 × 14 = 112.

Question: (7) 11 × 13 = _____

Answer: Step 1: Split 13 into 10 and 3.

Step 2: 11 × 10 = 110.

Step 3: 11 × 3 = 33.

Step 4: Add the two parts: 110 + 33 = 143.

Final answer: 11 × 13 = 143.

Question: (8) 14 × 16 = _____

Answer: Step 1: Split 16 into 10 and 6.

Step 2: 14 × 10 = 140.

Step 3: 14 × 6 = 84.

Step 4: Add the two parts: 140 + 84 = 224.

Final answer: 14 × 16 = 224.

Question: (9) 23 × 27 = _____

Answer: Step 1: Split 27 into 20 and 7.

Step 2: 23 × 20 = 460.

Step 3: 23 × 7 = 161.

Step 4: Add the two parts: 460 + 161 = 621.

Final answer: 23 × 27 = 621.

Question: (10) 29 × 32 = _____

Answer: Step 1: Split 32 into 30 and 2.

Step 2: 29 × 30 = 870.

Step 3: 29 × 2 = 58.

Step 4: Add the two parts: 870 + 58 = 928.

Final answer: 29 × 32 = 928.

Question: (11) 35 × 40 = _____

Answer: Step 1: First multiply 35 × 4 = 140.

Step 2: Because 40 is 4 tens, write one zero at the end: 140 × 10 = 1400.

Final answer: 35 × 40 = 1400.

Question: (12) 65 × 75 = _____

Answer: Step 1: Split 75 into 70 and 5.

Step 2: 65 × 70 = 4550.

Step 3: 65 × 5 = 325.

Step 4: Add the two parts: 4550 + 325 = 4875.

Final answer: 65 × 75 = 4875.

Question: (13) 45 × 27 = _____

Answer: Step 1: Split 27 into 20 and 7.

Step 2: 45 × 20 = 900.

Step 3: 45 × 7 = 315.

Step 4: Add the two parts: 900 + 315 = 1215.

Final answer: 45 × 27 = 1215.

Question: (14) 68 × 14 = _____

Answer: Step 1: Split 14 into 10 and 4.

Step 2: 68 × 10 = 680.

Step 3: 68 × 4 = 272.

Step 4: Add the two parts: 680 + 272 = 952.

Final answer: 68 × 14 = 952.

Question: (15) 79 × 21 = _____

Answer: Step 1: Split 21 into 20 and 1.

Step 2: 79 × 20 = 1580.

Step 3: 79 × 1 = 79.

Step 4: Add the two parts: 1580 + 79 = 1659.

Final answer: 79 × 21 = 1659.

Question: (16) 85 × 35 = _____

Answer: Step 1: Split 35 into 30 and 5.

Step 2: 85 × 30 = 2550.

Step 3: 85 × 5 = 425.

Step 4: Add the two parts: 2550 + 425 = 2975.

Final answer: 85 × 35 = 2975.

Question: (17) 92 × 25 = _____

Answer: Step 1: Split 25 into 20 and 5.

Step 2: 92 × 20 = 1840.

Step 3: 92 × 5 = 460.

Step 4: Add the two parts: 1840 + 460 = 2300.

Final answer: 92 × 25 = 2300.

Question: (18) 95 × 40 = _____

Answer: Step 1: First multiply 95 × 4 = 380.

Step 2: Because 40 is 4 tens, write one zero at the end: 380 × 10 = 3800.

Final answer: 95 × 40 = 3800.

EXERCISE 22 — Solve real life problems involving multiplication

Question: (1) There are 15 boxes. Each box has 6 motor bikes. How many motor bikes are there in all?

Answer: Step 1: Total motor bikes = 15 × 6.

Step 2: Multiply the ones: 5 × 6 = 30. Write 0 and carry 3.

Step 3: Multiply the tens: 1 × 6 = 6, plus the carried 3 makes 9.

Final answer: There are 90 motor bikes in all.

Question: (2) How many days are there in 40 weeks?

Answer: Step 1: One week has 7 days, so 40 weeks = 40 × 7 days.

Step 2: First multiply 4 × 7 = 28.

Step 3: Because 40 is 4 tens, write one zero at the end: 280.

Final answer: There are 280 days in 40 weeks.

Question: (3) There are 8 nests on a tree. Each nest has 6 birds. How many birds are there in all?

Answer: Step 1: Total birds = 8 × 6.

Step 2: Count in sixes 8 times: 6, 12, 18, 24, 30, 36, 42, 48.

Final answer: There are 48 birds in all.

Question: (4) A box contains 54 chocolates. How many chocolates are there in 7 such boxes?

Answer: Step 1: Total chocolates = 54 × 7.

Step 2: Multiply the ones: 4 × 7 = 28. Write 8 and carry 2.

Step 3: Multiply the tens: 5 × 7 = 35, plus the carried 2 makes 37.

Final answer: There are 378 chocolates in 7 boxes.

Question: (5) One cone has 3 scoops of ice cream. How many scoops are there in 48 cones?

Answer: Step 1: Total scoops = 48 × 3.

Step 2: Multiply the ones: 8 × 3 = 24. Write 4 and carry 2.

Step 3: Multiply the tens: 4 × 3 = 12, plus the carried 2 makes 14.

Final answer: There are 144 scoops in 48 cones.

Question: (6) One shelf has 25 books. How many books are there on 6 shelves?

Answer: Step 1: Total books = 25 × 6.

Step 2: Multiply the ones: 5 × 6 = 30. Write 0 and carry 3.

Step 3: Multiply the tens: 2 × 6 = 12, plus the carried 3 makes 15.

Final answer: There are 150 books on 6 shelves.

ACTIVITY — Name the dividend, divisor and quotient

Question: (1) In 8 ÷ 2 = 4, which number is the divisor?

Answer: Step 1: In a division sum, the number we divide by is called the divisor.

Step 2: Here we divide 8 by 2.

Final answer: The divisor is 2.

Question: (2) In 42 ÷ 7 = 6, which number is the dividend?

Answer: Step 1: The number which is divided is called the dividend.

Step 2: Here 42 is divided into 7 equal parts.

Final answer: The dividend is 42.

Question: (3) In 64 ÷ 8 = 8, which number is the quotient?

Answer: Step 1: The answer of a division sum is called the quotient.

Step 2: Here the answer of the division is 8.

Final answer: The quotient is 8.

EXERCISE 23 — (A) Divide the following

Question: (1) 56 ÷ 8 = _____

Answer: Step 1: Think of the table of 8.

Step 2: 8 × 7 = 56, so 56 divided into 8 equal parts gives 7 in each part.

Final answer: 56 ÷ 8 = 7.

Question: (2) 32 ÷ 4 = _____

Answer: Step 1: Think of the table of 4.

Step 2: 4 × 8 = 32, so 32 divided into 4 equal parts gives 8 in each part.

Final answer: 32 ÷ 4 = 8.

Question: (3) 80 ÷ 10 = _____

Answer: Step 1: Think of the table of 10.

Step 2: 10 × 8 = 80, so 80 divided into 10 equal parts gives 8 in each part.

Final answer: 80 ÷ 10 = 8.

Question: (4) 54 ÷ 6 = _____

Answer: Step 1: Think of the table of 6.

Step 2: 6 × 9 = 54, so 54 divided into 6 equal parts gives 9 in each part.

Final answer: 54 ÷ 6 = 9.

Question: (5) 24 ÷ 8 = _____

Answer: Step 1: Think of the table of 8.

Step 2: 8 × 3 = 24, so 24 divided into 8 equal parts gives 3 in each part.

Final answer: 24 ÷ 8 = 3.

Question: (6) 45 ÷ 5 = _____

Answer: Step 1: Think of the table of 5.

Step 2: 5 × 9 = 45, so 45 divided into 5 equal parts gives 9 in each part.

Final answer: 45 ÷ 5 = 9.

Question: (7) 72 ÷ 6 = _____

Answer: Step 1: Split 72 into 60 and 12.

Step 2: 60 ÷ 6 = 10.

Step 3: 12 ÷ 6 = 2.

Step 4: Add the two parts: 10 + 2 = 12.

Final answer: 72 ÷ 6 = 12.

Question: (8) 85 ÷ 5 = _____

Answer: Step 1: Split 85 into 50 and 35.

Step 2: 50 ÷ 5 = 10.

Step 3: 35 ÷ 5 = 7.

Step 4: Add the two parts: 10 + 7 = 17.

Final answer: 85 ÷ 5 = 17.

Question: (9) 76 ÷ 4 = _____

Answer: Step 1: Split 76 into 40 and 36.

Step 2: 40 ÷ 4 = 10.

Step 3: 36 ÷ 4 = 9.

Step 4: Add the two parts: 10 + 9 = 19.

Final answer: 76 ÷ 4 = 19.

Question: (10) 75 ÷ 5 = _____

Answer: Step 1: Split 75 into 50 and 25.

Step 2: 50 ÷ 5 = 10.

Step 3: 25 ÷ 5 = 5.

Step 4: Add the two parts: 10 + 5 = 15.

Final answer: 75 ÷ 5 = 15.

EXERCISE 23 — (B) Divide the following

Question: (1) 15 ÷ 3 = _____

Answer: Step 1: Think of the table of 3.

Step 2: 3 × 5 = 15.

Final answer: 15 ÷ 3 = 5.

Question: (2) 28 ÷ 4 = _____

Answer: Step 1: Think of the table of 4.

Step 2: 4 × 7 = 28.

Final answer: 28 ÷ 4 = 7.

Question: (3) 64 ÷ 8 = _____

Answer: Step 1: Think of the table of 8.

Step 2: 8 × 8 = 64.

Final answer: 64 ÷ 8 = 8.

Question: (4) 60 ÷ 6 = _____

Answer: Step 1: Think of the table of 6.

Step 2: 6 × 10 = 60.

Final answer: 60 ÷ 6 = 10.

Question: (5) 90 ÷ 10 = _____

Answer: Step 1: Think of the table of 10.

Step 2: 10 × 9 = 90.

Final answer: 90 ÷ 10 = 9.

Question: (6) 56 ÷ 7 = _____

Answer: Step 1: Think of the table of 7.

Step 2: 7 × 8 = 56.

Final answer: 56 ÷ 7 = 8.

Question: (7) 60 ÷ 4 = _____

Answer: Step 1: Split 60 into 40 and 20.

Step 2: 40 ÷ 4 = 10.

Step 3: 20 ÷ 4 = 5.

Step 4: Add the two parts: 10 + 5 = 15.

Final answer: 60 ÷ 4 = 15.

Question: (8) 99 ÷ 3 = _____

Answer: Step 1: Split 99 into 90 and 9.

Step 2: 90 ÷ 3 = 30.

Step 3: 9 ÷ 3 = 3.

Step 4: Add the two parts: 30 + 3 = 33.

Final answer: 99 ÷ 3 = 33.

Question: (9) 65 ÷ 5 = _____

Answer: Step 1: Split 65 into 50 and 15.

Step 2: 50 ÷ 5 = 10.

Step 3: 15 ÷ 5 = 3.

Step 4: Add the two parts: 10 + 3 = 13.

Final answer: 65 ÷ 5 = 13.

Question: (10) 77 ÷ 7 = _____

Answer: Step 1: Split 77 into 70 and 7.

Step 2: 70 ÷ 7 = 10.

Step 3: 7 ÷ 7 = 1.

Step 4: Add the two parts: 10 + 1 = 11.

Final answer: 77 ÷ 7 = 11.

Question: (11) 91 ÷ 7 = _____

Answer: Step 1: Split 91 into 70 and 21.

Step 2: 70 ÷ 7 = 10.

Step 3: 21 ÷ 7 = 3.

Step 4: Add the two parts: 10 + 3 = 13.

Final answer: 91 ÷ 7 = 13.

Question: (12) 50 ÷ 2 = _____

Answer: Step 1: Split 50 into 40 and 10.

Step 2: 40 ÷ 2 = 20.

Step 3: 10 ÷ 2 = 5.

Step 4: Add the two parts: 20 + 5 = 25.

Final answer: 50 ÷ 2 = 25.

EXERCISE 23 — (C) Name the circled part of each division sum (dividend, divisor or quotient)

Question: (1) In "15 bananas are divided into 3 equal groups", the circled number 15 is called the _____.

Answer: Step 1: The number which is divided is called the dividend.

Step 2: Here 15 bananas are being divided.

Final answer: 15 is the dividend.

Question: (2) In 10 ÷ 5 = 2, the circled number 2 is called the _____.

Answer: Step 1: The answer of a division sum is called the quotient.

Step 2: Here the answer of 10 ÷ 5 is 2.

Final answer: 2 is the quotient.

Question: (3) In 21 ÷ 7 = 3, the circled number 7 is called the _____.

Answer: Step 1: The number we divide by is called the divisor.

Step 2: Here we divide 21 by 7.

Final answer: 7 is the divisor.

Question: (4) In 72 ÷ 8 = 9, the circled number 9 is called the _____.

Answer: Step 1: The answer of a division sum is called the quotient.

Step 2: Here the answer of 72 ÷ 8 is 9.

Final answer: 9 is the quotient.

Question: (5) In 40 ÷ 8 = 5, the circled number 8 is called the _____.

Answer: Step 1: The number we divide by is called the divisor.

Step 2: Here we divide 40 by 8.

Final answer: 8 is the divisor.

Question: (6) In 49 ÷ 7 = 7, the circled number 49 is called the _____.

Answer: Step 1: The number which is divided is called the dividend.

Step 2: Here 49 is divided by 7.

Final answer: 49 is the dividend.

EXERCISE 24 — Solve real life problems involving division

Question: (1) 28 sweets are shared equally among 4 children. How many sweets does each child get?

Answer: Step 1: Sweets for each child = 28 ÷ 4.

Step 2: Think of the table of 4: 4 × 7 = 28.

Final answer: Each child gets 7 sweets.

Question: (2) 45 pencils are packed equally in 5 boxes. How many pencils

are there in each box?

Answer: Step 1: Pencils in each box = 45 ÷ 5.

Step 2: Think of the table of 5: 5 × 9 = 45.

Final answer: There are 9 pencils in each box.

Question: (3) 9 children share Rs 63 equally. How much money does each child get?

Answer: Step 1: Money for each child = 63 ÷ 9.

Step 2: Think of the table of 9: 9 × 7 = 63.

Final answer: Each child gets Rs 7.

Question: (4) A book has 100 pages. Ahmed reads 5 pages daily. In how many days will he finish the book?

Answer: Step 1: Number of days = 100 ÷ 5.

Step 2: Split 100 into 50 and 50.

Step 3: 50 ÷ 5 = 10 and 50 ÷ 5 = 10.

Step 4: Add the two parts: 10 + 10 = 20.

Final answer: He will finish the book in 20 days.

Question: (5) 49 notebooks are distributed equally among 7 students. How many notebooks does each student get?

Answer: Step 1: Notebooks for each student = 49 ÷ 7.

Step 2: Think of the table of 7: 7 × 7 = 49.

Final answer: Each student gets 7 notebooks.

Question: (6) 15 litres of milk is filled equally in 5 bottles. How many litres of milk are there in each bottle?

Answer: Step 1: Milk in each bottle = 15 ÷ 5.

Step 2: Think of the table of 5: 5 × 3 = 15.

Final answer: There are 3 litres of milk in each bottle.

Question: (7) 81 toffees are shared equally among 9 children. How many toffees does each child get?

Answer: Step 1: Toffees for each child = 81 ÷ 9.

Step 2: Think of the table of 9: 9 × 9 = 81.

Final answer: Each child gets 9 toffees.

EXERCISE 25 — Solve real life problems involving division

Question: (1) 80 candles are packed in packets of 4. How many packets are made?

Answer: Step 1: Number of packets = 80 ÷ 4.

Step 2: Split 80 into 40 and 40.

Step 3: 40 ÷ 4 = 10 and 40 ÷ 4 = 10.

Step 4: Add the two parts: 10 + 10 = 20.

Final answer: 20 packets are made.

Question: (2) 96 bottles are packed in boxes of 6. How many boxes are needed?

Answer: Step 1: Number of boxes = 96 ÷ 6.

Step 2: Split 96 into 60 and 36.

Step 3: 60 ÷ 6 = 10 and 36 ÷ 6 = 6.

Step 4: Add the two parts: 10 + 6 = 16.

Final answer: 16 boxes are needed.

Question: (3) 36 players are divided into teams of 12. How many teams are formed?

Answer: Step 1: Number of teams = 36 ÷ 12.

Step 2: Think of the table of 12: 12 × 3 = 36.

Final answer: 3 teams are formed.

Question: (4) 42 eggs are packed in boxes of 6. How many boxes are filled?

Answer: Step 1: Number of boxes = 42 ÷ 6.

Step 2: Think of the table of 6: 6 × 7 = 42.

Final answer: 7 boxes are filled.

Question: (5) 6 kg of sugar costs Rs 84. What is the cost of 1 kg of sugar?

Answer: Step 1: Cost of 1 kg = 84 ÷ 6.

Step 2: Split 84 into 60 and 24.

Step 3: 60 ÷ 6 = 10 and 24 ÷ 6 = 4.

Step 4: Add the two parts: 10 + 4 = 14.

Final answer: The cost of 1 kg of sugar is Rs 14.

Unit 3 FRACTIONS

ACTIVITY 1 — Write the fraction of the coloured part and write it in words

Question: (1) A shape is divided into 6 equal parts and 1 part is coloured. Write the fraction and its name in words.

Answer: Step 1: Total equal parts = 6. This number is written below the line; it is called the denominator.

Step 2: Coloured parts = 1. This number is written above the line; it is called the numerator.

Final answer: The fraction is 1/6, which is read as one-sixth.



1/6

Question: (2) A shape is divided into 5 equal parts and 3 parts are coloured. Write the fraction and its name in words.

Answer: Step 1: Total equal parts = 5, so the denominator is 5.

Step 2: Coloured parts = 3, so the numerator is 3.

Final answer: The fraction is 3/5, which is read as three-fifth.



3/5

Question: (3) A triangle is divided into 9 equal small triangles and 4 are

coloured. Write the fraction and its name in words.

Answer: Step 1: Total equal parts = 9, so the denominator is 9.

Step 2: Coloured parts = 4, so the numerator is 4.

Final answer: The fraction is 4/9, which is read as four-ninth.



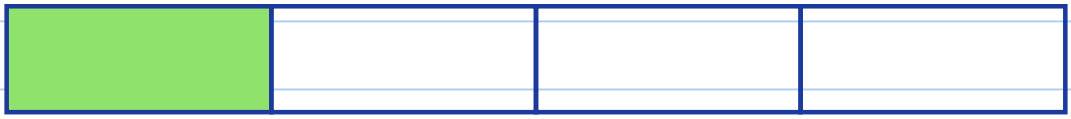
4/9

Question: (4) A square is divided into 4 equal parts and 1 part is coloured. Write the fraction and its name in words.

Answer: Step 1: Total equal parts = 4, so the denominator is 4.

Step 2: Coloured parts = 1, so the numerator is 1.

Final answer: The fraction is 1/4, which is read as one-fourth.



1/4

Question: (5) A shape is divided into 6 equal parts and 5 parts are coloured. Write the fraction and its name in words.

Answer: Step 1: Total equal parts = 6, so the denominator is 6.

Step 2: Coloured parts = 5, so the numerator is 5.

Final answer: The fraction is 5/6, which is read as five-sixth.



5/6

ACTIVITY 2 — Colour the given fraction of each shape

Question: (1) Colour 2/5 of a strip divided into 5 equal parts.

Answer: Step 1: The denominator 5 shows the strip has 5 equal parts.

Step 2: The numerator 2 shows we colour 2 parts.

Final answer: Colour any 2 of the 5 parts; the coloured fraction is 2/5.



2/5

Question: (2) Colour 1/3 of a triangle divided into 3 equal parts.

Answer: Step 1: The denominator 3 shows the triangle has 3 equal parts.

Step 2: The numerator 1 shows we colour 1 part.

Final answer: Colour any 1 of the 3 parts; the coloured fraction is 1/3.



1/3

Question: (3) Colour 3/8 of a square divided into 8 equal parts.

Answer: Step 1: The denominator 8 shows the square has 8 equal parts.

Step 2: The numerator 3 shows we colour 3 parts.

Final answer: Colour any 3 of the 8 parts; the coloured fraction is 3/8.



3/8

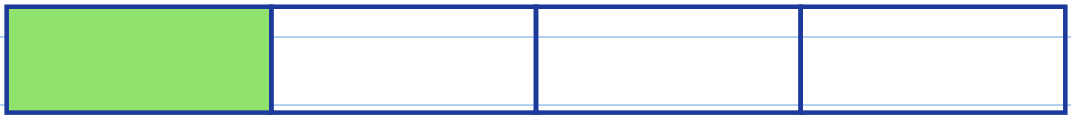
EXERCISE 26 — (A) Write the fraction of the coloured part of each figure

Question: (1) A figure divided into 4 equal parts with 1 part coloured.

Answer: Step 1: Total parts = 4.

Step 2: Coloured parts = 1.

Final answer: The coloured fraction is 1/4.



1/4

Question: (2) A figure divided into 10 equal parts with 7 parts coloured.

Answer: Step 1: Total parts = 10.

Step 2: Coloured parts = 7.

Final answer: The coloured fraction is 7/10.



7/10

Question: (3) A figure divided into 3 equal parts with 2 parts coloured.

Answer: Step 1: Total parts = 3.

Step 2: Coloured parts = 2.

Final answer: The coloured fraction is 2/3.



2/3

Question: (4) A figure divided into 7 equal parts with 3 parts coloured.

Answer: Step 1: Total parts = 7.

Step 2: Coloured parts = 3.

Final answer: The coloured fraction is 3/7.



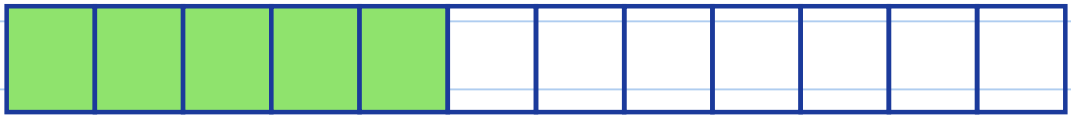
3/7

Question: (5) A figure divided into 12 equal parts with 5 parts coloured.

Answer: Step 1: Total parts = 12.

Step 2: Coloured parts = 5.

Final answer: The coloured fraction is 5/12.



5/12

Question: (6) A figure divided into 7 equal parts with 5 parts coloured.

Answer: Step 1: Total parts = 7.

Step 2: Coloured parts = 5.

Final answer: The coloured fraction is 5/7.



5/7

EXERCISE 26 — (B) Colour the given fraction in each figure

Question: (1) Colour 3/4 of a figure with 4 equal parts.

Answer: Step 1: The denominator is 4, so the figure has 4 equal parts.

Step 2: The numerator is 3, so colour 3 parts.

Final answer: Colour 3 out of the 4 parts.



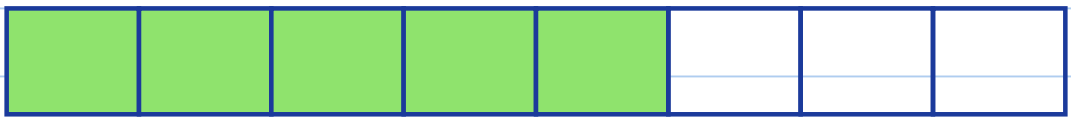
3/4

Question: (2) Colour 5/8 of a figure with 8 equal parts.

Answer: Step 1: The denominator is 8, so the figure has 8 equal parts.

Step 2: The numerator is 5, so colour 5 parts.

Final answer: Colour 5 out of the 8 parts.



5/8

Question: (3) Colour 1/4 of a figure with 4 equal parts.

Answer: Step 1: The denominator is 4, so the figure has 4 equal parts.

Step 2: The numerator is 1, so colour 1 part.

Final answer: Colour 1 out of the 4 parts.



1/4

Question: (4) Colour 6/7 of a figure with 7 equal strips.

Answer: Step 1: The denominator is 7, so the figure has 7 equal strips.

Step 2: The numerator is 6, so colour 6 strips.

Final answer: Colour 6 out of the 7 strips.



6/7

EXERCISE 26 — (C) Write the fraction shown by each figure

Question: (1) A figure divided into 11 equal parts with 4 parts coloured.

Answer: Step 1: Total parts = 11, so the denominator is 11.

Step 2: Coloured parts = 4, so the numerator is 4.

Final answer: The fraction is 4/11.



4/11

Question: (2) A figure divided into 8 equal parts with 7 parts coloured.

Answer: Step 1: Total parts = 8, so the denominator is 8.

Step 2: Coloured parts = 7, so the numerator is 7.

Final answer: The fraction is 7/8.



7/8

EXERCISE 27 — 1. Write the fraction of the coloured part of each figure and state which fractions are equivalent

Question: (i) Four figures show the fractions 2/4, 1/3, 3/6 and 1/2. Which of these fractions are equivalent?

Answer: Step 1: 2/4 means 2 out of 4 equal parts; half of the figure is coloured.

Step 2: 3/6 means 3 out of 6 equal parts; half of the figure is coloured.

Step 3: 1/2 also means half of the figure is coloured.

Step 4: 1/3 means only one third is coloured, so it is not equal to half.

Final answer: 2/4 = 3/6 = 1/2 are equivalent fractions.

Question: (ii) Four figures show the fractions 1/2, 2/4, 2/6 and 3/6. Which of these fractions are equivalent?

Answer: Step 1: 1/2 means half of the figure is coloured.

Step 2: 2/4 means 2 out of 4 parts, which is also half.

Step 3: 3/6 means 3 out of 6 parts, which is also half.

Step 4: 2/6 means 2 out of 6 parts, which is one third, not half.

Final answer: 1/2 = 2/4 = 3/6 are equivalent fractions.

Question: (iii) Four figures show the fractions 2/5, 3/6, 2/4 and 1/2. Which of these fractions are equivalent?

Answer: Step 1: 3/6 means 3 out of 6 parts, which is half.

Step 2: 2/4 means 2 out of 4 parts, which is also half.

Step 3: 1/2 is also half.

Step 4: 2/5 is less than half, so it is not equivalent.

Final answer: 3/6 = 2/4 = 1/2 are equivalent fractions.

EXERCISE 27 — 2. Write three equivalent fractions of each of the following

Question: (i) 1/4

Answer: Step 1: Multiply the numerator and the denominator by 2: 1 × 2 = 2 and 4 × 2 = 8, giving 2/8.

Step 2: Multiply both by 3: 1 × 3 = 3 and 4 × 3 = 12, giving 3/12.

Step 3: Multiply both by 4: 1 × 4 = 4 and 4 × 4 = 16, giving 4/16.

Final answer: Three equivalent fractions of 1/4 are 2/8, 3/12 and 4/16.

Question: (ii) 2/3

Answer: Step 1: Multiply the numerator and the denominator by 2: 2 × 2 = 4 and 3 × 2 = 6, giving 4/6.

Step 2: Multiply both by 3: 2 × 3 = 6 and 3 × 3 = 9, giving 6/9.

Step 3: Multiply both by 4: 2 × 4 = 8 and 3 × 4 = 12, giving 8/12.

Final answer: Three equivalent fractions of 2/3 are 4/6, 6/9 and 8/12.

Question: (iii) 3/4

Answer: Step 1: Multiply the numerator and the denominator by 2: 3 × 2 = 6 and 4 × 2 = 8, giving 6/8.

Step 2: Multiply both by 3: 3 × 3 = 9 and 4 × 3 = 12, giving 9/12.

Step 3: Multiply both by 4: 3 × 4 = 12 and 4 × 4 = 16, giving 12/16.

Final answer: Three equivalent fractions of 3/4 are 6/8, 9/12 and 12/16.

Question: (iv) 4/5

Answer: Step 1: Multiply the numerator and the denominator by 2: 4 × 2 =

8 and 5 × 2 = 10, giving 8/10.

Step 2: Multiply both by 3: 4 × 3 = 12 and 5 × 3 = 15, giving 12/15.

Step 3: Multiply both by 4: 4 × 4 = 16 and 5 × 4 = 20, giving 16/20.

Final answer: Three equivalent fractions of 4/5 are 8/10, 12/15 and 16/20.

Question: (v) 1/5

Answer: Step 1: Multiply the numerator and the denominator by 2: 1 × 2 = 2 and 5 × 2 = 10, giving 2/10.

Step 2: Multiply both by 3: 1 × 3 = 3 and 5 × 3 = 15, giving 3/15.

Step 3: Multiply both by 4: 1 × 4 = 4 and 5 × 4 = 20, giving 4/20.

Final answer: Three equivalent fractions of 1/5 are 2/10, 3/15 and 4/20.

Question: (vi) 2/6

Answer: Step 1: Multiply the numerator and the denominator by 2: 2 × 2 = 4 and 6 × 2 = 12, giving 4/12.

Step 2: Multiply both by 3: 2 × 3 = 6 and 6 × 3 = 18, giving 6/18.

Step 3: Multiply both by 4: 2 × 4 = 8 and 6 × 4 = 24, giving 8/24.

Final answer: Three equivalent fractions of 2/6 are 4/12, 6/18 and 8/24.

ACTIVITY — Sort the fractions into proper fractions and improper fractions: 2/9, 4/3, 2/5, 11/7, 1/3, 6/1, 9/9, 8/1, 12/15, 14/13

Question: Write the given fractions in two groups: proper fractions and improper fractions.

Answer: Step 1: A proper fraction has a smaller numerator than denominator: 2/9, 2/5, 1/3 and 12/15 are proper fractions.

Step 2: An improper fraction has a numerator equal to or bigger than the

denominator: 4/3, 11/7, 6/1, 9/9, 8/1 and 14/13 are improper fractions.

Final answer: Proper fractions: 2/9, 2/5, 1/3, 12/15. Improper fractions: 4/3, 11/7, 6/1, 9/9, 8/1, 14/13.

ACTIVITY — Put a tick (✓) for a true statement and a cross (✗) for a false statement

Question: (1) 3/4 < 7/4

Answer: Step 1: Both fractions have the same denominator 4.

Step 2: Compare the numerators: 3 is less than 7.

Final answer: 3/4 < 7/4 is true. ✓

Question: (2) 1/8 > 3/8

Answer: Step 1: Both fractions have the same denominator 8.

Step 2: Compare the numerators: 1 is less than 3, so 1/8 is smaller, not greater.

Final answer: 1/8 > 3/8 is false. ✗

Question: (3) 11/12 = 5/12

Answer: Step 1: Both fractions have the same denominator 12.

Step 2: Compare the numerators: 11 is not equal to 5.

Final answer: 11/12 = 5/12 is false. ✗

Question: (4) 7/13 = 7/13

Answer: Step 1: Both fractions have the same numerator and the same denominator.

Step 2: Equal fractions are written with the = sign.

Final answer: 7/13 = 7/13 is true. ✓

EXERCISE 28 — (A) Encircle the proper fractions: 2/3, 3/5, 7/5, 5/1, 4/9, 3/2, 11/7, 9/9

Question: Which of the given fractions are proper fractions?

Answer: Step 1: In a proper fraction the numerator is smaller than the denominator.

Step 2: 2/3 has 2 < 3, 3/5 has 3 < 5 and 4/9 has 4 < 9, so these are proper.

Step 3: 7/5, 5/1, 3/2, 11/7 and 9/9 have numerators equal to or bigger than the denominators, so they are improper.

Final answer: The proper fractions are 2/3, 3/5 and 4/9.

EXERCISE 28 — (B) Compare the following fractions using <, > or =

Question: (1) 1/6 _____ 3/6

Answer: Step 1: The denominators are the same (6).

Step 2: Compare the numerators: 1 is less than 3.

Final answer: 1/6 < 3/6.

Question: (2) 3/5 _____ 2/5

Answer: Step 1: The denominators are the same (5).

Step 2: Compare the numerators: 3 is greater than 2.

Final answer: 3/5 > 2/5.

Question: (3) 2/8 _____ 4/8

Answer: Step 1: The denominators are the same (8).

Step 2: Compare the numerators: 2 is less than 4.

Final answer: 2/8 < 4/8.

Question: (4) 7/9 _____ 6/9

Answer: Step 1: The denominators are the same (9).

Step 2: Compare the numerators: 7 is greater than 6.

Final answer: 7/9 > 6/9.

Question: (5) 2/7 _____ 2/7

Answer: Step 1: Both the numerators and the denominators are the same.

Step 2: The two fractions are exactly equal.

Final answer: 2/7 = 2/7.

Question: (6) 8/10 _____ 5/10

Answer: Step 1: The denominators are the same (10).

Step 2: Compare the numerators: 8 is greater than 5.

Final answer: 8/10 > 5/10.

Question: (7) 3/9 _____ 4/9

Answer: Step 1: The denominators are the same (9).

Step 2: Compare the numerators: 3 is less than 4.

Final answer: 3/9 < 4/9.

Question: (8) 5/6 _____ 2/6

Answer: Step 1: The denominators are the same (6).

Step 2: Compare the numerators: 5 is greater than 2.

Final answer: 5/6 > 2/6.

Question: (9) 7/11 _____ 7/11

Answer: Step 1: Both the numerators and the denominators are the same.

Step 2: The two fractions are exactly equal.

Final answer: 7/11 = 7/11.

ACTIVITY — Add the fractions with the help of the figures

Question: (1) 1/7 + 4/7 = _____

Answer: Step 1: The denominators are the same (7), so only the numerators are added.

Step 2: 1 + 4 = 5.

Final answer: 1/7 + 4/7 = 5/7.



5/7

Question: (2) 4/9 + 3/9 = _____

Answer: Step 1: The denominators are the same (9), so only the numerators are added.

Step 2: 4 + 3 = 7.

Final answer: 4/9 + 3/9 = 7/9.



7/9

Question: (3) 3/11 + 5/11 = _____

Answer: Step 1: The denominators are the same (11), so only the numerators are added.

Step 2: 3 + 5 = 8.

Final answer: 3/11 + 5/11 = 8/11.



8/11

EXERCISE 29 — (A) Add the following fractions

Question: (1) 1/3 + 1/3 = _____

Answer: Step 1: The denominators are the same (3).

Step 2: Add the numerators: 1 + 1 = 2.

Step 3: The denominator stays 3.

Final answer: 1/3 + 1/3 = 2/3.

Question: (2) 1/5 + 3/5 = _____

Answer: Step 1: The denominators are the same (5).

Step 2: Add the numerators: 1 + 3 = 4.

Step 3: The denominator stays 5.

Final answer: 1/5 + 3/5 = 4/5.

Question: (3) 2/4 + 1/4 = _____

Answer: Step 1: The denominators are the same (4).

Step 2: Add the numerators: 2 + 1 = 3.

Step 3: The denominator stays 4.

Final answer: 2/4 + 1/4 = 3/4.

Question: (4) 3/6 + 2/6 = _____

Answer: Step 1: The denominators are the same (6).

Step 2: Add the numerators: 3 + 2 = 5.

Step 3: The denominator stays 6.

Final answer: 3/6 + 2/6 = 5/6.

Question: (5) 2/7 + 4/7 = _____

Answer: Step 1: The denominators are the same (7).

Step 2: Add the numerators: 2 + 4 = 6.

Step 3: The denominator stays 7.

Final answer: 2/7 + 4/7 = 6/7.

Question: (6) 3/8 + 4/8 = _____

Answer: Step 1: The denominators are the same (8).

Step 2: Add the numerators: 3 + 4 = 7.

Step 3: The denominator stays 8.

Final answer: 3/8 + 4/8 = 7/8.

Question: (7) 5/9 + 4/9 = _____

Answer: Step 1: The denominators are the same (9).

Step 2: Add the numerators: 5 + 4 = 9.

Step 3: The denominator stays 9; 9/9 means one whole.

Final answer: 5/9 + 4/9 = 9/9 = 1 whole.

Question: (8) 3/11 + 6/11 = _____

Answer: Step 1: The denominators are the same (11).

Step 2: Add the numerators: 3 + 6 = 9.

Step 3: The denominator stays 11.

Final answer: 3/11 + 6/11 = 9/11.

Question: (9) 3/15 + 10/15 = _____

Answer: Step 1: The denominators are the same (15).

Step 2: Add the numerators: 3 + 10 = 13.

Step 3: The denominator stays 15.

Final answer: 3/15 + 10/15 = 13/15.

Question: (10) 11/18 + 6/18 = _____

Answer: Step 1: The denominators are the same (18).

Step 2: Add the numerators: 11 + 6 = 17.

Step 3: The denominator stays 18.

Final answer: 11/18 + 6/18 = 17/18.

Question: (11) 1/14 + 12/14 = _____

Answer: Step 1: The denominators are the same (14).

Step 2: Add the numerators: 1 + 12 = 13.

Step 3: The denominator stays 14.

Final answer: 1/14 + 12/14 = 13/14.

Question: (12) 9/20 + 10/20 = _____

Answer: Step 1: The denominators are the same (20).

Step 2: Add the numerators: 9 + 10 = 19.

Step 3: The denominator stays 20.

Final answer: 9/20 + 10/20 = 19/20.

EXERCISE 29 — (B) Add the following fractions

Question: (1) 3/7 + 2/7 = _____

Answer: Step 1: The denominators are the same (7).

Step 2: Add the numerators: 3 + 2 = 5.

Final answer: 3/7 + 2/7 = 5/7.

Question: (2) 4/12 + 5/12 = _____

Answer: Step 1: The denominators are the same (12).

Step 2: Add the numerators: 4 + 5 = 9.

Final answer: 4/12 + 5/12 = 9/12.

Question: (3) 6/15 + 7/15 = _____

Answer: Step 1: The denominators are the same (15).

Step 2: Add the numerators: 6 + 7 = 13.

Final answer: 6/15 + 7/15 = 13/15.

EXERCISE 29 — (C) Add the fractions with the help of the figures

Question: (1) 2/5 + 1/5 = _____

Answer: Step 1: The figures show fifths; the denominators are the same (5).

Step 2: Add the numerators: 2 + 1 = 3.

Final answer: 2/5 + 1/5 = 3/5.

Question: (2) 3/7 + 5/7 = _____

Answer: Step 1: The figures show sevenths; the denominators are the same (7).

Step 2: Add the numerators: 3 + 5 = 8.

Final answer: 3/7 + 5/7 = 8/7.

Question: (3) 8/10 + 1/10 = _____

Answer: Step 1: The figures show tenths; the denominators are the same (10).

Step 2: Add the numerators: 8 + 1 = 9.

Final answer: 8/10 + 1/10 = 9/10.

Question: (4) 1/12 + 1/12 = _____

Answer: Step 1: The figures show twelfths; the denominators are the same (12).

Step 2: Add the numerators: 1 + 1 = 2.

Final answer: 1/12 + 1/12 = 2/12.

Question: (5) 6/13 + 3/13 = _____

Answer: Step 1: The figures show thirteenths; the denominators are the same (13).

Step 2: Add the numerators: 6 + 3 = 9.

Final answer: 6/13 + 3/13 = 9/13.

Question: (6) 7/15 + 4/15 = _____

Answer: Step 1: The figures show fifteenths; the denominators are the same (15).

Step 2: Add the numerators: 7 + 4 = 11.

Final answer: 7/15 + 4/15 = 11/15.

ACTIVITY — Subtract the fractions with the help of the figures

Question: (1) 3/4 - 2/4 = _____

Answer: Step 1: The denominators are the same (4), so only the numerators are subtracted.

Step 2: 3 - 2 = 1.

Final answer: 3/4 - 2/4 = 1/4.



1/4

Question: (2) 6/7 - 1/7 = _____

Answer: Step 1: The denominators are the same (7), so only the numerators are subtracted.

Step 2: 6 - 1 = 5.

Final answer: 6/7 - 1/7 = 5/7.



5/7

Question: (3) 11/12 - 5/12 = _____

Answer: Step 1: The denominators are the same (12), so only the numerators are subtracted.

Step 2: 11 - 5 = 6.

Final answer: 11/12 - 5/12 = 6/12.



6/12

EXERCISE 30 — 1. Subtract the following fractions

Question: (1) 2/3 - 1/3 = _____

Answer: Step 1: The denominators are the same (3).

Step 2: Subtract the numerators: 2 - 1 = 1.

Step 3: The denominator stays 3.

Final answer: 2/3 - 1/3 = 1/3.

Question: (2) 4/5 - 2/5 = _____

Answer: Step 1: The denominators are the same (5).

Step 2: Subtract the numerators: 4 - 2 = 2.

Step 3: The denominator stays 5.

Final answer: 4/5 - 2/5 = 2/5.

Question: (3) 5/6 - 4/6 = _____

Answer: Step 1: The denominators are the same (6).

Step 2: Subtract the numerators: 5 - 4 = 1.

Step 3: The denominator stays 6.

Final answer: 5/6 - 4/6 = 1/6.

Question: (4) 7/8 - 4/8 = _____

Answer: Step 1: The denominators are the same (8).

Step 2: Subtract the numerators: 7 - 4 = 3.

Step 3: The denominator stays 8.

Final answer: 7/8 - 4/8 = 3/8.

Question: (5) 6/7 - 4/7 = _____

Answer: Step 1: The denominators are the same (7).

Step 2: Subtract the numerators: 6 - 4 = 2.

Step 3: The denominator stays 7.

Final answer: 6/7 - 4/7 = 2/7.

Question: (6) 8/9 - 6/9 = _____

Answer: Step 1: The denominators are the same (9).

Step 2: Subtract the numerators: 8 - 6 = 2.

Step 3: The denominator stays 9.

Final answer: 8/9 - 6/9 = 2/9.

Question: (7) 9/10 - 4/10 = _____

Answer: Step 1: The denominators are the same (10).

Step 2: Subtract the numerators: 9 - 4 = 5.

Step 3: The denominator stays 10.

Final answer: 9/10 - 4/10 = 5/10.

Question: (8) 11/13 - 7/13 = _____

Answer: Step 1: The denominators are the same (13).

Step 2: Subtract the numerators: 11 - 7 = 4.

Step 3: The denominator stays 13.

Final answer: 11/13 - 7/13 = 4/13.

Question: (9) 10/11 - 5/11 = _____

Answer: Step 1: The denominators are the same (11).

Step 2: Subtract the numerators: 10 - 5 = 5.

Step 3: The denominator stays 11.

Final answer: 10/11 - 5/11 = 5/11.

Question: (10) 9/12 - 2/12 = _____

Answer: Step 1: The denominators are the same (12).

Step 2: Subtract the numerators: 9 - 2 = 7.

Step 3: The denominator stays 12.

Final answer: 9/12 - 2/12 = 7/12.

Question: (11) 14/15 - 13/15 = _____

Answer: Step 1: The denominators are the same (15).

Step 2: Subtract the numerators: 14 - 13 = 1.

Step 3: The denominator stays 15.

Final answer: 14/15 - 13/15 = 1/15.

Question: (12) 17/18 - 12/18 = _____

Answer: Step 1: The denominators are the same (18).

Step 2: Subtract the numerators: 17 - 12 = 5.

Step 3: The denominator stays 18.

Final answer: 17/18 - 12/18 = 5/18.

EXERCISE 30 — 2. Subtract the following fractions

Question: (1) 2/4 - 1/4 = _____

Answer: Step 1: The denominators are the same (4).

Step 2: Subtract the numerators: 2 - 1 = 1.

Final answer: 2/4 - 1/4 = 1/4.

Question: (2) 13/14 - 2/14 = _____

Answer: Step 1: The denominators are the same (14).

Step 2: Subtract the numerators: 13 - 2 = 11.

Final answer: 13/14 - 2/14 = 11/14.

Question: (3) 17/19 - 16/19 = _____

Answer: Step 1: The denominators are the same (19).

Step 2: Subtract the numerators: 17 - 16 = 1.

Final answer: 17/19 - 16/19 = 1/19.

EXERCISE 30 — 3. Subtract the following fractions

Question: (1) 4/7 - 2/7 = _____

Answer: Step 1: The denominators are the same (7).

Step 2: Subtract the numerators: 4 - 2 = 2.

Final answer: 4/7 - 2/7 = 2/7.

Question: (2) 8/9 - 1/9 = _____

Answer: Step 1: The denominators are the same (9).

Step 2: Subtract the numerators: 8 - 1 = 7.

Final answer: 8/9 - 1/9 = 7/9.

Question: (3) 9/10 - 8/10 = _____

Answer: Step 1: The denominators are the same (10).

Step 2: Subtract the numerators: 9 - 8 = 1.

Final answer: 9/10 - 8/10 = 1/10.

Question: (4) 12/13 - 5/13 = _____

Answer: Step 1: The denominators are the same (13).

Step 2: Subtract the numerators: 12 - 5 = 7.

Final answer: 12/13 - 5/13 = 7/13.

Question: (5) 10/11 - 6/11 = _____

Answer: Step 1: The denominators are the same (11).

Step 2: Subtract the numerators: 10 - 6 = 4.

Final answer: 10/11 - 6/11 = 4/11.

Question: (6) 16/17 - 10/17 = _____

Answer: Step 1: The denominators are the same (17).

Step 2: Subtract the numerators: 16 - 10 = 6.

Final answer: 16/17 - 10/17 = 6/17.

Question: (7) 13/25 - 7/25 = _____

Answer: Step 1: The denominators are the same (25).

Step 2: Subtract the numerators: 13 - 7 = 6.

Final answer: 13/25 - 7/25 = 6/25.

Question: (8) 16/17 - 15/19 = _____

Answer: Step 1: This sum is printed in the book with different denominators, 17 and 19. First make both denominators the same: 17 ×

19 = 323.

Step 2: Change 16/17 into 323rds: 16 × 19 = 304, so 16/17 = 304/323.

Step 3: Change 15/19 into 323rds: 15 × 17 = 255, so 15/19 = 255/323.

Step 4: Subtract the numerators: 304 - 255 = 49.

Step 5: The denominator stays 323.

Final answer: 16/17 - 15/19 = 49/323.

Question: (9) 18/15 - 15/15 = _____

Answer: Step 1: The denominators are the same (15).

Step 2: Subtract the numerators: 18 - 15 = 3.

Final answer: 18/15 - 15/15 = 3/15.

Unit 4 MEASUREMENT

ACTIVITY — Write cm, m or km for measuring each of the following

Question: (1) Length of an ant is measured in _____.

Answer: Step 1: An ant is very small, only a few centimetres long.

Step 2: Small lengths are measured in centimetres.

Final answer: The length of an ant is measured in centimetres (cm).

Question: (2) Height of a tree is measured in _____.

Answer: Step 1: A tree is tall, much taller than a child.

Step 2: Such heights are measured in metres.

Final answer: The height of a tree is measured in metres (m).

Question: (3) Height of a mountain is measured in _____.

Answer: Step 1: A mountain is very tall.

Step 2: Its height is measured in metres.

Final answer: The height of a mountain is measured in metres (m).

Question: (4) Width of the Phuleli Canal is measured in _____.

Answer: Step 1: A canal is wide, wider than a road.

Step 2: Such widths are measured in metres.

Final answer: The width of the Phuleli Canal is measured in metres (m).

Question: (5) Length of a pen is measured in _____.

Answer: Step 1: A pen is small and fits in the hand.

Step 2: Small lengths are measured in centimetres.

Final answer: The length of a pen is measured in centimetres (cm).

Question: (6) Height of a door is measured in _____.

Answer: Step 1: A door is about as tall as a grown-up person.

Step 2: Such heights are measured in metres.

Final answer: The height of a door is measured in metres (m).

Question: (7) Distance from Karachi to Sukkur is measured in _____.

Answer: Step 1: Karachi and Sukkur are two far away cities.

Step 2: Long distances between cities are measured in kilometres.

Final answer: The distance from Karachi to Sukkur is measured in kilometres (km).

EXERCISE 31 — 1. Measure the length of the following objects with a ruler (in centimetres)

Question: (i) Length of the pen = _____ cm

Answer: Step 1: Place the zero of the ruler at one end of the pen.

Step 2: Read the mark at the other end of the pen; it comes to 6.

Final answer: The length of the pen is 6 cm.

Question: (ii) Length of the eraser = _____ cm

Answer: Step 1: Place the zero of the ruler at one end of the eraser.

Step 2: Read the mark at the other end; it comes to 3.

Final answer: The length of the eraser is 3 cm.

Question: (iii) Length of the carrot = _____ cm

Answer: Step 1: Place the zero of the ruler at one end of the carrot.

Step 2: Read the mark at the other end; it comes to 7.

Final answer: The length of the carrot is 7 cm.

Question: (iv) Length of the toothbrush = _____ cm

Answer: Step 1: Place the zero of the ruler at one end of the toothbrush.

Step 2: Read the mark at the other end; it comes to 10.

Final answer: The length of the toothbrush is 10 cm.

Question: (v) Length of the mobile phone = _____ cm

Answer: Step 1: Place the zero of the ruler at one end of the mobile phone.

Step 2: Read the mark at the other end; it comes to 8.

Final answer: The length of the mobile phone is 8 cm.

EXERCISE 31 — 2. Put a tick (✓) for the correct unit and a cross (✗) for the wrong unit

Question: (i) The length of a toothpaste tube is measured in metres.

Answer: Step 1: A toothpaste tube is small, only a few centimetres long.

Step 2: Metre is too big a unit for it; centimetre is used.

Final answer: The statement is wrong. ✗

Question: (ii) The length of a big window is measured in metres.

Answer: Step 1: A big window is about one or two metres long.

Step 2: Metre is the right unit for such a length.

Final answer: The statement is correct. ✓

Question: (iii) The distance between Karachi and Hyderabad is measured

in kilometres.

Answer: Step 1: Karachi and Hyderabad are two different cities, far from each other.

Step 2: Long distances are measured in kilometres.

Final answer: The statement is correct. ✓

Question: (iv) The height of a tower is measured in centimetres.

Answer: Step 1: A tower is very tall.

Step 2: Centimetre is too small a unit for a tower; metre is used.

Final answer: The statement is wrong. ✗

Question: (v) The length of a wall brick is measured in kilometres.

Answer: Step 1: A brick is small and fits in the hand.

Step 2: Kilometre is used only for very long distances, not for a brick.

Final answer: The statement is wrong. ✗

Question: (vi) The length of a hairbrush is measured in centimetres.

Answer: Step 1: A hairbrush is a small object.

Step 2: Small lengths are measured in centimetres.

Final answer: The statement is correct. ✓

EXERCISE 31 — 3. Fill in the blanks with the suitable unit (centimetre or metre)

Question: (i) The length of a science book is measured in _____.

Answer: Step 1: A book is a small object, less than one metre long.

Final answer: The length of a science book is measured in centimetres.

Question: (ii) The length of a piece of chalk is measured in _____.

Answer: Step 1: A piece of chalk is very small.

Final answer: The length of a piece of chalk is measured in centimetres.

Question: (iii) The length of a colour pencil is measured in _____.

Answer: Step 1: A colour pencil is small and fits in the hand.

Final answer: The length of a colour pencil is measured in centimetres.

Question: (iv) The length of a window is measured in _____.

Answer: Step 1: A window is big, about a metre or more.

Final answer: The length of a window is measured in metres.

Question: (v) The length of a door is measured in _____.

Answer: Step 1: A door is tall, about two metres.

Final answer: The length of a door is measured in metres.

EXERCISE 32 — (A) Add the following lengths

Question: (1) 67 cm + 21 cm = _____

Answer: Step 1: Add the ones: 7 + 1 = 8.

Step 2: Add the tens: 6 + 2 = 8.

Final answer: 67 cm + 21 cm = 88 cm.

Question: (2) 37 m + 85 m = _____

Answer: Step 1: Add the ones: 7 + 5 = 12. Write 2 and carry 1.

Step 2: Add the tens: 3 + 8 = 11, plus the carried 1 makes 12.

Final answer: 37 m + 85 m = 122 m.

Question: (3) 28 km + 15 km = _____

Answer: Step 1: Add the ones: 8 + 5 = 13. Write 3 and carry 1.

Step 2: Add the tens: 2 + 1 = 3, plus the carried 1 makes 4.

Final answer: 28 km + 15 km = 43 km.

Question: (4) 11 m 33 cm + 1 m 41 cm = _____

Answer: Step 1: Add the centimetres: 33 + 41 = 74 cm.

Step 2: Add the metres: 11 + 1 = 12 m.

Final answer: 11 m 33 cm + 1 m 41 cm = 12 m 74 cm.

Question: (5) 8 m 68 cm + 6 m 31 cm = _____

Answer: Step 1: Add the centimetres: 68 + 31 = 99 cm.

Step 2: Add the metres: 8 + 6 = 14 m.

Final answer: 8 m 68 cm + 6 m 31 cm = 14 m 99 cm.

Question: (6) 35 km 515 m + 12 km 436 m = _____

Answer: Step 1: Add the metres: 515 + 436 = 951 m.

Step 2: Add the kilometres: 35 + 12 = 47 km.

Final answer: 35 km 515 m + 12 km 436 m = 47 km 951 m.

EXERCISE 32 — (B) Add the following lengths

Question: (1) 66 cm + 23 cm = _____

Answer: Step 1: Add the ones: 6 + 3 = 9.

Step 2: Add the tens: 6 + 2 = 8.

Final answer: 66 cm + 23 cm = 89 cm.

Question: (2) 18 m + 9 m = _____

Answer: Step 1: Add the ones: 8 + 9 = 17. Write 7 and carry 1.

Step 2: Add the tens: 1 + 0 = 1, plus the carried 1 makes 2.

Final answer: 18 m + 9 m = 27 m.

Question: (3) 23 km + 18 km = _____

Answer: Step 1: Add the ones: 3 + 8 = 11. Write 1 and carry 1.

Step 2: Add the tens: 2 + 1 = 3, plus the carried 1 makes 4.

Final answer: 23 km + 18 km = 41 km.

Question: (4) 44 m 5 cm + 78 m 30 cm = _____

Answer: Step 1: Add the centimetres: 5 + 30 = 35 cm.

Step 2: Add the metres: 44 + 78 = 122 m (4 + 8 = 12 in the ones, write 2 carry 1; 4 + 7 = 11, plus the carried 1 makes 12).

Final answer: 44 m 5 cm + 78 m 30 cm = 122 m 35 cm.

EXERCISE 33 — Solve real life problems involving addition of length

Question: (1) Sana has a ribbon of 65 cm and Ayesha has a ribbon of 38 cm. What is the total length of both ribbons?

Answer: Step 1: Total length = 65 + 38.

Step 2: Add the ones: 5 + 8 = 13. Write 3 and carry 1.

Step 3: Add the tens: 6 + 3 = 9, plus the carried 1 makes 10.

Final answer: The total length of both ribbons is 103 cm.

Question: (2) The length of a road is 240 m. Another road of 105 m joins it. What is the total length of the road now?

Answer: Step 1: Total length = 240 + 105.

Step 2: Add the ones: 0 + 5 = 5.

Step 3: Add the tens: 4 + 0 = 4.

Step 4: Add the hundreds: 2 + 1 = 3.

Final answer: The total length of the road is 345 m.

Question: (3) The distance from Karachi to Hyderabad is 170 km and from Hyderabad to Moro is 165 km. What is the distance from Karachi to Moro?

Answer: Step 1: Total distance = 170 + 165.

Step 2: Add the ones: 0 + 5 = 5.

Step 3: Add the tens: 7 + 6 = 13. Write 3 and carry 1.

Step 4: Add the hundreds: 1 + 1 = 2, plus the carried 1 makes 3.

Final answer: The distance from Karachi to Moro is 335 km.

Question: (4) A boy walks 300 m to the shop, then 785 m to the park and then 500 m to his home. How much distance does he walk in all?

Answer: Step 1: Total distance = 300 + 785 + 500.

Step 2: First add 300 + 785 = 1085.

Step 3: Then add 1085 + 500 = 1585.

Final answer: He walks 1585 m in all.

EXERCISE 34 — (A) Subtract the following lengths

Question: (1) 81 cm - 42 cm = _____

Answer: Step 1: Subtract the ones: 1 cannot take away 2, so borrow 1 ten: 11 - 2 = 9.

Step 2: Subtract the tens: 8 tens became 7 tens; 7 - 4 = 3.

Final answer: 81 cm - 42 cm = 39 cm.

Question: (2) 48 km - 19 km = _____

Answer: Step 1: Subtract the ones: 8 cannot take away 9, so borrow 1 ten: 18 - 9 = 9.

Step 2: Subtract the tens: 4 tens became 3 tens; 3 - 1 = 2.

Final answer: 48 km - 19 km = 29 km.

Question: (3) 26 m - 8 m = _____

Answer: Step 1: Subtract the ones: 6 cannot take away 8, so borrow 1 ten: 16 - 8 = 8.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 - 0 = 1.

Final answer: 26 m - 8 m = 18 m.

Question: (4) 39 km 80 m - 17 km 48 m = _____

Answer: Step 1: Subtract the metres: 80 - 48 = 32 m.

Step 2: Subtract the kilometres: 39 - 17 = 22 km.

Final answer: 39 km 80 m - 17 km 48 m = 22 km 32 m.

Question: (5) 44 km 72 m - 26 km 50 m = _____

Answer: Step 1: Subtract the metres: 72 - 50 = 22 m.

Step 2: Subtract the kilometres: 44 - 26 = 18 km.

Final answer: 44 km 72 m - 26 km 50 m = 18 km 22 m.

Question: (6) 58 km 465 m - 29 km 287 m = _____

Answer: Step 1: Subtract the metres: ones 5 - 7 cannot be done, so borrow 1 ten: 15 - 7 = 8.

Step 2: Tens: 6 tens became 5 tens; 5 - 8 cannot be done, so borrow 1 hundred: 15 - 8 = 7. Hundreds: 4 became 3; 3 - 2 = 1. So 465 - 287 = 178

m.

Step 3: Subtract the kilometres: 58 - 29 = 29 km.

Final answer: 58 km 465 m - 29 km 287 m = 29 km 178 m.

EXERCISE 34 — (B) Subtract the following lengths

Question: (1) 76 cm - 36 cm = _____

Answer: Step 1: Subtract the ones: 6 - 6 = 0.

Step 2: Subtract the tens: 7 - 3 = 4.

Final answer: 76 cm - 36 cm = 40 cm.

Question: (2) 12 m - 9 m = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 9, so borrow 1 ten: 12 - 9 = 3.

Step 2: Subtract the tens: 1 ten became 0 tens.

Final answer: 12 m - 9 m = 3 m.

Question: (3) 95 km - 88 km = _____

Answer: Step 1: Subtract the ones: 5 cannot take away 8, so borrow 1 ten: 15 - 8 = 7.

Step 2: Subtract the tens: 9 tens became 8 tens; 8 - 8 = 0.

Final answer: 95 km - 88 km = 7 km.

Question: (4) 680 m - 494 m = _____

Answer: Step 1: Subtract the ones: 0 cannot take away 4, so borrow 1 ten: 10 - 4 = 6.

Step 2: Subtract the tens: 8 tens became 7 tens; 7 - 9 cannot be done, so borrow 1 hundred: 17 - 9 = 8.

Step 3: Subtract the hundreds: 6 hundreds became 5 hundreds; 5 - 4 = 1.

Final answer: 680 m - 494 m = 186 m.

Question: (5) 936 km 324 m - 864 km 150 m = _____

Answer: Step 1: Subtract the metres: 324 - 150 = 174 m.

Step 2: Subtract the kilometres: 936 - 864 = 72 km.

Final answer: 936 km 324 m - 864 km 150 m = 72 km 174 m.

EXERCISE 35 — Solve real life problems involving subtraction of length

Question: (1) A wire is 50 m long. 35 m of it is used. How much wire is left?

Answer: Step 1: Wire left = 50 - 35.

Step 2: Subtract the ones: 0 cannot take away 5, so borrow 1 ten: 10 - 5 = 5.

Step 3: Subtract the tens: 5 tens became 4 tens; 4 - 3 = 1.

Final answer: 15 m of wire is left.

Question: (2) A lace is 72 cm long. 47 cm is cut off from it. How much lace is left?

Answer: Step 1: Lace left = 72 - 47.

Step 2: Subtract the ones: 2 cannot take away 7, so borrow 1 ten: 12 - 7 = 5.

Step 3: Subtract the tens: 7 tens became 6 tens; 6 - 4 = 2.

Final answer: 25 cm of lace is left.

Question: (3) A ribbon is 1100 m long. 466 m is used. How much ribbon is left?

Answer: Step 1: Ribbon left = 1100 - 466.

Step 2: Subtract the ones: 0 cannot take away 6. The tens digit is also 0, so borrow 1 hundred: hundreds 1 becomes 0, tens become 10; the tens give 1 ten to the ones: tens become 9, ones become 10. Now 10 - 6 = 4.

Step 3: Subtract the tens: 9 - 6 = 3.

Step 4: Subtract the hundreds: 0 cannot take away 4, so borrow 1 thousand: thousands 1 becomes 0, hundreds become 10. Now 10 - 4 = 6.

Final answer: 634 m of ribbon is left.

Question: (4) The height of father is 2 m 50 cm and the height of his son is 1 m 22 cm. How much taller is the father than his son?

Answer: Step 1: Difference = 2 m 50 cm - 1 m 22 cm.

Step 2: Subtract the centimetres: 50 - 22 = 28 cm.

Step 3: Subtract the metres: 2 - 1 = 1 m.

Final answer: The father is 1 m 28 cm taller than his son.

ACTIVITY — Colour the correct unit, g or kg, for each of the following

Question: (1) Face cream (50 g): g or kg?

Answer: Step 1: Face cream is light, only 50 grams.

Step 2: Light things are weighed in grams.

Final answer: Face cream is weighed in grams (g).

Question: (2) An iron rod: g or kg?

Answer: Step 1: An iron rod is heavy.

Step 2: Heavy things are weighed in kilograms.

Final answer: An iron rod is weighed in kilograms (kg).

Question: (3) Toothpaste: g or kg?

Answer: Step 1: A toothpaste tube is light.

Step 2: Light things are weighed in grams.

Final answer: Toothpaste is weighed in grams (g).

Question: (4) A ring (5 g): g or kg?

Answer: Step 1: A ring is very light, only 5 grams.

Step 2: Very light things are weighed in grams.

Final answer: A ring is weighed in grams (g).

Question: (5) A biscuit pack (40 g): g or kg?

Answer: Step 1: A biscuit pack is light, only 40 grams.

Step 2: Light things are weighed in grams.

Final answer: A biscuit pack is weighed in grams (g).

Question: (6) A rice sack (50 kg): g or kg?

Answer: Step 1: A sack full of rice is very heavy, 50 kilograms.

Step 2: Heavy things are weighed in kilograms.

Final answer: A rice sack is weighed in kilograms (kg).

EXERCISE 36 — 1. Encircle the correct unit, g or kg, for each of the following

Question: (i) A bag of carrots weighing 2 _____.

Answer: Step 1: A bag of carrots is heavy, more than one kilogram.

Final answer: A bag of carrots weighs 2 kg.

Question: (ii) A packet of chillies weighing 100 _____.

Answer: Step 1: A small packet of chillies is light.

Final answer: A packet of chillies weighs 100 g.

Question: (iii) A bag of flour weighing 10 _____.

Answer: Step 1: A big bag of flour is heavy.

Final answer: A bag of flour weighs 10 kg.

Question: (iv) A tin of ghee weighing 500 _____.

Answer: Step 1: 500 is less than 1000, and a small tin of ghee is light.

Final answer: A tin of ghee weighs 500 g.

Question: (v) A packet of tea weighing 250 _____.

Answer: Step 1: A packet of tea is light.

Final answer: A packet of tea weighs 250 g.

Question: (vi) A baby weighing 4 _____.

Answer: Step 1: A baby weighs a few kilograms.

Final answer: A baby weighs 4 kg.

Question: (vii) A packet of garam masala weighing 50 _____.

Answer: Step 1: A small packet of garam masala is very light.

Final answer: A packet of garam masala weighs 50 g.

Question: (viii) A sack of potatoes weighing 30 _____.

Answer: Step 1: A sack of potatoes is very heavy.

Final answer: A sack of potatoes weighs 30 kg.

EXERCISE 36 — 2. Write the suitable unit, g or kg, for each of the following

Question: (i) An ice cream brick is weighed in _____.

Answer: Step 1: An ice cream brick is light.

Final answer: An ice cream brick is weighed in grams (g).

Question: (ii) A brick is weighed in _____.

Answer: Step 1: A building brick is heavy.

Final answer: A brick is weighed in kilograms (kg).

Question: (iii) A mobile phone is weighed in _____.

Answer: Step 1: A mobile phone is light.

Final answer: A mobile phone is weighed in grams (g).

Question: (iv) A log of wood is weighed in _____.

Answer: Step 1: A log of wood is heavy.

Final answer: A log is weighed in kilograms (kg).

Question: (v) A necklace is weighed in _____.

Answer: Step 1: A necklace is very light.

Final answer: A necklace is weighed in grams (g).

Question: (vi) A hat (cap) is weighed in _____.

Answer: Step 1: A cap is light.

Final answer: A hat is weighed in grams (g).

EXERCISE 36 — 3. Write the appropriate unit, g or kg, with each weight

Question: (i) Sugar cubes: 5 _____.

Answer: Step 1: A few sugar cubes are very light.

Final answer: Sugar cubes weigh 5 g.

Question: (ii) A bag of sugar: 3 _____.

Answer: Step 1: A bag of sugar is heavy.

Final answer: A bag of sugar weighs 3 kg.

Question: (iii) A packet of salt: 1 _____.

Answer: Step 1: A full packet of salt weighs about one kilogram.

Final answer: A packet of salt weighs 1 kg.

Question: (iv) A tin of ghee: 450 _____.

Answer: Step 1: 450 is less than 1000 grams, so the unit is grams.

Final answer: A tin of ghee weighs 450 g.

Question: (v) A cat: 6 _____.

Answer: Step 1: A cat weighs a few kilograms.

Final answer: A cat weighs 6 kg.

Question: (vi) A loaf of bread: 500 _____.

Answer: Step 1: A loaf of bread is light, half of 1000 grams.

Final answer: A loaf of bread weighs 500 g.

Question: (vii) A packet of corn flakes: 300 _____.

Answer: Step 1: A packet of corn flakes is light.

Final answer: A packet of corn flakes weighs 300 g.

Question: (viii) A boy: 38 _____.

Answer: Step 1: A boy is heavy, many kilograms.

Final answer: A boy weighs 38 kg.

Question: (ix) A goat: 25 _____.

Answer: Step 1: A goat is heavy.

Final answer: A goat weighs 25 kg.

EXERCISE 37 — (A) Add the following masses

Question: (1) 290 g + 305 g = _____

Answer: Step 1: Add the ones: 0 + 5 = 5.

Step 2: Add the tens: 9 + 0 = 9.

Step 3: Add the hundreds: 2 + 3 = 5.

Final answer: 290 g + 305 g = 595 g.

Question: (2) 504 g + 313 g = _____

Answer: Step 1: Add the ones: 4 + 3 = 7.

Step 2: Add the tens: 0 + 1 = 1.

Step 3: Add the hundreds: 5 + 3 = 8.

Final answer: 504 g + 313 g = 817 g.

Question: (3) 325 g + 477 g = _____

Answer: Step 1: Add the ones: 5 + 7 = 12. Write 2 and carry 1.

Step 2: Add the tens: 2 + 7 = 9, plus the carried 1 makes 10. Write 0 and carry 1.

Step 3: Add the hundreds: 3 + 4 = 7, plus the carried 1 makes 8.

Final answer: 325 g + 477 g = 802 g.

Question: (4) 35 kg + 46 kg = _____

Answer: Step 1: Add the ones: 5 + 6 = 11. Write 1 and carry 1.

Step 2: Add the tens: 3 + 4 = 7, plus the carried 1 makes 8.

Final answer: 35 kg + 46 kg = 81 kg.

Question: (5) 1214 kg + 2453 kg = _____

Answer: Step 1: Add the ones: 4 + 3 = 7.

Step 2: Add the tens: 1 + 5 = 6.

Step 3: Add the hundreds: 2 + 4 = 6.

Step 4: Add the thousands: 1 + 2 = 3.

Final answer: 1214 kg + 2453 kg = 3667 kg.

Question: (6) 2396 kg + 5725 kg = _____

Answer: Step 1: Add the ones: 6 + 5 = 11. Write 1 and carry 1.

Step 2: Add the tens: 9 + 2 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 3 + 7 = 10, plus the carried 1 makes 11. Write 1 and carry 1.

Step 4: Add the thousands: 2 + 5 = 7, plus the carried 1 makes 8.

Final answer: 2396 kg + 5725 kg = 8121 kg.

EXERCISE 37 — (B) Add the following masses

Question: (1) 350 g + 200 g = _____

Answer: Step 1: Add the ones: 0 + 0 = 0.

Step 2: Add the tens: 5 + 0 = 5.

Step 3: Add the hundreds: 3 + 2 = 5.

Final answer: 350 g + 200 g = 550 g.

Question: (2) 758 kg + 265 kg = _____

Answer: Step 1: Add the ones: 8 + 5 = 13. Write 3 and carry 1.

Step 2: Add the tens: 5 + 6 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 7 + 2 = 9, plus the carried 1 makes 10.

Final answer: 758 kg + 265 kg = 1023 kg.

Question: (3) 375 g + 900 g = _____

Answer: Step 1: Add the ones: 5 + 0 = 5.

Step 2: Add the tens: 7 + 0 = 7.

Step 3: Add the hundreds: 3 + 9 = 12.

Final answer: 375 g + 900 g = 1275 g.

Question: (4) 485 g + 7312 g = _____

Answer: Step 1: Add the ones: 5 + 2 = 7.

Step 2: Add the tens: 8 + 1 = 9.

Step 3: Add the hundreds: 4 + 3 = 7.

Step 4: Add the thousands: 0 + 7 = 7.

Final answer: 485 g + 7312 g = 7797 g.

EXERCISE 38 — Solve real life problems involving addition of mass

Question: (1) Ahmed bought 250 g of red chillies and 350 g of turmeric.

How many grams of spices did he buy in all?

Answer: Step 1: Total spices = 250 + 350.

Step 2: Add the ones: 0 + 0 = 0.

Step 3: Add the tens: 5 + 5 = 10. Write 0 and carry 1.

Step 4: Add the hundreds: 2 + 3 = 5, plus the carried 1 makes 6.

Final answer: He bought 600 g of spices in all.

Question: (2) A shopkeeper sold 10 kg of flour in the morning and 10 kg in the evening. How much flour did he sell in all?

Answer: Step 1: Total flour = 10 + 10.

Step 2: Add the ones: 0 + 0 = 0. Add the tens: 1 + 1 = 2.

Final answer: He sold 20 kg of flour in all.

Question: (3) Maria bought 2 kg of potatoes, 7 kg of onions and 6 kg of tomatoes. How many kilograms of vegetables did she buy in all?

Answer: Step 1: Total vegetables = 2 + 7 + 6.

Step 2: First add 2 + 7 = 9.

Step 3: Then add 9 + 6 = 15.

Final answer: She bought 15 kg of vegetables in all.

Question: (4) A farmer picked 28 kg of cotton on Monday and 31 kg on Tuesday. How much cotton did he pick in all?

Answer: Step 1: Total cotton = 28 + 31.

Step 2: Add the ones: 8 + 1 = 9.

Step 3: Add the tens: 2 + 3 = 5.

Final answer: He picked 59 kg of cotton in all.

EXERCISE 39 — (A) Subtract the following masses

Question: (1) 32 kg - 31 kg = _____

Answer: Step 1: Subtract the ones: 2 - 1 = 1.

Step 2: Subtract the tens: 3 - 3 = 0.

Final answer: 32 kg - 31 kg = 1 kg.

Question: (2) 496 g - 382 g = _____

Answer: Step 1: Subtract the ones: 6 - 2 = 4.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 4 - 3 = 1.

Final answer: 496 g - 382 g = 114 g.

Question: (3) 812 g - 653 g = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 3, so borrow 1 ten: 12 - 3 = 9.

Step 2: Subtract the tens: 1 ten became 0 tens; 0 cannot take away 5, so borrow 1 hundred: 10 - 5 = 5.

Step 3: Subtract the hundreds: 8 hundreds became 7 hundreds; 7 - 6 = 1.

Final answer: 812 g - 653 g = 159 g.

Question: (4) 724 kg - 345 kg = _____

Answer: Step 1: Subtract the ones: 4 cannot take away 5, so borrow 1 ten: 14 - 5 = 9.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 cannot take away 4, so borrow 1 hundred: 11 - 4 = 7.

Step 3: Subtract the hundreds: 7 hundreds became 6 hundreds; 6 - 3 = 3.

Final answer: 724 kg - 345 kg = 379 kg.

Question: (5) 823 kg - 712 kg = _____

Answer: Step 1: Subtract the ones: 3 - 2 = 1.

Step 2: Subtract the tens: 2 - 1 = 1.

Step 3: Subtract the hundreds: 8 - 7 = 1.

Final answer: 823 kg - 712 kg = 111 kg.

Question: (6) 986 g - 397 g = _____

Answer: Step 1: Subtract the ones: 6 cannot take away 7, so borrow 1 ten: 16 - 7 = 9.

Step 2: Subtract the tens: 8 tens became 7 tens; 7 cannot take away 9, so borrow 1 hundred: 17 - 9 = 8.

Step 3: Subtract the hundreds: 9 hundreds became 8 hundreds; 8 - 3 = 5.

Final answer: 986 g - 397 g = 589 g.

EXERCISE 39 — (B) Subtract the following masses

Question: (1) 600 g - 348 g = _____

Answer: Step 1: Subtract the ones: 0 cannot take away 8. The tens digit is also 0, so borrow 1 hundred: hundreds 6 becomes 5, tens become 10; the tens give 1 ten to the ones: tens become 9, ones become 10. Now 10 - 8 = 2.

Step 2: Subtract the tens: 9 - 4 = 5.

Step 3: Subtract the hundreds: 5 - 3 = 2.

Final answer: 600 g - 348 g = 252 g.

Question: (2) 428 kg - 175 kg = _____

Answer: Step 1: Subtract the ones: 8 - 5 = 3.

Step 2: Subtract the tens: 2 cannot take away 7, so borrow 1 hundred: 12 - 7 = 5.

Step 3: Subtract the hundreds: 4 hundreds became 3 hundreds; 3 - 1 = 2.

Final answer: 428 kg - 175 kg = 253 kg.

Question: (3) 345 g - 125 g = _____

Answer: Step 1: Subtract the ones: 5 - 5 = 0.

Step 2: Subtract the tens: 4 - 2 = 2.

Step 3: Subtract the hundreds: 3 - 1 = 2.

Final answer: 345 g - 125 g = 220 g.

Question: (4) 20 kg - 12 kg = _____

Answer: Step 1: Subtract the ones: 0 cannot take away 2, so borrow 1 ten: 10 - 2 = 8.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 - 1 = 0.

Final answer: 20 kg - 12 kg = 8 kg.

Question: (5) 370 g - 150 g = _____

Answer: Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 7 - 5 = 2.

Step 3: Subtract the hundreds: 3 - 1 = 2.

Final answer: 370 g - 150 g = 220 g.

Question: (6) 200 kg - 120 kg = _____

Answer: Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 0 cannot take away 2, so borrow 1 hundred: 10 -

2 = 8.

Step 3: Subtract the hundreds: 2 hundreds became 1 hundred; 1 - 1 = 0.

Final answer: 200 kg - 120 kg = 80 kg.

EXERCISE 40 — Solve real life problems involving subtraction of mass

Question: (1) A sack has 68 kg of rice. 53 kg is taken out. How much rice is left in the sack?

Answer: Step 1: Rice left = 68 - 53.

Step 2: Subtract the ones: 8 - 3 = 5.

Step 3: Subtract the tens: 6 - 5 = 1.

Final answer: 15 kg of rice is left in the sack.

Question: (2) A shopkeeper had 250 kg of sugar. He sold 150 kg. How much sugar is left?

Answer: Step 1: Sugar left = 250 - 150.

Step 2: Subtract the ones: 0 - 0 = 0.

Step 3: Subtract the tens: 5 - 5 = 0.

Step 4: Subtract the hundreds: 2 - 1 = 1.

Final answer: 100 kg of sugar is left.

Question: (3) A truck can carry 5000 kg of load. 3550 kg of load is already put in it. How much more load can be put in the truck?

Answer: Step 1: More load = 5000 - 3550.

Step 2: Subtract the ones: 0 - 0 = 0.

Step 3: Subtract the tens: 0 cannot take away 5, so borrow from the

hundreds/thousands: 10 - 5 = 5.

Step 4: Subtract the hundreds: after borrowing, 9 - 5 = 4.

Step 5: Subtract the thousands: 4 - 3 = 1.

Final answer: 1450 kg more load can be put in the truck.

Question: (4) Ikram weighs 30 kg and his little brother weighs 8 kg. How much heavier is Ikram than his brother?

Answer: Step 1: Difference = 30 - 8.

Step 2: Subtract the ones: 0 cannot take away 8, so borrow 1 ten: 10 - 8 = 2.

Step 3: Subtract the tens: 3 tens became 2 tens; 2 - 0 = 2.

Final answer: Ikram is 22 kg heavier than his brother.

ACTIVITY — Encircle the correct unit, ml or l, for the capacity of each of the following

Question: (1) A teacup: ml or l?

Answer: Step 1: A teacup holds only a small quantity of tea.

Step 2: Small quantities of liquid are measured in millilitres.

Final answer: The capacity of a teacup is measured in millilitres (ml).

Question: (2) A jug: ml or l?

Answer: Step 1: A jug holds a large quantity of water, about a litre or more.

Step 2: Large quantities of liquid are measured in litres.

Final answer: The capacity of a jug is measured in litres (l).

Question: (3) A shampoo bottle: ml or l?

Answer: Step 1: A shampoo bottle holds a small quantity of shampoo.

Final answer: The capacity of a shampoo bottle is measured in millilitres (ml).

Question: (4) A spoonful of syrup: ml or l?

Answer: Step 1: A spoon holds only a few drops of syrup.

Final answer: A spoonful of syrup is measured in millilitres (ml).

Question: (5) A tub (bucket): ml or l?

Answer: Step 1: A bucket holds many litres of water.

Final answer: The capacity of a tub is measured in litres (l).

Question: (6) A lotion bottle: ml or l?

Answer: Step 1: A lotion bottle holds a small quantity of lotion.

Final answer: The capacity of a lotion bottle is measured in millilitres (ml).

Question: (7) A drink bottle: ml or l?

Answer: Step 1: A big drink bottle holds about a litre or more.

Final answer: The capacity of a drink bottle is measured in litres (l).

Question: (8) An oil drum: ml or l?

Answer: Step 1: An oil drum holds a very large quantity of oil.

Final answer: The capacity of an oil drum is measured in litres (l).

ACTIVITY — Read and write the volume of liquid in each container

Question: (1) A container whose liquid level reaches the 20 mark on the

scale in litres.

Answer: Step 1: See where the top of the liquid reaches on the scale.

Step 2: The level stops at 20 and the scale is in litres.

Final answer: The volume is 20 l.

Question: (2) An ink pot whose level reaches the 10 mark in millilitres.

Answer: Step 1: See where the top of the ink reaches on the scale.

Step 2: The level stops at 10 and the scale is in millilitres.

Final answer: The volume is 10 ml.

Question: (3) A water bottle whose level reaches the 4 mark in litres.

Answer: Step 1: See where the top of the water reaches on the scale.

Step 2: The level stops at 4 and the scale is in litres.

Final answer: The volume is 4 l.

Question: (4) A bucket whose level reaches the 5 mark in litres.

Answer: Step 1: See where the top of the water reaches on the scale.

Step 2: The level stops at 5 and the scale is in litres.

Final answer: The volume is 5 l.

Question: (5) A glass of juice whose level reaches the 200 mark in millilitres.

Answer: Step 1: See where the top of the juice reaches on the scale.

Step 2: The level stops at 200 and the scale is in millilitres.

Final answer: The volume is 200 ml.

Question: (6) A jar whose level reaches the 100 mark in millilitres.

Answer: Step 1: See where the top of the liquid reaches on the scale.

Step 2: The level stops at 100 and the scale is in millilitres.

Final answer: The volume is 100 ml.

Question: (7) A bottle whose level reaches the 3 mark in litres.

Answer: Step 1: See where the top of the liquid reaches on the scale.

Step 2: The level stops at 3 and the scale is in litres.

Final answer: The volume is 3 l.

Question: (8) A syringe whose level reaches the 5 mark in millilitres.

Answer: Step 1: See where the top of the liquid reaches on the scale of the syringe.

Step 2: The level stops at 5 and the scale is in millilitres.

Final answer: The volume is 5 ml.

Question: (9) A small vial whose level reaches the 5 mark in millilitres.

Answer: Step 1: See where the top of the liquid reaches on the scale.

Step 2: The level stops at 5 and the scale is in millilitres.

Final answer: The volume is 5 ml.

EXERCISE 41 — (A) Encircle the correct capacity for each of the following

Question: (1) A juice carton: 90 ml, 900 ml or 2900 ml?

Answer: Step 1: A juice carton holds nearly one litre of juice.

Step 2: 90 ml is too little and 2900 ml is too much for a carton.

Final answer: The correct capacity is 900 ml.

Question: (2) A bleach bottle: 30 ml, 300 ml or 3 l?

Answer: Step 1: A bleach bottle is a big bottle holding a large quantity.

Step 2: It holds about 3 litres.

Final answer: The correct capacity is 3 l.

Question: (3) A shampoo bottle: 500 ml, 50 ml or 5 l?

Answer: Step 1: A shampoo bottle holds a medium quantity of shampoo.

Step 2: 50 ml is too little and 5 l is too much.

Final answer: The correct capacity is 500 ml.

EXERCISE 41 — (B) Choose the correct unit, l or ml, for each quantity

Question: (1) Milk in a jug: 9 _____.

Answer: Step 1: A jug holds a large quantity of milk.

Final answer: Milk in a jug is 9 l.

Question: (2) Syrup in a spoon: 5 _____.

Answer: Step 1: A spoon holds only a tiny quantity of syrup.

Final answer: Syrup in a spoon is 5 ml.

Question: (3) Water in a water tank: 100 _____.

Answer: Step 1: A water tank holds a very large quantity of water.

Final answer: Water in a water tank is 100 l.

Question: (4) Juice in a juice pack: 250 _____.

Answer: Step 1: A small juice pack holds a small quantity of juice.

Final answer: Juice in a juice pack is 250 ml.

Question: (5) Medicine in a medicine bottle: 200 _____.

Answer: Step 1: A medicine bottle holds a small quantity of medicine.

Final answer: Medicine in a medicine bottle is 200 ml.

EXERCISE 42 — (A) Add the following capacities

Question: (1) 635 ml + 223 ml = _____

Answer: Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 3 + 2 = 5.

Step 3: Add the hundreds: 6 + 2 = 8.

Final answer: 635 ml + 223 ml = 858 ml.

Question: (2) 562 ml + 458 ml = _____

Answer: Step 1: Add the ones: 2 + 8 = 10. Write 0 and carry 1.

Step 2: Add the tens: 6 + 5 = 11, plus the carried 1 makes 12. Write 2 and carry 1.

Step 3: Add the hundreds: 5 + 4 = 9, plus the carried 1 makes 10.

Final answer: 562 ml + 458 ml = 1020 ml.

Question: (3) 898 l + 101 l = _____

Answer: Step 1: Add the ones: 8 + 1 = 9.

Step 2: Add the tens: 9 + 0 = 9.

Step 3: Add the hundreds: 8 + 1 = 9.

Final answer: 898 l + 101 l = 999 l.

Question: (4) 729 l + 271 l = _____

Answer: Step 1: Add the ones: 9 + 1 = 10. Write 0 and carry 1.

Step 2: Add the tens: 2 + 7 = 9, plus the carried 1 makes 10. Write 0 and

carry 1.

Step 3: Add the hundreds: 7 + 2 = 9, plus the carried 1 makes 10.

Final answer: 729 l + 271 l = 1000 l.

Question: (5) 999 ml + 666 ml = _____

Answer: Step 1: Add the ones: 9 + 6 = 15. Write 5 and carry 1.

Step 2: Add the tens: 9 + 6 = 15, plus the carried 1 makes 16. Write 6 and carry 1.

Step 3: Add the hundreds: 9 + 6 = 15, plus the carried 1 makes 16.

Final answer: 999 ml + 666 ml = 1665 ml.

Question: (6) 3 l 218 ml + 2 l 551 ml = _____

Answer: Step 1: Add the millilitres: 218 + 551 = 769 ml.

Step 2: Add the litres: 3 + 2 = 5 l.

Final answer: 3 l 218 ml + 2 l 551 ml = 5 l 769 ml.

EXERCISE 42 — (B) Add the following capacities

Question: (1) 320 ml + 175 ml = _____

Answer: Step 1: Add the ones: 0 + 5 = 5.

Step 2: Add the tens: 2 + 7 = 9.

Step 3: Add the hundreds: 3 + 1 = 4.

Final answer: 320 ml + 175 ml = 495 ml.

Question: (2) 608 l + 747 l = _____

Answer: Step 1: Add the ones: 8 + 7 = 15. Write 5 and carry 1.

Step 2: Add the tens: 0 + 4 = 4, plus the carried 1 makes 5.

Step 3: Add the hundreds: 6 + 7 = 13.

Final answer: 608 l + 747 l = 1355 l.

Question: (3) 340 ml + 5 l 650 ml = _____

Answer: Step 1: Add the millilitres: 340 + 650 = 990 ml.

Step 2: There are 0 litres in the first quantity, so the litres stay 5.

Final answer: 340 ml + 5 l 650 ml = 5 l 990 ml.

Question: (4) 36 l + 23 l 825 ml = _____

Answer: Step 1: Add the litres: 36 + 23 = 59 l.

Step 2: The millilitres are only in the second quantity, so they stay 825 ml.

Final answer: 36 l + 23 l 825 ml = 59 l 825 ml.

EXERCISE 43 — Solve real life problems involving addition of capacity

Question: (1) A bottle has 350 ml of oil and another bottle has 200 ml of oil. How much oil is there in both bottles?

Answer: Step 1: Total oil = 350 + 200.

Step 2: Add the ones: 0 + 0 = 0.

Step 3: Add the tens: 5 + 0 = 5.

Step 4: Add the hundreds: 3 + 2 = 5.

Final answer: There is 550 ml of oil in both bottles.

Question: (2) A tank has 15 l of water. 18 l more water is poured into it. How much water is in the tank now?

Answer: Step 1: Total water = 15 + 18.

Step 2: Add the ones: 5 + 8 = 13. Write 3 and carry 1.

Step 3: Add the tens: 1 + 1 = 2, plus the carried 1 makes 3.

Final answer: There is 33 l of water in the tank now.

Question: (3) Three drums have 1320 l, 1215 l and 830 l of petrol. How much petrol is there in all the drums?

Answer: Step 1: Total petrol = 1320 + 1215 + 830.

Step 2: First add 1320 + 1215 = 2535.

Step 3: Then add 2535 + 830 = 3365.

Final answer: There is 3365 l of petrol in all the drums.

Question: (4) A tanker carried 550 l of water on Monday and 375 l on Tuesday. How much water did it carry in the two days?

Answer: Step 1: Total water = 550 + 375.

Step 2: Add the ones: 0 + 5 = 5.

Step 3: Add the tens: 5 + 7 = 12. Write 2 and carry 1.

Step 4: Add the hundreds: 5 + 3 = 8, plus the carried 1 makes 9.

Final answer: It carried 925 l of water in the two days.

EXERCISE 44 — (A) Subtract the following capacities

Question: (1) 375 ml - 250 ml = _____

Answer: Step 1: Subtract the ones: 5 - 0 = 5.

Step 2: Subtract the tens: 7 - 5 = 2.

Step 3: Subtract the hundreds: 3 - 2 = 1.

Final answer: 375 ml - 250 ml = 125 ml.

Question: (2) 723 l - 259 l = _____

Answer: Step 1: Subtract the ones: 3 cannot take away 9, so borrow 1 ten: 13 - 9 = 4.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 cannot take away 5, so borrow 1 hundred: 11 - 5 = 6.

Step 3: Subtract the hundreds: 7 hundreds became 6 hundreds; 6 - 2 = 4.

Final answer: 723 l - 259 l = 464 l.

Question: (3) 5326 l - 3152 l = _____

Answer: Step 1: Subtract the ones: 6 - 2 = 4.

Step 2: Subtract the tens: 2 - 5 cannot be done, so borrow 1 hundred: 12 - 5 = 7.

Step 3: Subtract the hundreds: 3 hundreds became 2 hundreds; 2 - 1 = 1.

Step 4: Subtract the thousands: 5 - 3 = 2.

Final answer: 5326 l - 3152 l = 2174 l.

Question: (4) 553 ml - 312 ml = _____

Answer: Step 1: Subtract the ones: 3 - 2 = 1.

Step 2: Subtract the tens: 5 - 1 = 4.

Step 3: Subtract the hundreds: 5 - 3 = 2.

Final answer: 553 ml - 312 ml = 241 ml.

Question: (5) 944 ml - 856 ml = _____

Answer: Step 1: Subtract the ones: 4 cannot take away 6, so borrow 1 ten: 14 - 6 = 8.

Step 2: Subtract the tens: 4 tens became 3 tens; 3 cannot take away 5, so

borrow 1 hundred: 13 - 5 = 8.

Step 3: Subtract the hundreds: 9 hundreds became 8 hundreds; 8 - 8 = 0.

Final answer: 944 ml - 856 ml = 88 ml.

Question: (6) 700 ml - 685 ml = _____

Answer: Step 1: Subtract the ones: 0 cannot take away 5. The tens digit is also 0, so borrow 1 hundred: hundreds 7 becomes 6, tens become 10; the tens give 1 ten to the ones: tens become 9, ones become 10. Now 10 - 5 = 5.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 6 - 6 = 0.

Final answer: 700 ml - 685 ml = 15 ml.

EXERCISE 44 — (B) Subtract the following capacities

Question: (1) 598 ml - 483 ml = _____

Answer: Step 1: Subtract the ones: 8 - 3 = 5.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 5 - 4 = 1.

Final answer: 598 ml - 483 ml = 115 ml.

Question: (2) 1000 l - 560 l = _____

Answer: Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 0 cannot take away 6, so borrow through the zeros from the thousands: tens become 10 after borrowing; 10 - 6 = 4.

Step 3: Subtract the hundreds: after borrowing, 9 - 5 = 4.

Step 4: Subtract the thousands: 1 became 0.

Final answer: 1000 l - 560 l = 440 l.

Question: (3) 630 ml - 450 ml = _____

Answer: Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 3 cannot take away 5, so borrow 1 hundred: 13 - 5 = 8.

Step 3: Subtract the hundreds: 6 hundreds became 5 hundreds; 5 - 4 = 1.

Final answer: 630 ml - 450 ml = 180 ml.

Question: (4) 490 l - 380 l = _____

Answer: Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 4 - 3 = 1.

Final answer: 490 l - 380 l = 110 l.

EXERCISE 45 — Solve real life problems involving subtraction of capacity

Question: (1) A jug has 550 ml of water. 220 ml is poured out. How much water is left in the jug?

Answer: Step 1: Water left = 550 - 220.

Step 2: Subtract the ones: 0 - 0 = 0.

Step 3: Subtract the tens: 5 - 2 = 3.

Step 4: Subtract the hundreds: 5 - 2 = 3.

Final answer: 330 ml of water is left in the jug.

Question: (2) A drum has 240 l of oil. 198 l is sold. How much oil is left in

the drum?

Answer: Step 1: Oil left = 240 - 198.

Step 2: Subtract the ones: 0 cannot take away 8, so borrow 1 ten: 10 - 8 = 2.

Step 3: Subtract the tens: 4 tens became 3 tens; 3 cannot take away 9, so borrow 1 hundred: 13 - 9 = 4.

Step 4: Subtract the hundreds: 2 hundreds became 1 hundred; 1 - 1 = 0.

Final answer: 42 l of oil is left in the drum.

Question: (3) A tank has 1200 l of water. 1185 l is used. How much water is left in the tank?

Answer: Step 1: Water left = 1200 - 1185.

Step 2: Subtract the ones: 0 cannot take away 5; borrowing through the zeros, ones become 10: 10 - 5 = 5.

Step 3: Subtract the tens: after borrowing, 9 - 8 = 1.

Step 4: Subtract the hundreds: after borrowing, 1 - 1 = 0, and the thousands 1 - 1 = 0.

Final answer: 15 l of water is left in the tank.

Question: (4) A water tanker has 1400 l of water. It supplies 750 l to a colony. How much water is left in the tanker?

Answer: Step 1: Water left = 1400 - 750.

Step 2: Subtract the ones: 0 - 0 = 0.

Step 3: Subtract the tens: 0 cannot take away 5, so borrow 1 hundred: hundreds 4 becomes 3 and tens become 10. Now 10 - 5 = 5.

Step 4: Subtract the hundreds: 3 cannot take away 7, so borrow 1

thousand: thousands 1 becomes 0 and hundreds become 13. Now 13 - 7 = 6.

Final answer: 650 l of water is left in the tanker.

Unit 5 TIME

ACTIVITY — Fill in the blanks

Question: (1) 1 hour = _____ minutes.

Answer: Step 1: The minute hand takes 60 small steps to go round the clock once, and that is one hour.

Final answer: 1 hour = 60 minutes.

Question: (2) Quarter of an hour = _____ minutes.

Answer: Step 1: One hour has 60 minutes.

Step 2: One quarter of 60 is 60 ÷ 4 = 15.

Final answer: Quarter of an hour = 15 minutes.

Question: (3) Half an hour = _____ minutes.

Answer: Step 1: One hour has 60 minutes.

Step 2: Half of 60 is 60 ÷ 2 = 30.

Final answer: Half an hour = 30 minutes.

Question: (4) 1 day = _____ hours.

Answer: Step 1: A full day and night has 24 hours: 12 hours of the day and 12 hours of the night together make 24.

Final answer: 1 day = 24 hours.

Question: (5) 2 days = _____ hours.

Answer: Step 1: 1 day = 24 hours.

Step 2: 2 days = 24 + 24 = 48 hours.

Final answer: 2 days = 48 hours.

Question: (6) 3 weeks = _____ days.

Answer: Step 1: 1 week = 7 days.

Step 2: 3 weeks = 7 + 7 + 7 = 21 days.

Final answer: 3 weeks = 21 days.

Question: (7) Half month = _____ days.

Answer: Step 1: One month has about 30 days.

Step 2: Half of 30 is 30 ÷ 2 = 15.

Final answer: Half month = 15 days.

Question: (8) 2 years = _____ months.

Answer: Step 1: 1 year = 12 months.

Step 2: 2 years = 12 + 12 = 24 months.

Final answer: 2 years = 24 months.

EXERCISE 46 — 1. Read the time shown on each clock (a.m. or p.m. is written with each clock)

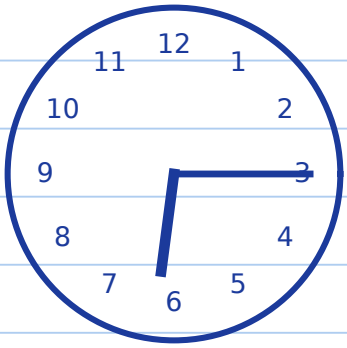
Question: (i) What time does the clock show? (a.m.)

Answer: Step 1: The short hour hand is just past 6.

Step 2: The long minute hand is on 3, which means 3 × 5 = 15 minutes.

Step 3: The label says a.m., so it is morning time.

Final answer: The time is 6:15 a.m.



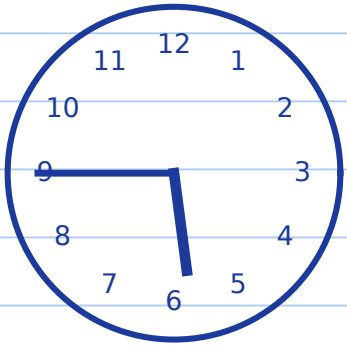
Question: (ii) What time does the clock show? (p.m.)

Answer: Step 1: The short hour hand is between 5 and 6, close to 6.

Step 2: The long minute hand is on 9, which means 9 × 5 = 45 minutes.

Step 3: The label says p.m., so it is evening time.

Final answer: The time is 5:45 p.m.



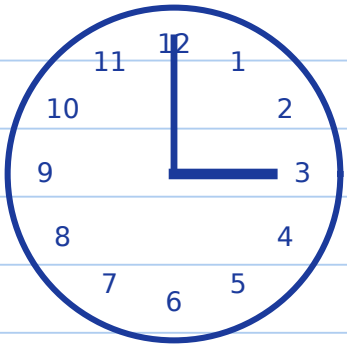
Question: (iii) What time does the clock show? (p.m.)

Answer: Step 1: The short hour hand is exactly on 3.

Step 2: The long minute hand is on 12, which means 0 minutes.

Step 3: The label says p.m.

Final answer: The time is 3:00 p.m.



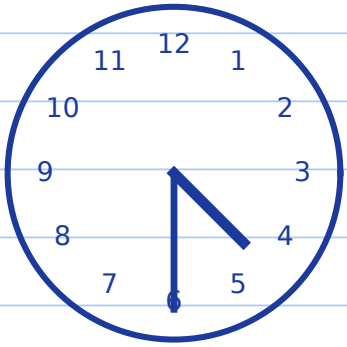
Question: (iv) What time does the clock show? (a.m.)

Answer: Step 1: The short hour hand is halfway between 4 and 5.

Step 2: The long minute hand is on 6, which means 6 × 5 = 30 minutes.

Step 3: The label says a.m.

Final answer: The time is 4:30 a.m.



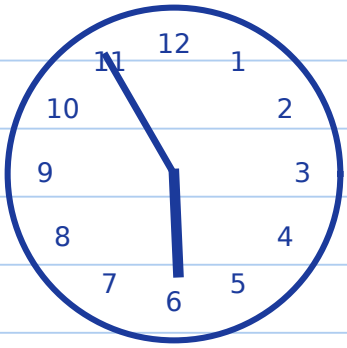
Question: (v) What time does the clock show? (p.m.)

Answer: Step 1: The short hour hand is just before 6.

Step 2: The long minute hand is on 11, which means 11 × 5 = 55 minutes.

Step 3: The label says p.m.

Final answer: The time is 5:55 p.m.



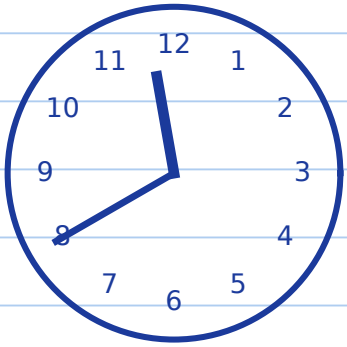
Question: (vi) What time does the clock show? (p.m.)

Answer: Step 1: The short hour hand is between 11 and 12.

Step 2: The long minute hand is on 8, which means 8 × 5 = 40 minutes.

Step 3: The label says p.m., so it is night time.

Final answer: The time is 11:40 p.m.



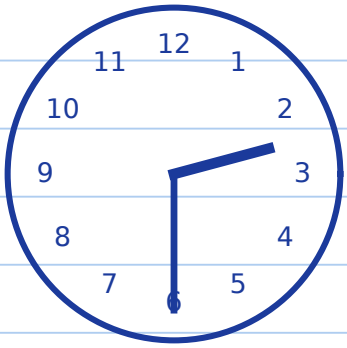
Question: (vii) What time does the clock show? (p.m.)

Answer: Step 1: The short hour hand is halfway between 2 and 3.

Step 2: The long minute hand is on 6, which means 30 minutes.

Step 3: The label says p.m.

Final answer: The time is 2:30 p.m.



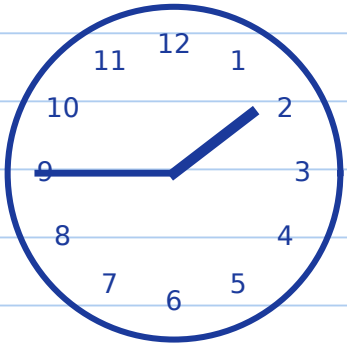
Question: (viii) What time does the clock show? (a.m.)

Answer: Step 1: The short hour hand is between 1 and 2, close to 2.

Step 2: The long minute hand is on 9, which means 45 minutes.

Step 3: The label says a.m.

Final answer: The time is 1:45 a.m.



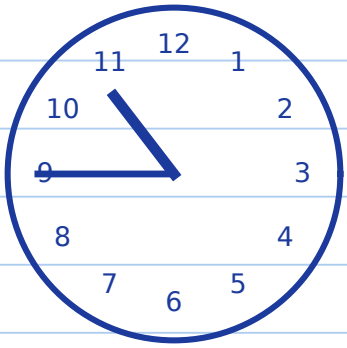
Question: (ix) What time does the clock show? (p.m.)

Answer: Step 1: The short hour hand is between 10 and 11, close to 11.

Step 2: The long minute hand is on 9, which means 45 minutes.

Step 3: The label says p.m., so it is night time.

Final answer: The time is 10:45 p.m.



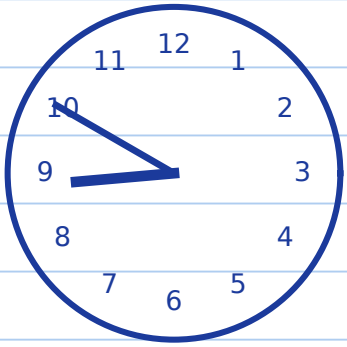
EXERCISE 46 — 2. Write the time of each daily activity shown in the pictures

Question: (i) Wake up time (clock shows the time with a.m.)

Answer: Step 1: The hour hand is between 8 and 9.

Step 2: The minute hand is on 10, which means 10 × 5 = 50 minutes.

Final answer: Wake up time is 8:50 a.m.

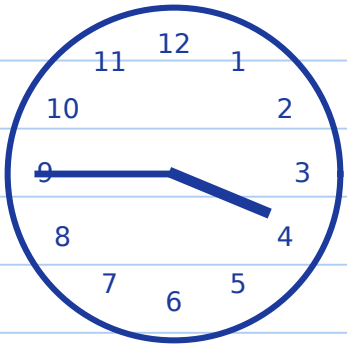


Question: (ii) Playing time (p.m.)

Answer: Step 1: The hour hand is between 3 and 4.

Step 2: The minute hand is on 9, which means 45 minutes.

Final answer: Playing time is 3:45 p.m.

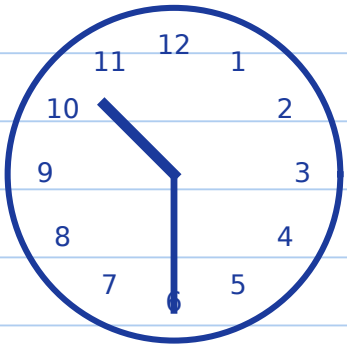


Question: (iii) Sleeping time (p.m.)

Answer: Step 1: The hour hand is between 10 and 11.

Step 2: The minute hand is on 6, which means 30 minutes.

Final answer: Sleeping time is 10:30 p.m.

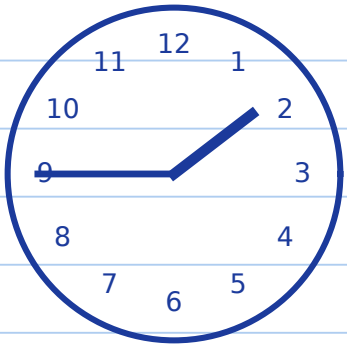


Question: (iv) Fajr prayer time (a.m.)

Answer: Step 1: The hour hand is between 1 and 2, close to 2.

Step 2: The minute hand is on 9, which means 45 minutes.

Final answer: Fajr prayer time is 1:45 a.m.

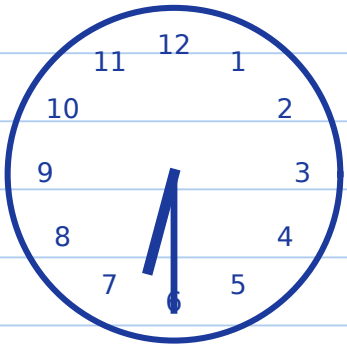


Question: (v) Time shown on the clock (a.m.)

Answer: Step 1: The hour hand is between 6 and 7.

Step 2: The minute hand is on 6, which means 30 minutes.

Final answer: The time is 6:30 a.m.

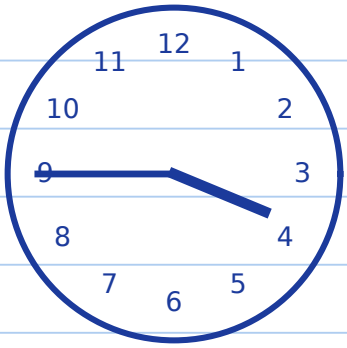


Question: (vi) Time shown on the clock (a.m.)

Answer: Step 1: The hour hand is between 3 and 4.

Step 2: The minute hand is on 9, which means 45 minutes.

Final answer: The time is 3:45 a.m.



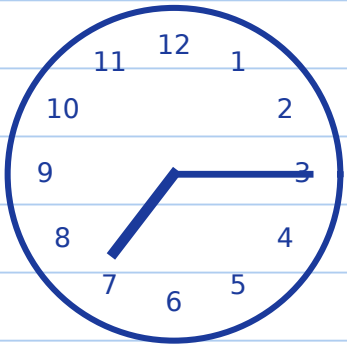
EXERCISE 46 — 3. Write the time in words (quarter past / half past / quarter to)

Question: (ii) The clock shows 7:15 in the evening. Write it in words.

Answer: Step 1: 15 minutes make a quarter of an hour.

Step 2: The minute hand on 3 means quarter past the hour.

Final answer: Quarter past 7 o'clock in the evening.

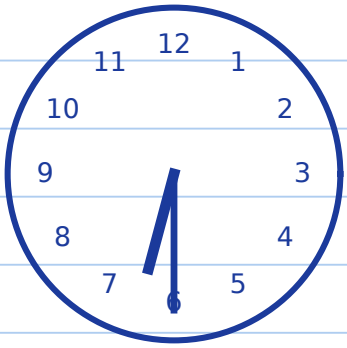


Question: (iii) The clock shows 6:30 in the morning. Write it in words.

Answer: Step 1: 30 minutes make half an hour.

Step 2: The minute hand on 6 means half past the hour.

Final answer: Half past 6 o'clock in the morning.

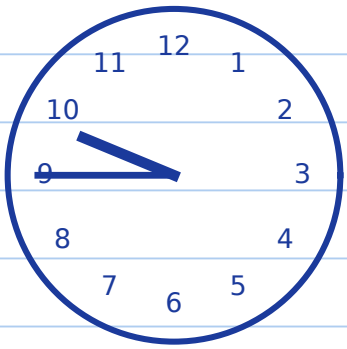


Question: (iv) The clock shows 9:45 in the night. Write it in words.

Answer: Step 1: 45 minutes means only 15 minutes are left to reach the next hour.

Step 2: 15 minutes before 10 is called quarter to 10.

Final answer: Quarter to 10 o'clock in the night.



ACTIVITY — Complete the sentences using the daily routine time table

Question: (1) I go to school at _____.

Answer: Step 1: Look at the "go to school" row of the table; the clock shows the hour hand on 8 and the minute hand on 12.

Step 2: School time is in the morning, so it is a.m.

Final answer: I go to school at 8:00 a.m.

Question: (2) I return from school at _____.

Answer: Step 1: Look at the "return from school" row; the clock shows 2 o'clock.

Step 2: It is after noon, so it is p.m.

Final answer: I return from school at 2:00 p.m.

Question: (3) I brush my teeth at _____.

Answer: Step 1: Look at the "brush teeth" row; the clock shows 7 o'clock.

Step 2: It is morning time, so it is a.m.

Final answer: I brush my teeth at 7:00 a.m.

Question: (4) I eat lunch at _____.

Answer: Step 1: Look at the "lunch" row; the clock shows half past 2.

Step 2: It is after noon, so it is p.m.

Final answer: I eat lunch at 2:30 p.m.

Question: (5) I eat breakfast at _____.

Answer: Step 1: Look at the "breakfast" row; the clock shows half past 7.

Step 2: It is morning time, so it is a.m.

Final answer: I eat breakfast at 7:30 a.m.

Question: (6) I eat dinner at _____.

Answer: Step 1: Look at the "dinner" row; the clock shows half past 9.

Step 2: It is night time, so it is p.m.

Final answer: I eat dinner at 9:30 p.m.

EXERCISE 47 — Write the time in the form "_____ minutes past / to _____"

Question: (1) 12:55

Answer: Step 1: The minutes are 55, which is more than half an hour.

Step 2: Minutes left to reach 1 o'clock = 60 - 55 = 5.

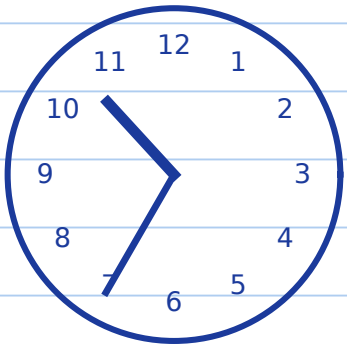
Final answer: 12:55 is 5 minutes to 1.

Question: (2) The clock shows 10:35.

Answer: Step 1: The hour hand is between 10 and 11, so the hour is 10.

Step 2: The minute hand shows 35 minutes, which is counted as minutes past 10.

Final answer: The time is 35 minutes past 10.

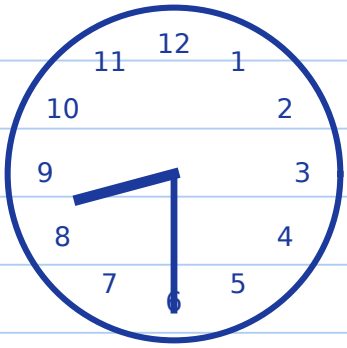


Question: (3) The clock shows 8:30.

Answer: Step 1: The hour hand is halfway between 8 and 9, so the hour is 8.

Step 2: The minute hand shows 30 minutes past 8.

Final answer: The time is 30 minutes past 8.

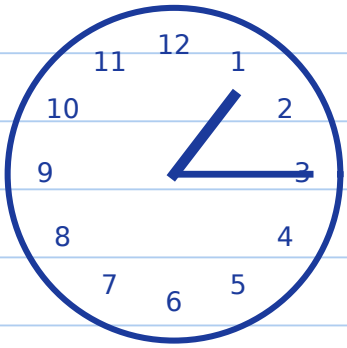


Question: (4) The clock shows 1:15.

Answer: Step 1: The hour hand is just past 1, so the hour is 1.

Step 2: The minute hand shows 15 minutes past 1.

Final answer: The time is 15 minutes past 1.



Question: (5) 9:25 a.m.

Answer: Step 1: The hour is 9 and the minutes are 25.

Step 2: a.m. means the time is in the morning.

Final answer: 9:25 a.m. is 25 minutes past 9 in the morning.

Question: (6) 11:30 a.m.

Answer: Step 1: The hour is 11 and the minutes are 30.

Step 2: a.m. means the time is in the morning.

Final answer: 11:30 a.m. is 30 minutes past 11 in the morning.

Question: (7) 3:40 a.m.

Answer: Step 1: The hour is 3 and the minutes are 40.

Step 2: a.m. means the time is in the morning.

Final answer: 3:40 a.m. is 40 minutes past 3 in the morning.

Question: (8) 5:15 a.m.

Answer: Step 1: The hour is 5 and the minutes are 15.

Step 2: a.m. means the time is in the morning.

Final answer: 5:15 a.m. is 15 minutes past 5 in the morning.

ACTIVITY — Use the September calendar given in the book and answer

Question: (1) Colour all the Sundays in blue. Which dates are Sundays?

Answer: Step 1: In the September calendar, 1 September is Monday, so Sundays fall on 7, 14, 21 and 28.

Step 2: Colour these four dates blue.

Final answer: The Sundays are 7, 14, 21 and 28 September.

Question: (2) Colour all the Wednesdays in yellow. Which dates are Wednesdays?

Answer: Step 1: The first Wednesday is 3 September.

Step 2: Keep adding 7: 3, 10, 17, 24. Colour these dates yellow.

Final answer: The Wednesdays are 3, 10, 17 and 24 September.

Question: (3) Colour all the Fridays in green. Which dates are Fridays?

Answer: Step 1: The first Friday is 5 September.

Step 2: Keep adding 7: 5, 12, 19, 26. Colour these dates green.

Final answer: The Fridays are 5, 12, 19 and 26 September.

Question: (4) 8 September is a _____.

Answer: Step 1: 7 September is Sunday.

Step 2: The next day, 8 September, is the day after Sunday.

Final answer: 8 September is a Monday.

Question: (5) 12 September is a _____.

Answer: Step 1: 12 September is in the Friday column of the calendar (5, 12, 19, 26).

Final answer: 12 September is a Friday.

Question: (6) 23 September is a _____.

Answer: Step 1: 21 September is Sunday and 22 September is Monday.

Step 2: The next day, 23 September, comes after Monday.

Final answer: 23 September is a Tuesday.

EXERCISE 48 — (A) Using the model calendar given in the book, write the day on the following dates

Question: (1) 23 March

Answer: Step 1: Find March in the model calendar.

Step 2: Look at the column of 23 March; it falls under Sunday.

Final answer: 23 March is a Sunday.

Question: (2) 11 September

Answer: Step 1: Find September in the model calendar.

Step 2: Look at the column of 11 September; it falls under Thursday.

Final answer: 11 September is a Thursday.

Question: (3) 31 March

Answer: Step 1: Find March in the model calendar.

Step 2: Look at the column of 31 March; it falls under Monday.

Final answer: 31 March is a Monday.

Question: (4) 6 September

Answer: Step 1: Find September in the model calendar.

Step 2: Look at the column of 6 September; it falls under Saturday.

Final answer: 6 September is a Saturday.

Question: (5) 27 April

Answer: Step 1: Find April in the model calendar.

Step 2: Look at the column of 27 April; it falls under Sunday.

Final answer: 27 April is a Sunday.

Question: (6) 9 November

Answer: Step 1: Find November in the model calendar.

Step 2: Look at the column of 9 November; it falls under Sunday.

Final answer: 9 November is a Sunday.

EXERCISE 48 — (B) Answer the following questions using the model calendar

Question: (1) On which day does the current calendar year begin (1 January)?

Answer: Step 1: Look at 1 January in the model calendar.

Step 2: It falls under Wednesday.

Final answer: The year begins on Wednesday, 1 January.

Question: (2) 14 August is celebrated as Independence Day. On which day does it fall this year?

Answer: Step 1: Find August in the model calendar.

Step 2: Look at the column of 14 August; it falls under Thursday.

Final answer: 14 August falls on a Thursday.

Question: (3) How many Fridays are there in the month of June?

Answer: Step 1: Find June in the model calendar.

Step 2: Read all the dates under Friday: 6, 13, 20 and 27.

Step 3: Count them: there are 4 Fridays.

Final answer: There are 4 Fridays in June.

Question: (4) On which day does your birthday fall this year?

Answer: Step 1: Find your birth month in the model calendar.

Step 2: Find your birth date in that month and look at the day written at the top of its column.

Final answer: The day at the top of your birth date's column is the day of your birthday this year. (Every student's answer will be according to his or her own birthday.)

Question: (5) Write the names of the months which have 31 days.

Answer: Step 1: Check the last date of every month in the model calendar.

Step 2: The months ending at 31 are January, March, May, July, August, October and December.

Final answer: The months with 31 days are January, March, May, July, August, October and December.

Question: (6) On which day does the current calendar year end (31 December)?

Answer: Step 1: Look at 31 December in the model calendar.

Step 2: It falls under Wednesday.

Final answer: The year ends on Wednesday, 31 December.

EXERCISE 49 — (A) Add the following hours

Question: (1) 3 hours + 4 hours = _____

Answer: Step 1: Add the numbers: 3 + 4 = 7.

Step 2: The unit stays hours.

Final answer: 3 hours + 4 hours = 7 hours.

Question: (2) 12 hours + 17 hours = _____

Answer: Step 1: Add the ones: 2 + 7 = 9.

Step 2: Add the tens: 1 + 1 = 2.

Final answer: 12 hours + 17 hours = 29 hours.

Question: (3) 25 hours + 16 hours = _____

Answer: Step 1: Add the ones: 5 + 6 = 11. Write 1 and carry 1.

Step 2: Add the tens: 2 + 1 = 3, plus the carried 1 makes 4.

Final answer: 25 hours + 16 hours = 41 hours.

Question: (4) 32 hours + 18 hours = _____

Answer: Step 1: Add the ones: 2 + 8 = 10. Write 0 and carry 1.

Step 2: Add the tens: 3 + 1 = 4, plus the carried 1 makes 5.

Final answer: 32 hours + 18 hours = 50 hours.

Question: (5) 36 hours + 25 hours = _____

Answer: Step 1: Add the ones: 6 + 5 = 11. Write 1 and carry 1.

Step 2: Add the tens: 3 + 2 = 5, plus the carried 1 makes 6.

Final answer: 36 hours + 25 hours = 61 hours.

Question: (6) 29 hours + 38 hours = _____

Answer: Step 1: Add the ones: 9 + 8 = 17. Write 7 and carry 1.

Step 2: Add the tens: 2 + 3 = 5, plus the carried 1 makes 6.

Final answer: 29 hours + 38 hours = 67 hours.

EXERCISE 49 — (B) Add the following hours

Question: (1) 32 hours + 22 hours = _____

Answer: Step 1: Add the ones: 2 + 2 = 4.

Step 2: Add the tens: 3 + 2 = 5.

Final answer: 32 hours + 22 hours = 54 hours.

Question: (2) 14 hours + 12 hours = _____

Answer: Step 1: Add the ones: 4 + 2 = 6.

Step 2: Add the tens: 1 + 1 = 2.

Final answer: 14 hours + 12 hours = 26 hours.

Question: (3) 108 hours + 117 hours = _____

Answer: Step 1: Add the ones: 8 + 7 = 15. Write 5 and carry 1.

Step 2: Add the tens: 0 + 1 = 1, plus the carried 1 makes 2.

Step 3: Add the hundreds: 1 + 1 = 2.

Final answer: 108 hours + 117 hours = 225 hours.

Question: (4) 125 hours + 135 hours = _____

Answer: Step 1: Add the ones: 5 + 5 = 10. Write 0 and carry 1.

Step 2: Add the tens: 2 + 3 = 5, plus the carried 1 makes 6.

Step 3: Add the hundreds: 1 + 1 = 2.

Final answer: 125 hours + 135 hours = 260 hours.

Question: (5) 385 hours + 323 hours = _____

Answer: Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 8 + 2 = 10. Write 0 and carry 1.

Step 3: Add the hundreds: 3 + 3 = 6, plus the carried 1 makes 7.

Final answer: 385 hours + 323 hours = 708 hours.

Question: (6) 480 hours + 342 hours = _____

Answer: Step 1: Add the ones: 0 + 2 = 2.

Step 2: Add the tens: 8 + 4 = 12. Write 2 and carry 1.

Step 3: Add the hundreds: 4 + 3 = 7, plus the carried 1 makes 8.

Final answer: 480 hours + 342 hours = 822 hours.

EXERCISE 50 — Solve real life problems involving addition of time (hours)

Question: (1) Ali studies 13 hours in a week and plays for 4 hours. How many hours does he spend on both activities?

Answer: Step 1: Total hours = 13 + 4.

Step 2: Add the ones: 3 + 4 = 7, and the tens stay 1.

Final answer: He spends 17 hours on both activities.

Question: (2) A worker works 53 hours in one week and 68 hours in the next week. How many hours does he work in the two weeks?

Answer: Step 1: Total hours = 53 + 68.

Step 2: Add the ones: 3 + 8 = 11. Write 1 and carry 1.

Step 3: Add the tens: 5 + 6 = 11, plus the carried 1 makes 12.

Final answer: He works 121 hours in the two weeks.

Question: (3) Fatima reads a book for 3 hours and writes for 4 hours. How many hours does she spend in all?

Answer: Step 1: Total hours = 3 + 4 = 7.

Final answer: She spends 7 hours in all.

Question: (4) A boy spends 2 hours at school work, 4 hours playing and 3 hours in other work. How many hours does he spend in all these activities?

Answer: Step 1: Total hours = 2 + 4 + 3.

Step 2: First add 2 + 4 = 6.

Step 3: Then add 6 + 3 = 9.

Final answer: He spends 9 hours in all these activities.

EXERCISE 51 — (A) Subtract the following hours

Question: (1) 5 hours - 4 hours = _____

Answer: Step 1: Subtract: 5 - 4 = 1.

Step 2: The unit stays hours.

Final answer: 5 hours - 4 hours = 1 hour.

Question: (2) 12 hours - 7 hours = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 7, so borrow 1 ten: 12 - 7 = 5.

Final answer: 12 hours - 7 hours = 5 hours.

Question: (3) 22 hours - 18 hours = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 8, so borrow 1 ten: 12 - 8 = 4.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 - 1 = 0.

Final answer: 22 hours - 18 hours = 4 hours.

Question: (4) 37 hours - 22 hours = _____

Answer: Step 1: Subtract the ones: 7 - 2 = 5.

Step 2: Subtract the tens: 3 - 2 = 1.

Final answer: 37 hours - 22 hours = 15 hours.

Question: (5) 33 hours - 14 hours = _____

Answer: Step 1: Subtract the ones: 3 cannot take away 4, so borrow 1 ten: 13 - 4 = 9.

Step 2: Subtract the tens: 3 tens became 2 tens; 2 - 1 = 1.

Final answer: 33 hours - 14 hours = 19 hours.

Question: (6) 49 hours - 39 hours = _____

Answer: Step 1: Subtract the ones: 9 - 9 = 0.

Step 2: Subtract the tens: 4 - 3 = 1.

Final answer: 49 hours - 39 hours = 10 hours.

EXERCISE 51 — (B) Subtract the following hours

Question: (1) 12 hours - 6 hours = _____

Answer: Step 1: Subtract the ones: 2 cannot take away 6, so borrow 1 ten: 12 - 6 = 6.

Final answer: 12 hours - 6 hours = 6 hours.

Question: (2) 48 hours - 6 hours = _____

Answer: Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Subtract the tens: 4 - 0 = 4.

Final answer: 48 hours - 6 hours = 42 hours.

Question: (3) 165 hours - 145 hours = _____

Answer: Step 1: Subtract the ones: 5 - 5 = 0.

Step 2: Subtract the tens: 6 - 4 = 2.

Step 3: Subtract the hundreds: 1 - 1 = 0.

Final answer: 165 hours - 145 hours = 20 hours.

Question: (4) 300 hours - 195 hours = _____

Answer: Step 1: Subtract the ones: 0 cannot take away 5. Borrow through the zeros: ones become 10; 10 - 5 = 5.

Step 2: Subtract the tens: after borrowing, 9 - 9 = 0.

Step 3: Subtract the hundreds: after borrowing, 2 - 1 = 1.

Final answer: 300 hours - 195 hours = 105 hours.

Question: (5) 723 hours - 486 hours = _____

Answer: Step 1: Subtract the ones: 3 cannot take away 6, so borrow 1 ten: 13 - 6 = 7.

Step 2: Subtract the tens: 2 tens became 1 ten; 1 cannot take away 8, so borrow 1 hundred: 11 - 8 = 3.

Step 3: Subtract the hundreds: 7 hundreds became 6 hundreds; 6 - 4 = 2.

Final answer: 723 hours - 486 hours = 237 hours.

Question: (6) 995 hours - 578 hours = _____

Answer: Step 1: Subtract the ones: 5 cannot take away 8, so borrow 1 ten: 15 - 8 = 7.

Step 2: Subtract the tens: 9 tens became 8 tens; 8 - 7 = 1.

Step 3: Subtract the hundreds: 9 - 5 = 4.

Final answer: 995 hours - 578 hours = 417 hours.

EXERCISE 52 — Solve real life problems involving subtraction of time (hours)

Question: (1) A train takes 8 hours to reach Lahore and a car takes 6 hours for the same journey. Which is faster and by how many hours?

Answer: Step 1: The car takes less time, so the car is faster.

Step 2: Difference = 8 - 6 = 2 hours.

Final answer: The car is faster; it takes 2 hours less than the train.

Question: (2) Ahmed studies for 22 hours in a week and his brother studies for 15 hours. How many more hours does Ahmed study?

Answer: Step 1: Difference = 22 - 15.

Step 2: Subtract the ones: 2 cannot take away 5, so borrow 1 ten: 12 - 5 = 7.

Step 3: Subtract the tens: 2 tens became 1 ten; 1 - 1 = 0.

Final answer: Ahmed studies 7 hours more.

Question: (3) Azeem reads for 9 hours and Aslam reads for 5 hours. How many more hours does Azeem read?

Answer: Step 1: Difference = 9 - 5 = 4.

Final answer: Azeem reads 4 hours more than Aslam.

Question: (4) Student A spends 14 hours on homework and student B spends 20 hours. How many more hours does B spend than A?

Answer: Step 1: Difference = 20 - 14.

Step 2: Subtract the ones: 0 cannot take away 4, so borrow 1 ten: 10 - 4 = 6.

Step 3: Subtract the tens: 2 tens became 1 ten; 1 - 1 = 0.

Final answer: B spends 6 hours more than A.

Question: (5) It is 5:00 p.m. Sara starts studying and studies for 2 hours and 30 minutes. For how long does she study, and till what time?

Answer: Step 1: The time she spends in studying is given as 2 hours and 30 minutes.

Step 2: Count forward from 5:00 p.m.: after 2 hours it is 7:00 p.m., and after 30 more minutes it is 7:30 p.m.

Final answer: Sara studies for 2 hours and 30 minutes, till 7:30 p.m.

Unit 6 GEOMETRY

ACTIVITY — Name each figure: line, line segment or ray

Question: Name the figures ST, BC, AB, LM, MN, PQ, XY and EF shown in the book.

Answer: Step 1: A line segment has two end points; a ray has one end point and an arrow on the other side.

Step 2: ST has two end points, so ST is a line segment.

Step 3: BC has an arrow on one side, so BC is a ray.

Step 4: AB has an arrow on one side, so AB is a ray.

Step 5: LM has two end points, so LM is a line segment.

Step 6: MN has an arrow on one side, so MN is a ray.

Step 7: The figure with end point P and arrow at Q is ray PQ.

Step 8: XY has two end points, so XY is a line segment.

Step 9: EF has two end points, so EF is a line segment.

Final answer: ST = line segment, BC = ray, AB = ray, LM = line segment, MN = ray, PQ = ray, XY = line segment, EF = line segment.

EXERCISE 53 — Line, line segment and ray

Question: (1) Put a tick (✓) on the line segment and a cross (✗) on the ray in the four figures (a) a curved arc, (b) a straight figure with two end points, (c) a wavy line, (d) a figure with an arrow.

Answer: Step 1: A line segment is straight and has two end points; figure (b) is like this, so it gets a tick (✓).

Step 2: A ray has one end point and an arrow; figure (d) is a ray, so in this question it gets a cross (✗) because we tick only the line segment.

Step 3: The curved arc (a) and the wavy line (c) are neither line segments nor rays.

Final answer: (a) neither, (b) line segment ✓, (c) neither, (d) ray ✗.

Question: (2)(a) How many end points does the line segment MN have? Name them.

Answer: Step 1: A line segment stops at both ends.

Step 2: Segment MN stops at M on one side and at N on the other side.

Final answer: Line segment MN has two end points, M and N.

Question: (2)(b) How many end points does the ray PQ have? Name it.

Answer: Step 1: A ray starts at one point and goes on with an arrow.

Step 2: Ray PQ starts at P, so P is its only end point.

Final answer: Ray PQ has one end point, P.

Question: (3) The line segment in the figure can be named in two ways. Write both names.

Answer: Step 1: A line segment can be read from either end.

Step 2: Reading from P to Q gives PQ, and reading from Q to P gives QP.

Final answer: The segment can be named PQ or QP.

Question: (4) In the figure, some rays meet at a point marked with a letter. Name that point.

Answer: Step 1: Look at the figure and find the point where the rays start and meet.

Step 2: That point is marked G.

Final answer: The point is G.

Question: (5) In the figure, four line segments have the same end point A. Name those segments.

Answer: Step 1: Find point A in the figure.

Step 2: The segments starting from A go to B, D, E and C.

Final answer: The four segments with common end point A are AB, AD, AE and AC.

Question: (6) In the figure with points P, Q and R, name the line segments and the rays.

Answer: Step 1: Join every pair of points: P with Q, Q with R and P with R; these are the segments.

Step 2: The figures with arrows on the same pairs are the rays.

Final answer: Line segments: PQ, QR and PR. Rays: PQ, QR and PR (each shown with an arrow).

ACTIVITY — Name the shapes (a) to (f) shown in the book

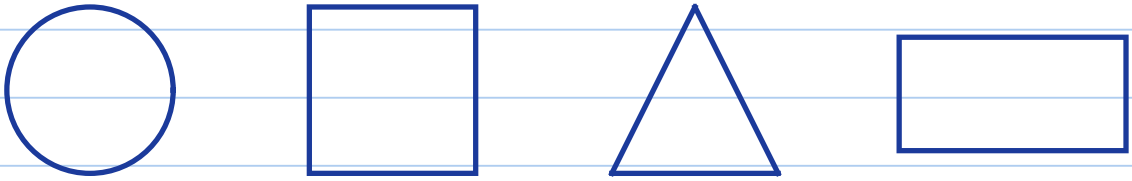
Question: Name each of the six shapes a, b, c, d, e and f.

Answer: Step 1: Count the sides of each shape and check if its sides and corners are equal.

Step 2: (a) has 4 equal sides, so it is a Square. (b) has 4 sides of different lengths, so it is a Quadrilateral. (c) has opposite sides equal, so it is a Rectangle.

Step 3: (d) and (e) each have 3 sides, so they are Triangles. (f) has 4 equal sides, so it is a Square.

Final answer: (a) Square, (b) Quadrilateral, (c) Rectangle, (d) Triangle, (e) Triangle, (f) Square.



Square Quadrilateral Rectangle Triangle

EXERCISE 54 — 1. Name the following shapes

Question: (a) A shape with 4 equal sides and 4 square corners.

Answer: Step 1: All four sides are equal.

Final answer: It is a Square.

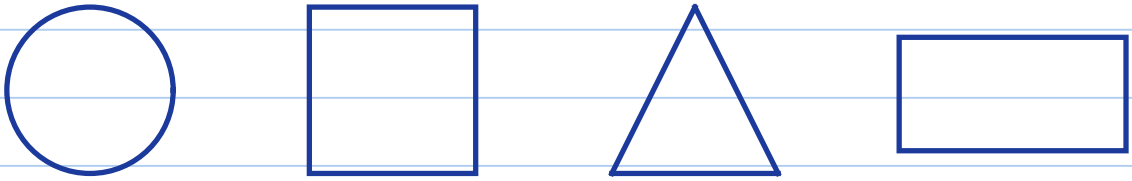


Square

Question: (b) A shape with 4 sides.

Answer: Step 1: Any closed shape with 4 sides is called a quadrilateral.

Final answer: It is a Quadrilateral.



Quadrilateral

Question: (c) A shape with opposite sides equal and 4 square corners.

Answer: Step 1: Opposite sides are equal but all sides are not equal.

Final answer: It is a Rectangle.



Rectangle

Question: (d) A shape with 3 sides.

Answer: Step 1: Count the sides: there are 3 sides.

Final answer: It is a Triangle.



Triangle

Question: (e) A shape with 3 sides.

Answer: Step 1: Count the sides: there are 3 sides.

Final answer: It is a Triangle.



Triangle

Question: (f) A shape with 4 equal sides.

Answer: Step 1: Count the sides: 4 sides, and all are equal.

Final answer: It is a Square.



Square

EXERCISE 54 — 2. Draw on the dot grid: two rectangles and a triangle (one rectangle is drawn as an example)

Question: How will you draw two rectangles and a triangle on the dot grid?

Answer: Step 1: For a rectangle, join 4 dots to make a shape whose opposite sides are equal; make two such rectangles of different sizes.

Step 2: For a triangle, join 3 dots with three straight lines.

Final answer: Draw two rectangles (opposite sides equal) and one triangle (3 sides) on the dot grid, as shown in the example.



Rectangle Rectangle Triangle

EXERCISE 54 — 3. Complete each sentence with Square, Rectangle or Triangle

Question: (i) A shape with 4 equal sides is a _____.

Answer: Step 1: A shape with 4 equal sides and square corners is a square.

Final answer: A shape with 4 equal sides is a Square.

Question: (ii) A shape with 3 sides is a _____.

Answer: Step 1: "Tri" means three; a 3-sided shape is a triangle.

Final answer: A shape with 3 sides is a Triangle.

Question: (iii) A shape with opposite sides equal is a _____.

Answer: Step 1: In a rectangle, the opposite sides are equal.

Final answer: A shape with opposite sides equal is a Rectangle.

Question: (iv) All the sides of a _____ are equal.

Answer: Step 1: In a square, all four sides are equal.

Final answer: All the sides of a Square are equal.

Question: (v) A _____ has 4 sides and its longer sides are opposite each other.

Answer: Step 1: A rectangle has 4 sides and its opposite sides are equal.

Final answer: A Rectangle has 4 sides with opposite sides equal.

EXERCISE 55 — 1. For each circle, name the centre, the diameter and the radius

Question: (i) The circle with centre O and the line EF passing through O.

Answer: Step 1: The point in the middle of the circle is its centre; here it is O.

Step 2: The line passing through the centre and touching the circle on both sides is the diameter; here it is EF.

Step 3: The line from the centre to the circle is the radius; EO and OF are radii.

Final answer: Centre = O, Diameter = EF, Radius = EO and OF.

Question: (ii) The circle with centre M and the line SR passing through M.

Answer: Step 1: The middle point of this circle is M.

Step 2: SR passes through M and touches the circle on both sides, so it is

the diameter.

Step 3: MS and MR join the centre to the circle, so they are radii.

Final answer: Centre = M, Diameter = SR, Radius = MS and MR.

Question: (iii) The circle with centre C and the line AB passing through C.

Answer: Step 1: The middle point of this circle is C.

Step 2: AB passes through C and touches the circle on both sides, so it is the diameter.

Step 3: CA and CB join the centre to the circle, so they are radii.

Final answer: Centre = C, Diameter = AB, Radius = CA and CB.

EXERCISE 55 — 2. In each of the three circles shown, name the diameter and the radius (the circles have no letters printed on them)

Question: (i) First circle: which line is the diameter and which is the radius?

Answer: Step 1: The straight line that passes through the centre of the circle and joins two points on the circle is the diameter.

Step 2: The straight line that joins the centre to any point on the circle is the radius.

Final answer: In this circle, the line through the centre is the diameter, and the line from the centre to the circle is the radius.

Question: (ii) Second circle: which line is the diameter and which is the radius?

Answer: Step 1: Look for the line passing through the centre; it is the longest line in the circle.

Step 2: The line from the centre to the edge is half of the diameter.

Final answer: The line through the centre is the diameter, and the line from the centre to the circle is the radius.

Question: (iii) Third circle: which line is the diameter and which is the radius?

Answer: Step 1: The diameter always passes through the centre and touches the circle at two points.

Step 2: The radius touches the circle at only one point and starts from the centre.

Final answer: The line through the centre is the diameter, and the line from the centre to the circle is the radius.

EXERCISE 56 — Find the perimeter of the following figures

Question: (1) A rectangle ZYXW with length 6 cm and width 3 cm.

Answer: Step 1: Perimeter means the total length all around the figure.

Step 2: Add all four sides: 6 + 3 + 6 + 3.

Step 3: 6 + 3 = 9 and 9 + 6 = 15 and 15 + 3 = 18.

Final answer: The perimeter of the rectangle is 18 cm.

Question: (2) A square with each side 3 cm.

Answer: Step 1: A square has 4 equal sides.

Step 2: Add all four sides: 3 + 3 + 3 + 3.

Step 3: 3 + 3 = 6, 6 + 3 = 9 and 9 + 3 = 12.

Final answer: The perimeter of the square is 12 cm.

Question: (3) A triangle DEF with sides DE = 6 cm, EF = 4 cm and FD = 5

cm.

Answer: Step 1: Add all three sides: 6 + 4 + 5.

Step 2: 6 + 4 = 10 and 10 + 5 = 15.

Final answer: The perimeter of the triangle is 15 cm.

Question: (4) A square table with each side 38 cm.

Answer: Step 1: A square table has 4 equal sides of 38 cm.

Step 2: Perimeter = 38 + 38 + 38 + 38, which is the same as 38 × 4.

Step 3: 38 × 4 = 152 (8 × 4 = 32, write 2 carry 3; 3 × 4 = 12, plus the carried 3 makes 15).

Final answer: The perimeter of the square table is 152 cm.

Question: (5) A window with length 75 cm and width 50 cm.

Answer: Step 1: A window is a rectangle, so its sides are 75, 50, 75 and 50 cm.

Step 2: Add them: 75 + 50 = 125, 125 + 75 = 200 and 200 + 50 = 250.

Final answer: The perimeter of the window is 250 cm.

Unit 7 DATA REPRESENTATION

ACTIVITY — This picture graph represents the number of different animals in a farm. (One cow picture represents 2 cows, one goat picture represents 5 goats, one horse picture represents 2 horses and one camel picture represents 5 camels.)

Read the picture graph and answer the following:

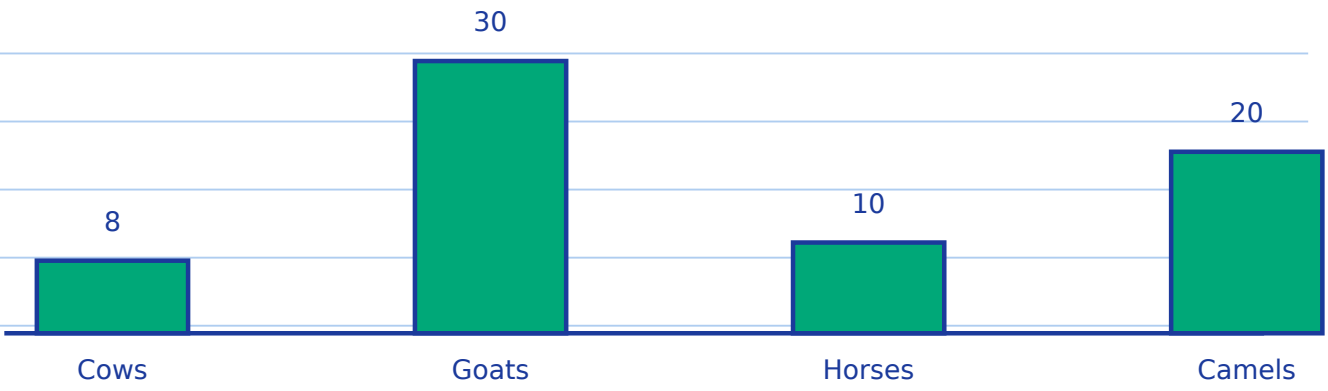
Question: (1) How many cows are there in the farm? (4 cow pictures)

Answer: Step 1: Count the cow pictures: there are 4 pictures.

Step 2: One cow picture represents 2 cows.

Step 3: Multiply: 4 × 2 = 8.

Final answer: There are 8 cows in the farm.



Question: (2) How many goats are there in the farm? (6 goat pictures)

Answer: Step 1: Count the goat pictures: there are 6 pictures.

Step 2: One goat picture represents 5 goats.

Step 3: Multiply: 6 × 5 = 30 (count in fives: 5, 10, 15, 20, 25, 30).

Final answer: There are 30 goats in the farm.

Question: (3) How many horses are there in the farm? (5 horse pictures)

Answer: Step 1: Count the horse pictures: there are 5 pictures.

Step 2: One horse picture represents 2 horses.

Step 3: Multiply: 5 × 2 = 10.

Final answer: There are 10 horses in the farm.

Question: (4) How many camels are there in the farm? (4 camel pictures)

Answer: Step 1: Count the camel pictures: there are 4 pictures.

Step 2: One camel picture represents 5 camels.

Step 3: Multiply: 4 × 5 = 20 (count in fives: 5, 10, 15, 20).

Final answer: There are 20 camels in the farm.

Question: (5) What is the total number of animals in the farm?

Answer: Step 1: Add all the animals: cows 8 + goats 30 + horses 10 + camels 20.

Step 2: 8 + 30 = 38.

Step 3: 38 + 10 = 48.

Step 4: 48 + 20 = 68.

Final answer: The total number of animals in the farm is 68.

EXERCISE 57 — (A) The picture graph shows the number of eggs sold in 4 days (one picture represents 5 eggs): 1st day 8 pictures, 2nd day 10 pictures, 3rd day 15 pictures, 4th day 11 pictures. Answer the following questions:

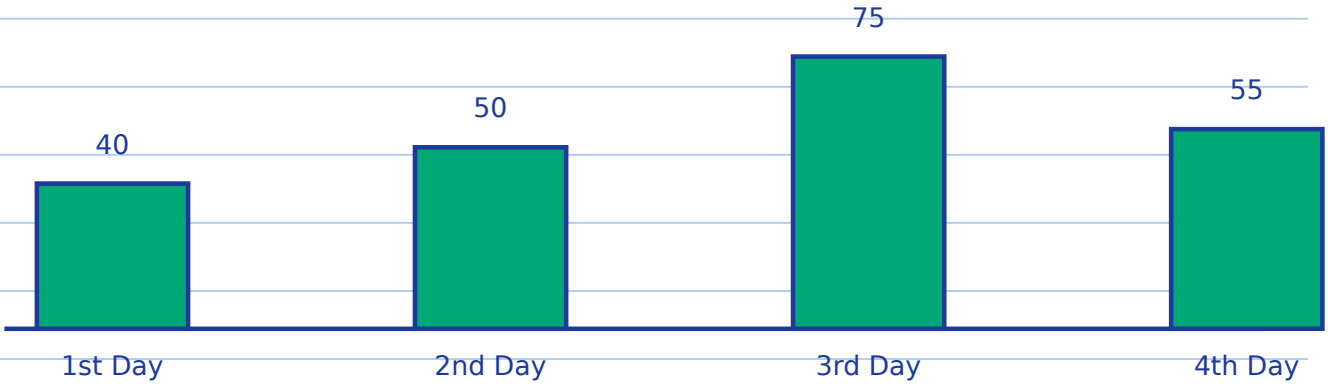
Question: (1) How many eggs were sold on the last (4th) day?

Answer: Step 1: Count the pictures for the 4th day: 11 pictures.

Step 2: One picture represents 5 eggs.

Step 3: Multiply: 11 × 5 = 55 (10 × 5 = 50 and 1 × 5 = 5; 50 + 5 = 55).

Final answer: 55 eggs were sold on the 4th day.



Question: (2) On which day were the least eggs sold?

Answer: Step 1: Find the eggs for each day: 1st day = 8 × 5 = 40, 2nd day = 10 × 5 = 50, 3rd day = 15 × 5 = 75, 4th day = 11 × 5 = 55.

Step 2: The smallest number is 40, which is the 1st day.

Final answer: The least eggs were sold on the 1st day.

Question: (3) On which day were the highest eggs sold?

Answer: Step 1: Compare the four days: 40, 50, 75 and 55 eggs.

Step 2: The biggest number is 75, which is the 3rd day.

Final answer: The highest eggs were sold on the 3rd day.

Question: (4) How many eggs were sold in the first two days?

Answer: Step 1: Eggs on the 1st day = 8 × 5 = 40.

Step 2: Eggs on the 2nd day = 10 × 5 = 50.

Step 3: Add them: 40 + 50 = 90.

Final answer: 90 eggs were sold in the first two days.

Question: (5) What is the difference between the eggs sold on the last two days?

Answer: Step 1: Eggs on the 3rd day = 15 × 5 = 75.

Step 2: Eggs on the 4th day = 11 × 5 = 55.

Step 3: Subtract: 75 - 55 = 20.

Final answer: The difference is 20 eggs.

EXERCISE 57 — (B) The picture graph shows the animals in a zoo (one picture represents 2 animals): Lion 2 pictures, Deer 3 pictures, Monkey 4 pictures, Fox 3 pictures, Rabbit 5 pictures, Elephant 1 picture. Answer the following questions:

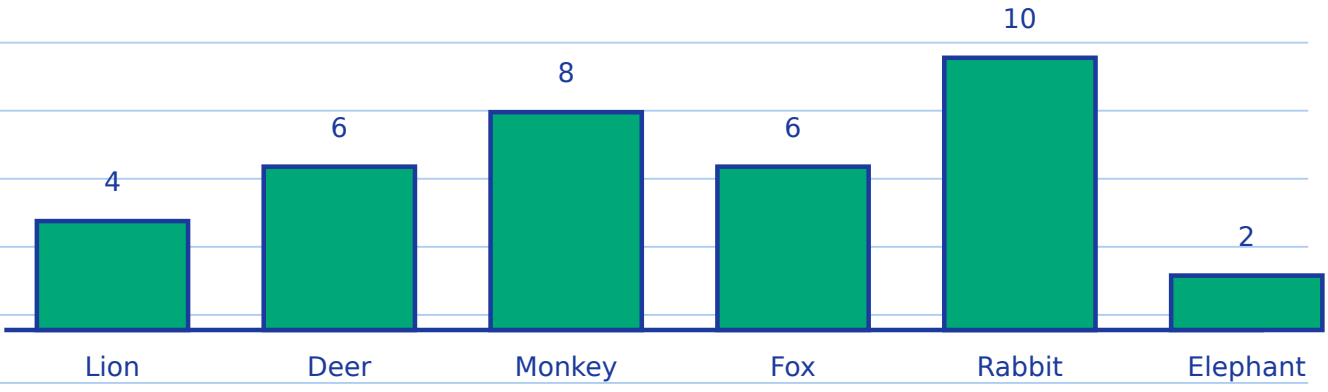
Question: (1) How many lions are there in the zoo?

Answer: Step 1: Count the lion pictures: 2 pictures.

Step 2: One picture represents 2 animals.

Step 3: Multiply: 2 × 2 = 4.

Final answer: There are 4 lions in the zoo.



Question: (2) What is the total number of animals in the zoo?

Answer: Step 1: Find each animal: lions = 2 × 2 = 4, deer = 3 × 2 = 6, monkeys = 4 × 2 = 8, foxes = 3 × 2 = 6, rabbits = 5 × 2 = 10, elephants

= 1 × 2 = 2.

Step 2: Add them one by one: 4 + 6 = 10, 10 + 8 = 18, 18 + 6 = 24, 24 + 10 = 34, 34 + 2 = 36.

Final answer: The total number of animals in the zoo is 36.

Question: (3) How many more foxes are there than elephants?

Answer: Step 1: Foxes = 3 × 2 = 6.

Step 2: Elephants = 1 × 2 = 2.

Step 3: Subtract: 6 - 2 = 4.

Final answer: There are 4 more foxes than elephants.

Question: (4) How many types of animals are there in the zoo?

Answer: Step 1: Count the rows of the picture graph: Lion, Deer, Monkey, Fox, Rabbit and Elephant.

Step 2: Count them: 1, 2, 3, 4, 5, 6.

Final answer: There are 6 types of animals in the zoo.

Question: (5) Which animal is the most in number?

Answer: Step 1: Compare the numbers: Lion 4, Deer 6, Monkey 8, Fox 6, Rabbit 10, Elephant 2.

Step 2: The biggest number is 10, which is the rabbits.

Final answer: Rabbits are the most in number.

Question: (6) Which animal is the least in number?

Answer: Step 1: Compare the numbers: Lion 4, Deer 6, Monkey 8, Fox 6, Rabbit 10, Elephant 2.

Step 2: The smallest number is 2, which is the elephants.

Final answer: Elephants are the least in number.

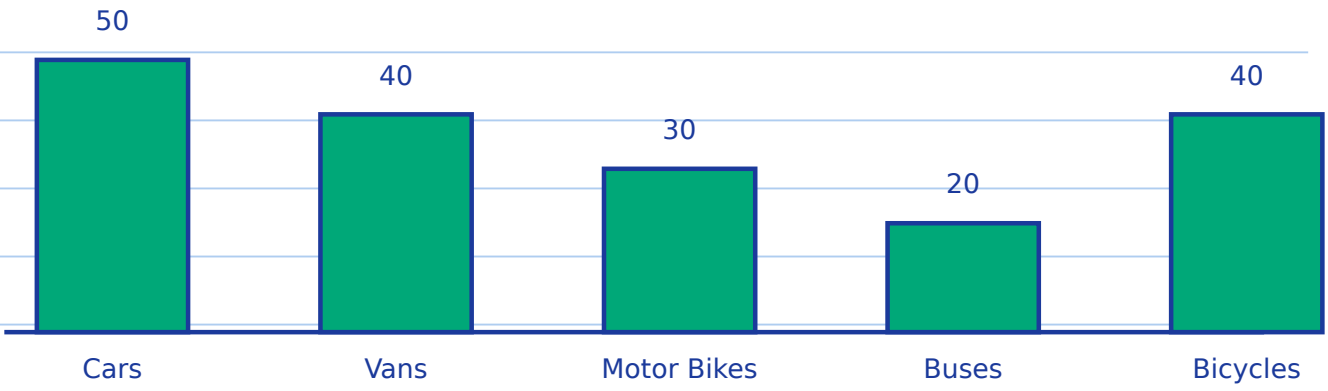
EXERCISE 57 — (C) The picture graph shows the vehicles passing on a road (one picture represents 10 vehicles): Cars 5 pictures, Vans 4 pictures, Motor Bikes 3 pictures, Buses 2 pictures, Bicycles 4 pictures. Answer the following questions:

Question: (1) Which vehicle passed the most?

Answer: Step 1: Find each vehicle: cars = 5 × 10 = 50, vans = 4 × 10 = 40, motor bikes = 3 × 10 = 30, buses = 2 × 10 = 20, bicycles = 4 × 10 = 40.

Step 2: The biggest number is 50, which is the cars.

Final answer: Cars passed the most.



Question: (2) Which vehicle passed the least?

Answer: Step 1: Compare the numbers: Cars 50, Vans 40, Motor Bikes 30, Buses 20, Bicycles 40.

Step 2: The smallest number is 20, which is the buses.

Final answer: Buses passed the least.

Question: (3) How many buses passed on the road?

Answer: Step 1: Count the bus pictures: 2 pictures.

Step 2: One picture represents 10 vehicles.

Step 3: Multiply: 2 × 10 = 20.

Final answer: 20 buses passed on the road.

Question: (4) How many motor bikes passed on the road?

Answer: Step 1: Count the motor bike pictures: 3 pictures.

Step 2: One picture represents 10 vehicles.

Step 3: Multiply: 3 × 10 = 30.

Final answer: 30 motor bikes passed on the road.

Question: (5) How many vans and bicycles passed in all?

Answer: Step 1: Vans = 4 × 10 = 40.

Step 2: Bicycles = 4 × 10 = 40.

Step 3: Add them: 40 + 40 = 80.

Final answer: 80 vans and bicycles passed in all.

Question: (6) How many more cars passed than buses?

Answer: Step 1: Cars = 5 × 10 = 50.

Step 2: Buses = 2 × 10 = 20.

Step 3: Subtract: 50 - 20 = 30.

Final answer: 30 more cars passed than buses.